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Exploring pattern formation in neurobiological networks with time-dependent synaptic connections

by

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Abstract

This thesis uses mathematical models of the brain to investigate localized neural activity, called bumps, which are crucial in working memory. The investigation begins with the classic Wilson and Cowan neural field model extended to include short-term synaptic plasticity (STSP). To address the limitations of the classic model, we transition to a next-generation neural field model derived from networks of quadratic integrate-and-fire neurons. This advanced model treats the firing rate as an emergent property and allows for a dual macroscopic and microscopic perspective. The central aim of this work is the integration of STSP dynamics into this next-generation model, which has not been previously discussed in literature. We perform stability analysis on this combined model, revealing how stability regions are altered. Finally, the model is applied to study wave entrainment in response to apparently moving stimuli. We show some nuances that are implicated in the oscillatory dynamics inherent in the next-generation model.

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Popular Summary

The human brain contains 86 billion neurons [2]. When questioning how to model the brain, perhaps the most fundamental starting point would be with the neurons themselves. They transmit and process electrical signals throughout the brain. The connections between individual neurons are called synapses. The strength at which two neurons are connected is not a static property, but can instead dynamically change in an activity-dependent manner. This idea is referred to as synaptic plasticity. We can distinguish between short-term and long-term synaptic plasticity. The former is of our main interest during this text, and is important on the timescale of seconds. Within short-term synaptic plasticity, we can distinguish between depression, the weakening of connections between neurons, and facilitation, the strengthening of such connections.

Considering the inherent complexity of network of neurons, it is not trivial how to discuss brain phenomena resulting from them. Therefore, developments were made mainly by Wilson and Cowan to not model individual neurons but rather view the brain as a field on which activity occurs [19]. This established neural field theory. Their work has been expanded to include short-term synaptic plasticity. Such models allow for capturing the dynamics when a specific group of neurons located together collectively turn on, referred to as a ‘bump’ of activity. This principle underpins the mechanism of working memory.

However, it must be noted that the Wilson and Cowan-type model is derived from phenomenological arguments, meaning it lacks a biophysical backbone. In addition to this are some further drawbacks. First, the obvious fact that we completely lost sight of *individual* neurons. Moreover, there are some missing features such as correlations of neurons. To that end, so called next-generation neural models were proposed to address the aforementioned matters. We highlight the work by Montbrió, Pazó, and Roxin [36], which uses an individual neuron model to derive mean field equations. This means we can simultaneously consider the microscopic and macroscopic properties.

What has not been well described in literature, is how the next-generation neural model behaves under short-term synaptic plasticity effects. In this thesis we study precisely that. Using mainly numerical methods and bifurcation analysis, we have looked at exploring the system and considering the stability of neural activity states. We describe how depression and facilitation cause activity to be elevated (dominant facilitation) or lowered (dominant depression). This has consequences when repeatedly exposed to an external stimulus. In one of our

simulations, we considered a stimulus which turned on and off multiple times. In the case of dominant depression and mild facilitation, we found that the high neural activity state could no longer be achieved after four stimulus pulses. But upon only turning off the mild effect of facilitation, the high state could not be reached after already three pulses.

Moreover, we look at the behavior of the system when a moving visual stimulus⁽ⁱ⁾ is introduced. A moving bump of neural activity can either successfully follow such a stimulus (called entrainment) or, whenever, say, the stimulus moves too fast, the bump loses the stimulus (failed entrainment). After reviewing wave entrainment with the classical model, we looked at the next-generation case. We conclude that the oscillatory behavior of the system variables can lead to partial entrainment failures.

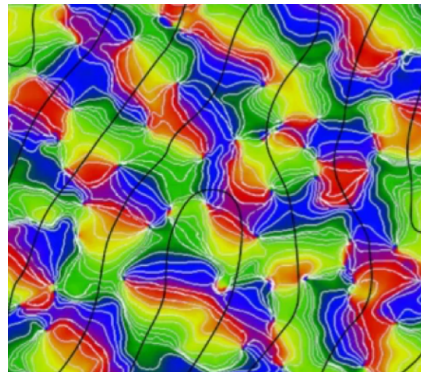
Overall, this thesis demonstrates that combining the next-generation neural models with short-term synaptic plasticity is a way to more accurately represent neural dynamics. However, there are various consequences to be aware of that arise as a result of the combination, predominantly the effect on steady state values, their stability, and how previous neural activity influences current dynamics. These are further examined in the main text.

⁽ⁱ⁾As will become clear, we look at an *apparently* moving stimulus, which flashes at a specific position, then turn off and reappears slightly spatially shifted, *et cetera*.

Introduction

This thesis is in the field of mathematical neuroscience, an exciting branch that, broadly speaking, applies mathematical tools towards models from the field of neuroscience. Neuroscience itself investigates how the nervous system of humans and other animals is organized to generate, for instance, responses in muscles and organs, but also higher-order functions such as attention, memory, thinking, and language [2, 20]. The brain, part of the nervous system, consists of many neurons that together form intricate networks and show complex patterns. These neurons are connected primarily by chemical synapses, through which they are able to communicate [60, 2]. We continuously receive external stimuli, and synaptic processes are involved in responding to such stimuli [34]. In order to store and process information, the brain relies on activity-dependent changes in strength within these synapses, called synaptic plasticity [30]. This plasticity can occur over different timescales. In this text we consider short-term synaptic plasticity, acting on the timescale of seconds, where synapses can transiently strengthen or weaken as a result of previous activity.

Short-term synaptic plasticity plays a crucial role in working memory (WM). WM refers to the ability to temporarily maintain and manipulate information [16]. It is involved in higher cognitive functions such as planning, reasoning, decision making, and language comprehension [28]. Studies using non-human primates and rodents have revealed that in a oculomotor delayed response task. In these experiments, they record neural activity while the animal is presented with a visual stimulus, which must then be remembered, typically for 1 to 5 seconds. The results show that specific neurons are elevated during the entire delay, so even when the stimulus is removed. For the aforementioned we reference [16, 53, 15, 28]. These neurons showing persistent activity are tuned for specific features. For instance, some might be tuned to location and direction of motions, while others are tuned to more abstract concepts [16].



Cortical orientation map. Taken from [39].

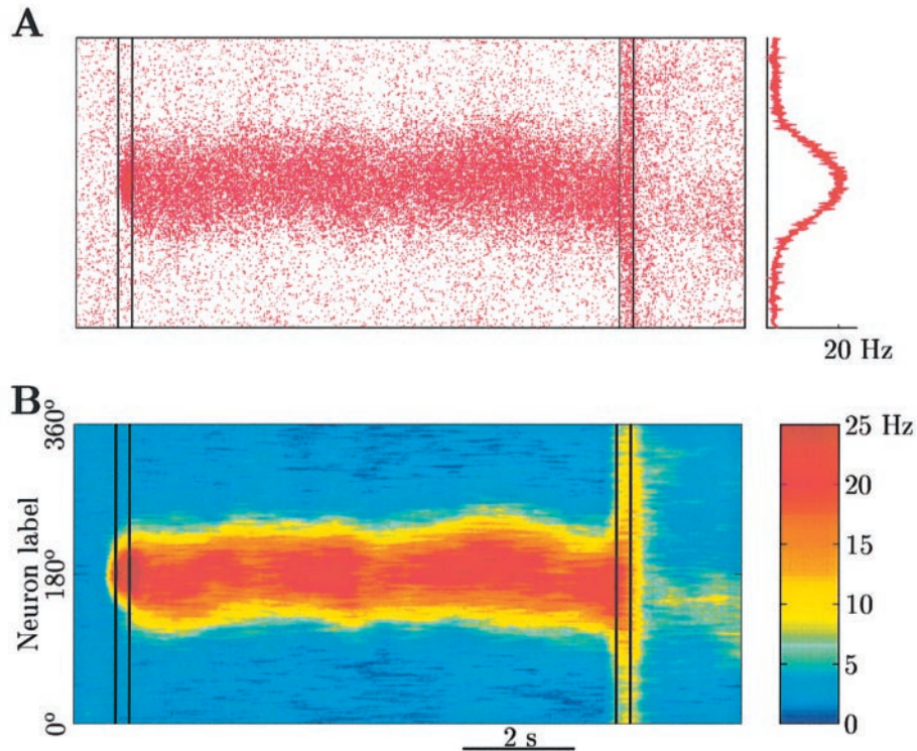
Since Imagine a visual stimulus consisting of a set of bright horizontal bars against a dark background. Such a stimulus reaches the retinal photoreceptors

in the eye, and that information is transmitted via the optic nerve to the primary visual cortex. It has been found that neurons in this area are selectively tuned to stimulus orientation, meaning distinct populations of neurons respond maximally to stimuli at specific angles [39, 49]. Moreover, neurons sharing the same orientation preference are spatially clustered into *orientation columns*, see figure above. As a consequence, when presenting horizontal bars, one is to observe elevated spiking activity in aligned groups of neurons. During this process, these neurons excite their neighbors thereby reinforcing the pattern. However, neurons that are tuned for a different orientation preference are inhibited. This leads to the creation of a localized bump of neural activity. Also see the figure below, which shows WM maintained by a localized bump. In mathematical models, we often consider bumps as idealized descriptions of the persistent neural activity seen in WM [15, 14]. In these models, bumps can exhibit instability of various types, such as drift instability, where the bump can wander from its initial position during the delay period. This can then be related to losing (forgetting) the memory cue [16]. We emphasize that mathematical models of neural activity provide a valuable general framework for investigating phenomena such as, but not limited to, the mechanisms governing WM.

Note that other mechanisms of WM include synaptic plasticity, which can also stabilize or destabilize bumps [14, 35]. Papers have even shown that many relevant neurons do not fire persistently during the delay period, but rather display dynamic activity patterns and periods of inactivity, thereby challenging the earlier idea and underlining the importance of synaptic plasticity [30]. This lead to models of WM where information is stored by facilitating neurons, and retrieved by activity propagation through the network [30].

The main aim of this thesis is to explore models of the cortex with short-term synaptic plasticity dynamics, generally looking at pattern formation. Due to the sheer complexity networks of neurons exhibit, it quickly becomes a challenge to describe the dynamics governing larger-scale phenomena, such as the output or function of large subsets of neurons, or waves moving across the brain. A particularly successful approach in circumventing this issue has been the development of neural field models, pioneered by Wilson and Cowan in 1972 [19]. Instead of tracking the activity of individual neurons, this ‘classic model’ describes the activity of a population of neurons. This macroscopic perspective has been explored and expanded upon, such as the inclusion of short-term synaptic plasticity [14, 54].

A more fundamental limitation of the type of models by Wilson and Cowan is the phenomenological basis. More recently, a next-generation of neural models has been brought forward. Specifically, a model developed by Montbrió, Pazó, and Roxin [36] has offered a macroscopic description of neuron networks. By applying statistical methods to large networks of individual quadratic integrate-and-fire neurons, they *derived* macroscopic equations. The benefit is that the obtained



Working memory represented by a bump. Taken from [15].

system is derived from the microscopic states, in turn allowing us to consider individual neuron spikes, i.e., to have a microscopic view.

The aforementioned has been well described in literature. However, the inclusion of short-term synaptic plasticity dynamics in the next-generation models has to our knowledge not been considered yet. The objective of this text is therefore to explore the impact of these dynamics in the next-generation setting.

In doing so, we first review the basics of neuroscience to better understand the classic neural field model that will subsequently be discussed. We study the classic model initially without and later with synaptic plasticity, looking mainly at the existence and stability of activity bumps. Afterwards, we move on to the next-generation neural models. We progressively enhance the model by Montbrió *et al.*, in the sense that synaptic plasticity is added, and we also consider how to spatially extend the model, as treated in [48]. In the various sections, we use numerical methods to develop simulations, as analytical progress becomes unfeasible with increasing complexity of the models.

Chapter 1

Preliminaries from Neuroscience

In this thesis, we will be looking mainly at models of the brain, and almost exclusively in specific the cortex. To better understand and appreciate these models, as well as to have a better overview of the corresponding fundamental biological concepts, we shall review some background theory of neuroscience. Below we follow [2] as a main source of inspiration.

The brain is part of the central nervous system and consists of two primary types of cells: neurons (nerve cells) and neuroglia (glial cells). The glial cells⁽ⁱⁱ⁾ mainly play a crucial role in providing support and maintenance to neurons, thereby helping to maintain homeostasis. They themselves do not conduct any electrical signals throughout the brain. The neurons, on the other hand, are responsible for receiving, processing, and transmitting signals. This in turn allows for a network of neurons to function together.

A schematic of a neuron is seen to the right, *Figure 1.1*. Extending from the cell body is the axon, which extends down and ultimately divides in branches. Such a branch then leads to a terminal bouton. This is where the neuron stops, and a small gap, or cleft, is found, after which the next neuron's dendrites are located. This junction where nerve cells communicate with one another is referred to as the synapse.

Signals sent across the brain can roughly be divided in two types. First, electrical signals, which are conducted along dendrites and axons of neurons. When such a signal reaches the synapse, typically⁽ⁱⁱⁱ⁾ another means of communication

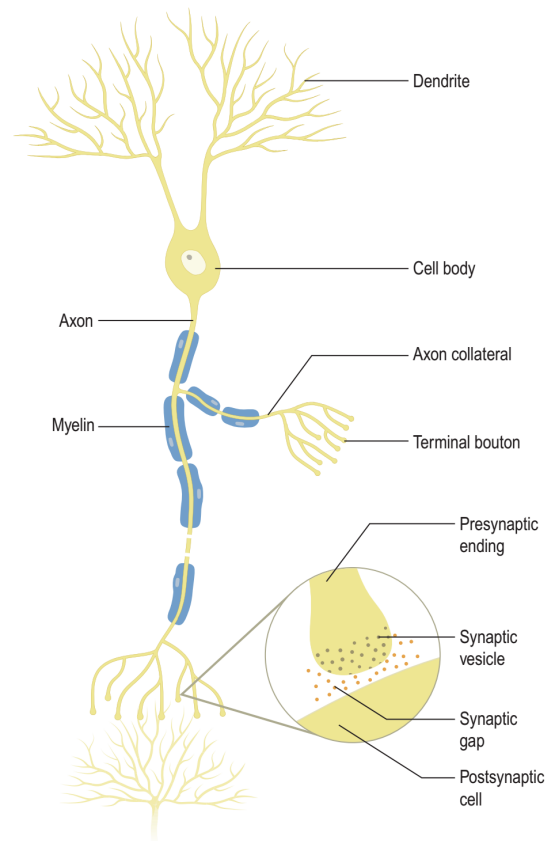


Figure 1.1: Schematic of a nerve cell. Taken from [20].

⁽ⁱⁱ⁾‘Glia’ is a Greek word that translates to ‘glue’. The glial cells can indeed, in a sense, be seen as a type of glue in which the neurons are embedded and maintained. However, there is no evidence that they actually *bind* neurons together [2].

⁽ⁱⁱⁱ⁾There also exist *electrical synapses*, often called *gap junctions*. They are relatively abundant during the development of the brain, and less so for a mature one. During development, the interactions between this type of synapse are important in establishing neural circuitry [44].

takes over, namely chemical signals are generally used in synapses. These chemicals are called neurotransmitters, and are released by the pre-synaptic bouton and captured at the post-synaptic site.

Below, we will zoom in on two aspects relevant for what is to come. We look at single electrical pulses, called action potentials, and moreover investigate synapses in some further detail.

1.1 Action Potentials

Neurons can generate and conduct electrical signals. A single signal is referred to as an action potential, or often a *spike* for brevity. Inside neurons, there is a negative resting potential, usually around -70 mV. This resting potential is created and maintained by ion pumps and ion channels which are situated in the neuronal membrane. The pumps can actively move particular ions from one side of the membrane to the other, thereby creating an ion concentration gradient. This process requires energy to be inputted. Next to this, the channels are selectively permeable to specific ions and can let a particular ion pass through the membrane by letting it follow the concentration gradient, meaning no energy needs to be inputted. There are various types of ion channels. Some ion channels are voltage-gated, meaning they are opened or closed depending on the membrane potential. But other types are controlled by the binding of certain ligands, or yet other mechanisms. The interplay between pumps and channels leads to a certain equilibrium which is indeed the resting potential, but the mechanisms of pumps and channels are also important in creating action potentials.

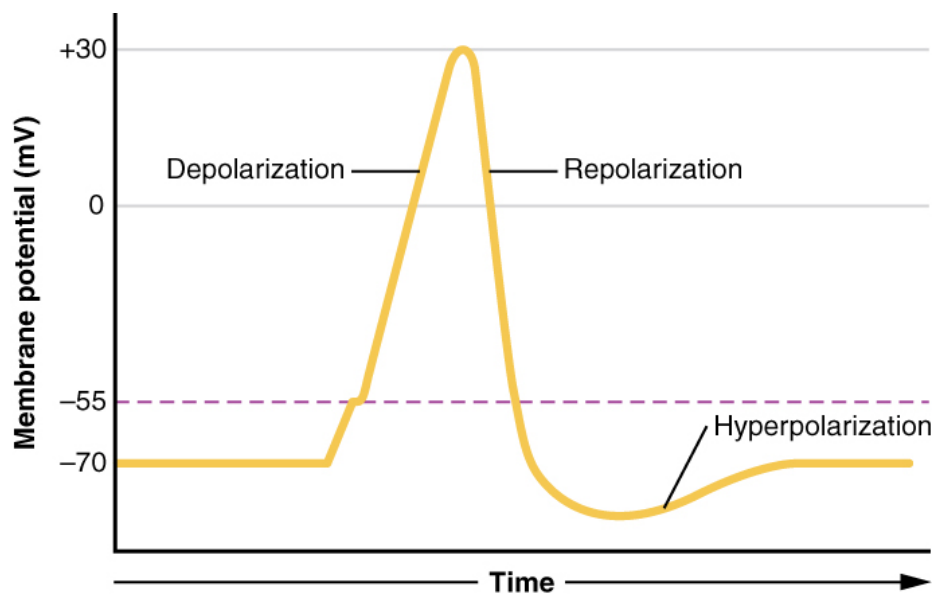


Figure 1.2: The membrane potential during an action potential (spike). Here, the resting membrane potential is -70 mV and the threshold is -55 mV (purple dashed line). At some time, a stimulus is applied causing the potential to rise beyond the threshold, after which the effects of depolarization and repolarization are seen, respectively. The repolarization undershoots the resting membrane potential, which we call hyperpolarization, after eventually returning to its value. From: [56].

For an action potential to be produced, we are to focus on the voltage-gated ion channels. More specifically, those selectively permeable to sodium (Na^+) and potassium (K^+). We mentioned that

their opening position depends on the membrane potential. If a stimulus reaches the axon of a neuron, it can bring the membrane potential up, i.e., cause depolarization. If the depolarization is sufficient enough to reach a critical threshold, then this leads to opening of the Na^+ channels. The concentration of Na^+ ions is much lower inside the axon compared to outside, as a result of the ion pump, so when the Na^+ channels open this will cause the Na^+ ions to rapidly enter the neuron (influx) which causes strong depolarization. After some slight delay, the K^+ channels open allowing K^+ ions to exit the neuron (efflux), in turn yielding to repolarization. Moreover, the Na^+ channels close again, leading the potential back to the its normal resting value. This process is schematically outlined in *Figure 1.2*. Here, one may observe that the resting membrane potential is undershot at the end of an action potential (hyperpolarization). This is because the K^+ channels start to close only after reaching the resting membrane potential. The closing of these channels does not happen instantaneously, which accounts for the action potential curve to undershoot. Notice that during this hyperpolarization period, which generally lasts for two milliseconds, it is much harder to reach the threshold again to create another action potential. This is why it is also called the refractory period.

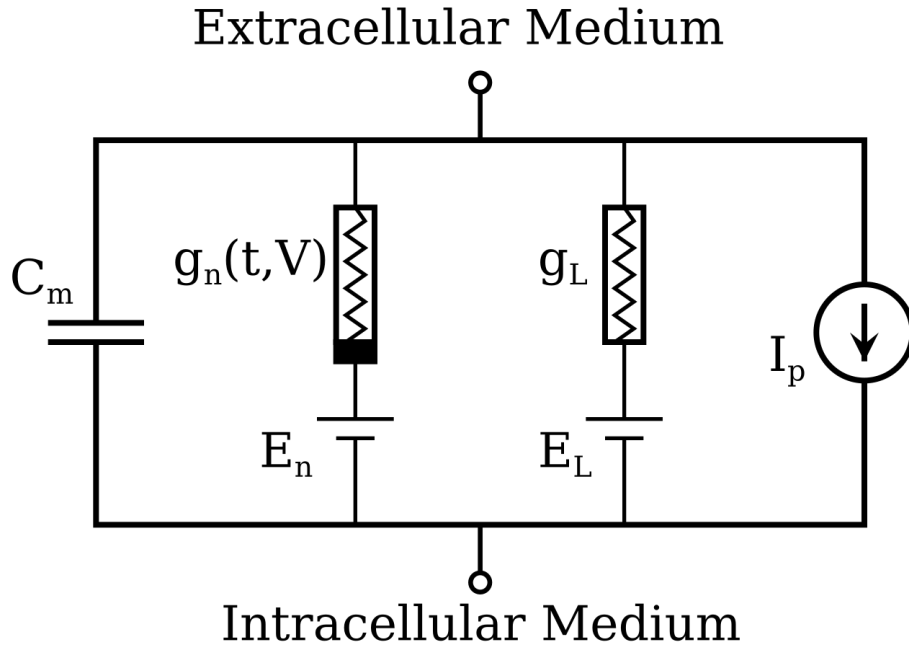


Figure 1.3: Schematic outline of the Hodgkin-Huxley model. From [58].

Many of the aforementioned mechanisms behind the construction of action potentials were first discovered by Alan Hodgkin and Andrew Huxley. They moreover formulated a mathematical model of action potentials in terms of ion channels and pumps, for which they obtained a Nobel prize in 1963. The model is readily referred to as the Hodgkin-Huxley model. For our purposes, we do not include the mathematical description of the dynamical system here, instead we refer to [50, 25]. A schematic outline of the model's components is seen in *Figure 1.3*.

Like we mentioned, action potentials propagate through the axon, ultimately arriving at the synapse. By far the majority of these synapses are *chemical* synapses. They do not continue to carry an action potential electrically. How an action potential is processed at such a synaptic site is further outlined below.

1.2 Synapses

When an action potential invades a chemical synapse at the presynaptic ending, it is not continued electrically but instead the signal can cause synaptic vesicles, which are full of neurotransmitters, to be released into the synaptic gap (ref. *Figures 1.1 and 1.4*). These neurotransmitters can subsequently bind to receptors in the postsynaptic membrane, causing postsynaptic ion channels to open or close depending on the type of neurotransmitter and type of receptor. This leads to a certain postsynaptic current to be created. For example, if an ion channel opens that is permeable to, say, K^+ ions, then this causes depolarization. This means it is more likely for the subsequent neuron to spike as it is pushed closer to the threshold. Hence, in such an instance we speak of an *excitatory* postsynaptic potential. Analogously, Cl^- ions lead to hyperpolarization or stabilization, in which case we speak of an *inhibitory* postsynaptic potential.

The above process is referred to as synaptic transmission. The efficacy of synaptic transmission is not a constant property. Instead, the efficacy of synaptic transmission depends on previous neural activity in the synapse. That is, synapses have the ability to become weaker or stronger over time in response to changes in its activity. This concept is called synaptic plasticity. In this thesis, we shall be greatly concerned with short-term synaptic plasticity.

This form of plasticity is characterized by activity-dependent and transient changes in synaptic strength on the timescale of milliseconds to minutes [22]. Changes also occur on longer timescales, say, hours to even years, upon long-term synaptic plasticity. Generally, STSP is pre-synaptically organized, meaning this effect is due to changes at the presynaptic neuron, while LTSP is post-synaptically organized [2].

In light of the goal being to model STSP, let's look at some more detail how it works. A synapse has about 200 synaptic vesicles, see *Figure 1.5*. The presynaptic ending is densely packed with these vesicles, only a fraction of which are available for immediate release; those are in the readily releasable pool (RRP). In general, RRP size and probability of release of each vesicle in RRP together determine synapse strength [31].

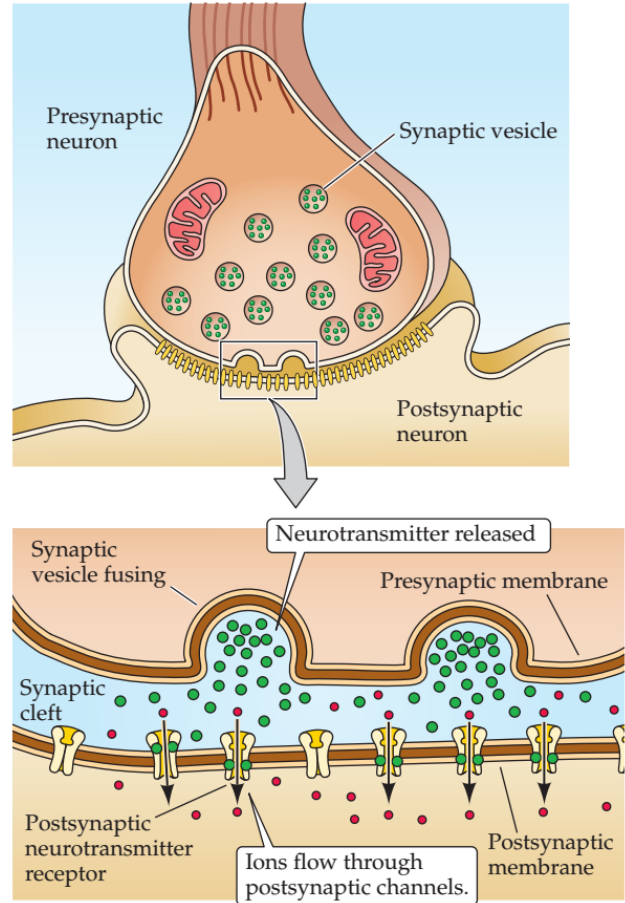


Figure 1.4: Schematic of a synapse. Amended from [2].

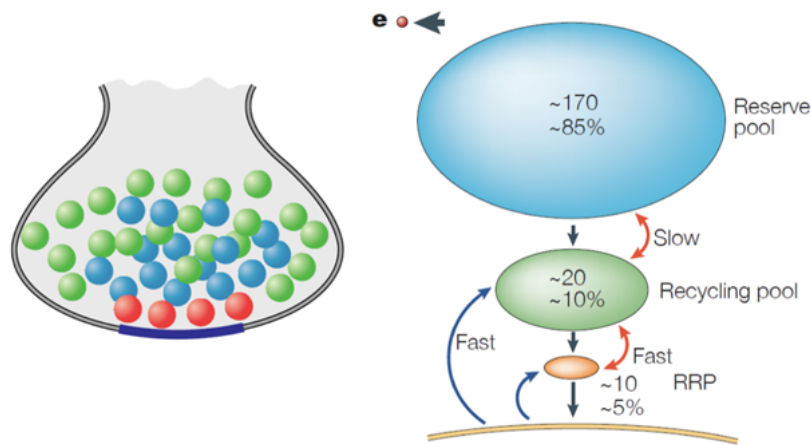


Figure 1.5: Overview of synaptic vesicles in various pools. Vesicles in the RRP are available for immediate release.

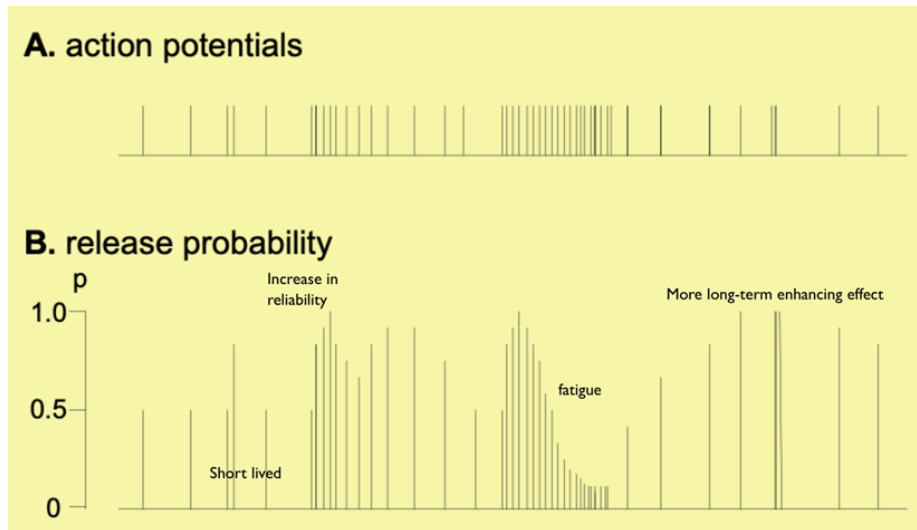


Figure 1.6: Changes in release probability with invading action potentials.

Whenever an action potential invades the presynapse, this in principle triggers vesicle release, *but not always*. That is to say, synaptic transmission is unreliable. For instance, in the large auditory calyceal synapse a mean release probability of 0.6 was observed, but some connections showed a probability less than 0.3, while others approached values near 1 [41]. This large variability in release probabilities is what allows for synaptic plasticity. Signals can be strengthened or weakened. With short-term synaptic plasticity, the release probabilities can change over time, depending on neural activity. This is reflected in *Figure 1.6*. If multiple action potentials arrive in quick succession, the reliability of vesicles release increases. After a while, fatigue can be seen. The dynamics accountable for the weakening and strengthening are called short-term synaptic depression and facilitation, respectively.

Definition 1.1 (Short-term synaptic depression). [2, 26]. Short-term synaptic depression (STSD) is caused by progressive depletion of the RRP leading to suppressed rates of vesicle release. Recovery of synaptic strength occurs once the RRP is replenished through refilling from the reserve pool, see *Figures 1.5* and *1.7*.

Definition 1.2 (Short-term synaptic facilitation). [2, 26]. Whenever action potentials invade the presynaptic ending at high frequency, a transient increase in synaptic strength is observed, termed short-term synaptic facilitation (STSF). Also see *Figure 1.7*.

These dynamics interplay with each other. These forms of short-term synaptic plasticity are not mutually exclusive. A synapse may, for example, exhibit strong depression while also showing facilitation. In general, neurons display both forms, with sometimes one dominating, while in other cases, both mechanisms may be equally prominent. We refer to [26, 23] for more information.

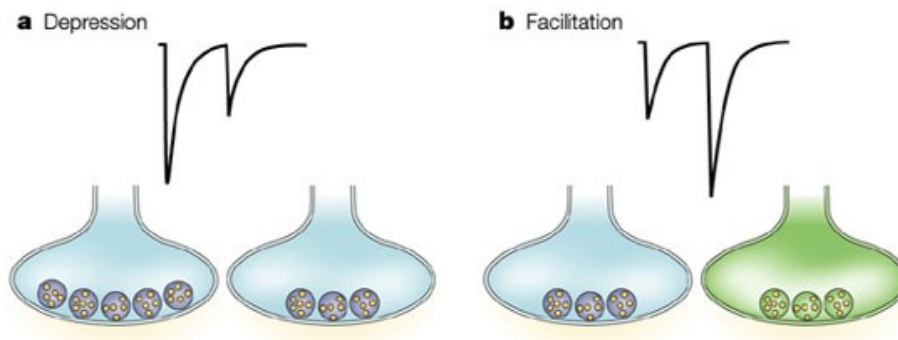


Figure 1.7: Upon STSD, vesicles are depleted causing suppressed rates of release. With STSF, synaptic strength is transiently elevated due to multiple action potentials arriving. Taken from [5].

1.3 Neural Field Model

It has become evident that the microscopic view above is quite elaborate and complex. Beyond the processes that were discussed there are *many* more that were not even touched upon. This complexity hints that we can expect great difficulty when trying to describe the entire brain from the bottom up. Take for instance the Hodgkin-Huxley model discussed earlier (*Figure 1.3*). Although it proves to be helpful at describing action potential in a single or a small group of neurons, it is not trivial how it allows for insight at the level of the entire cortex. Such scaling would indeed lead to staggering complexities and inevitable computational difficulties.

So, instead of trying to map out the intrinsic complexities inherent with billions of interacting neurons, one may turn their attention to another approach. Namely that of *neural field theory*, where a focus is put on the large-scale population properties rather than the particularities at the discrete level. The general aim, historically, has been to better understand the dynamics of cortical tissue [19]. The cortex is the outer layer of the cerebrum, characterized by its convoluted nature consisting of ridges and grooves (called gyri and sulci, respectively), see *Figure 1.8*. These convolutions serve to maximize the cortical surface area, which amounts to approximately 1 m^2 per hemisphere [20]. Conceptually, if this folded structure were to be flattened, it would essentially form a thin sheet of tissue. This sheet would be densely packed with approximately ten billion neurons [52]. Such a density means that taking even a small subsection, say, a $1 \times 1 \times 1 \text{ mm}$ cube of the cortex contains thousands of neurons. The specific goal of neural field theory, then, is to model the evolution of activity of such a patch, disregarding the notion of individual neurons. Such an approach is naturally well-suited to investigate larger-scale phenomena such as traveling waves, which are difficult to explain from a single-neuron perspective.

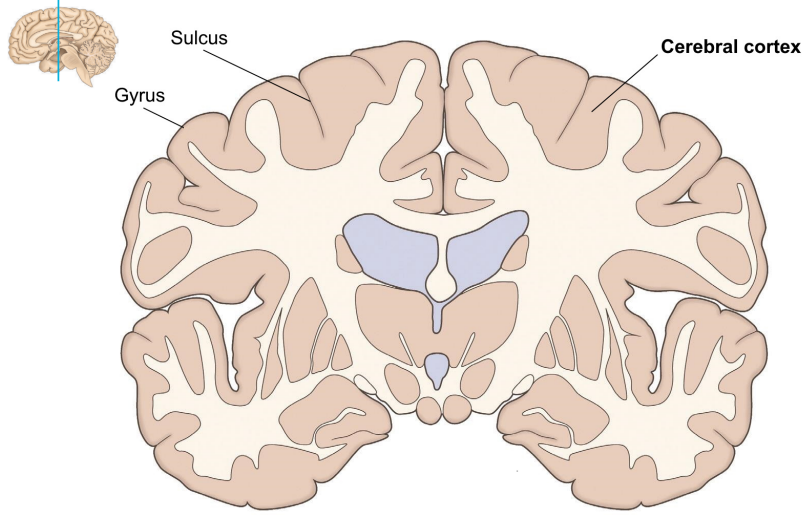


Figure 1.8: Coronal view of the brain (see inset). The outer, darker layer is the cerebral cortex, which consists of sulci and gyri. Amended from [20].



Figure 1.9: Sketch of connections in the cortex: (a) Rough outline of the brain, where an intracortical connection is drawn between position r' and r . (b) Viewing the connection from a piece of flattened cortex.

Neural fields were initially developed by Wilson and Cowan in 1972, and later further advanced by Amari^(iv). The idea is that we model the cortex as a continuum, i.e., $\Omega \subset \mathbb{R}^3$, in which there are infinitely many neurons. The model is given by [25, 19]:

$$\begin{aligned} \partial_t u(r, t) &= -u(r, t) + \int_{\Omega} w(r, r') f[u(r', t)] dr' + I(r, t), \quad (x, t) \in \Omega \times \mathbb{R}_{>0}, \\ u(x, 0) &= u_0(x), \quad x \in \Omega. \end{aligned} \quad (1.1)$$

Here, $u(r, t)$ represents the neural activity at position $r \in \Omega$ at time t . The part $\partial_t u(r, t) = -u(r, t)$ are the first order synaptic dynamics. It is seen that this is responsible for a natural decay of neural activity upon the absence of any other dynamics, leading u to a baseline resting state. Next, the term $I(r, t)$ lastly appended is the external input applied to position r at time t . This allows us to consider what happens when, for instance, a visual stimulus is applied. Notice the time dependence of the external stimulus, meaning we could explore what happens to neural activity if one is to observe a black screen with occasionally bright white flashes appearing.

^(iv)Amari had realized that taking a Heaviside firing function allows for analytical progress. In this Heaviside case, commonly referred to as the ‘Heaviside world’, he was able to provide explicit solutions.

Finally, then, *Equation (1.1)* also contains a convolution term, where we have two more functions: w and f , the synaptic kernel and the firing function, respectively, further explained below.

Synaptic Kernel Function

In modeling the dynamics of neural populations, it is crucial to describe how neurons are connected. More specifically, if we have a look at neurons at position r , we want to know how it is connected to neurons at position $r'^{(\text{v})}$. To this end, we define the synaptic kernel, or weight function, $w(r, r')$, as the synaptic connection strength from the neuron population at r to the one at r' . Generally speaking, cortical connection strength is said to be distant dependent, whereby most cortico-cortical connections project locally, meaning that at a specific location of neurons, most connections are short-range and relatively few are long-range projections that link distant regions of the cortex [24, 42, 4]. This principle forms the basis of two primary assumptions that are often made for w [19].

- **Homogeneity.** The synaptic kernel satisfies

$$w(r, r') := W(r - r'),$$

for all $r, r' \in \Omega$. That is, the function is said to be translationally invariant. In subsequent sequences we often neglect the difference in capitalization.

- **Even parity.** The synaptic kernel is even, so

$$W(-z) = W(z),$$

for all $z \in \Omega$. Whenever Ω is one-dimensional, this implies isotropy of the kernel function, whereby it is purely distance-dependent.

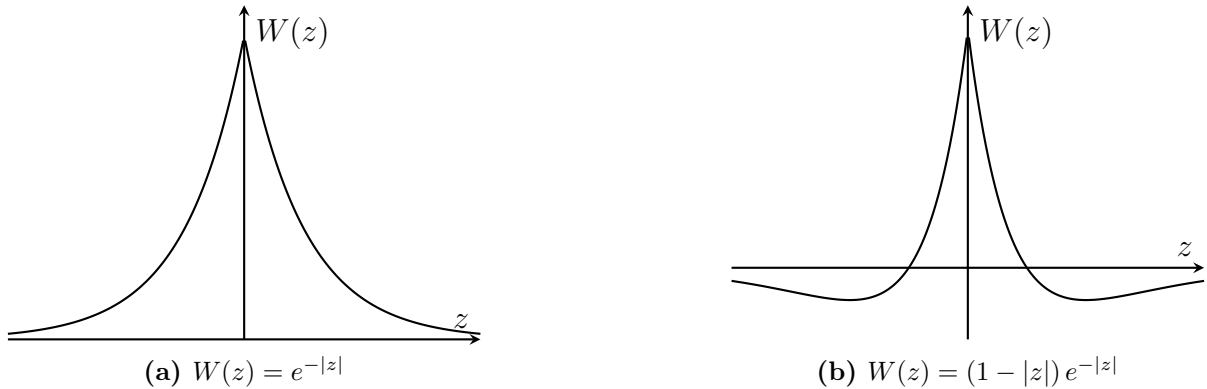


Figure 1.10: Some examples for the synaptic kernel function.

Some other important regulatory assumption we make is that w is continuous, and must be integrable, so

$$\int_{\Omega} w(z) dz < \infty.$$

^(v)Indeed, w depends only on spatial locations. However, we know that synaptic plasticity implies w is not constant with time. This will be dealt with later, when we introduce the dynamics for STSP.

Lastly, we assume that $\lim_{z \rightarrow \pm\infty} w(z) = 0$. Now, let us consider some functions that are often picked for w , see *Figure 1.11*. In (a), a relatively straight-forward function is chosen: $W(z) = \exp(-|z|)$. It is immediately clear that both assumptions are satisfied. Connections are strongest between nearby positions and become weaker with increasing distance. Another choice could be the one made in (b): $W(z) = (1 - |z|) \exp(-|z|)$. Around the center this initially corresponds to the previous, but we see the function become negative. The shape is often called a ‘Mexican hat’. This synaptic kernel reflects the idea that excited neurons tend to excite their neighbors, while inhibiting those farther away.

There are yet many more examples for W imaginable. Often we shall choose (b), but as will become clear, sometimes another synaptic kernel form is picked.

Firing Rate Function

The output of the neurons is given by f , referred to as the firing rate function. Its image is typically between zero and one, and any specific output represents the fraction of neurons that are active [46]. Experimental data suggests that f should approximately follow a sigmoid function [46, 59]:

$$f(u) = \frac{1}{1 + \exp[-\mu(u - \theta)]},$$

where μ controls the steepness (‘gain’), and θ satisfies $f(\theta) = 0.5$ (‘threshold’).

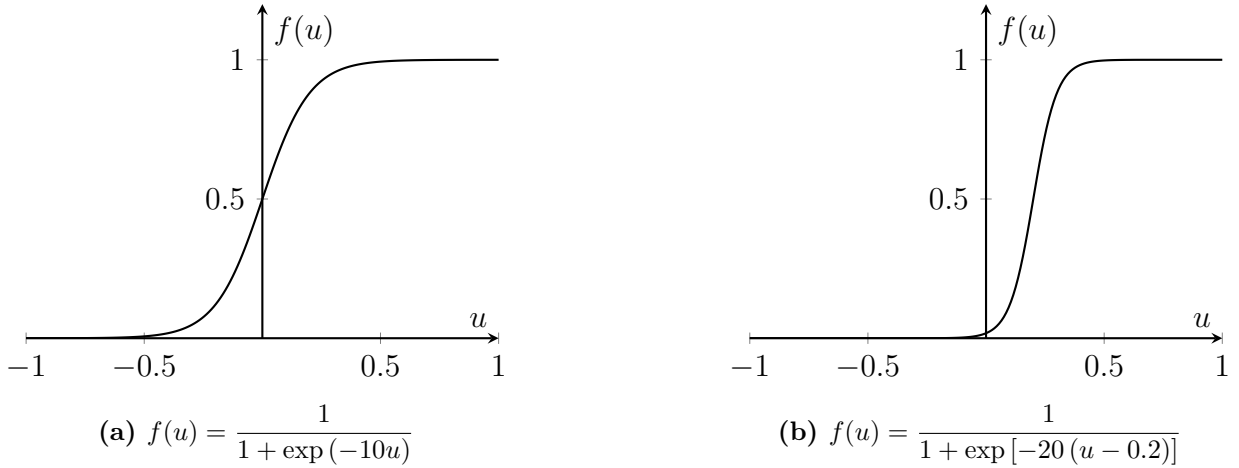


Figure 1.11: A sigmoid as the firing rate function: **(a)** We have a sigmoid where $\mu = 10$ and $\theta = 0$. It is seen that $f(\theta) = 0.5$. **(b)** A sigmoid where $\mu = 20$ and $\theta = 0.2$. A higher gain leads to the sigmoid looking more like a Heaviside function.

Note that θ shifted the graph to the right.

As previously noted, in the ‘Heaviside world’, brought forward by Amari to enable explicit solutions, it is assumed that the firing function follows a Heaviside:

$$f(u) := \mathcal{H}(u - \theta) = \begin{cases} 1 & u > \theta, \\ 0 & u \leq \theta. \end{cases}$$

This is equivalent to considering a sigmoidal function in the high gain limit, $\mu \rightarrow \infty$.

Chapter 2

Methods

The goal is to investigate the neural field equation through computational methods. All simulations presented in this work were carried out using the `Python` programming language, largely with the `Numpy` and `Scipy` libraries. Outlined below are the numerical methods to arrive at these simulations. We mainly follow the thought of [21].

Having reviewed the structure of the cortex, we know that it can be viewed as a large, thin sheet. Hence, in general, an easy and effective way to model the cortex is to pick a ring of some width $2L$, where $L \in \mathbb{R}_{>0}$:

$$\Omega := \mathbb{R} \setminus 2L\mathbb{Z}.$$

Here, one must pick $L \gg 1$ to make sure that characteristic length scale of the synaptic kernel w is sufficiently small compared to the spatial domain [14, 48, 21]. Unless specified otherwise, we choose the domain $\Omega := [-L, L)$ with periodic boundary conditions^(vi). Naturally, the synaptic kernel function is amended to become a $2L$ -periodic extension of w where $w_p(z) = w(z)$ for all $z \in [-L, L)$. We assume $I \equiv 0$, so now the neural field equation from before, *Equation (1.1)*, translates into the following system:

$$\begin{aligned} \partial_t u(r, t) &= -u(r, t) + \int_{\Omega} w_p(r, r') f[u(r', t)] dr', & (r, t) \in \Omega \times [0, T], \\ u(r, 0) &= u_0(r), & r \in \Omega, \end{aligned} \tag{2.1}$$

where $T \in \mathbb{R}_{>0}$ is some time at which the simulation terminates. To proceed with the numerical simulation, the continuous domain Ω must first be discretized. We divide the interval $\Omega = [-L, L)$ into n evenly spaced points. We let n be even, and moreover it should be sufficiently large to ensure a good approximation of the neural field model. Typically, we choose $n = 2^{10}$. The grid points are then defined as

$$r_j = -L + (j - 1)h,$$

where $j \in \{1, \dots, n\}$ and $h = 2L/n$ is the grid size (the spatial step size between grid points). Notice that the grid size h is determined by discretizing the interval $[-L, L]$, requiring $n+1$ evenly spaced points. However, since the objective is to approximate $u(r, t)$, and $u(L, t) = u(-L, t)$ by periodicity, the point at $r = L$ is redundant and hence dropped. That is, we approximate $u(r, t)$ just for $r \in [-L, L)$ meaning there are indeed n grid points r_j .

^(vi)Or, *mutatis mutandis*, the domain $\Omega := (-L, L]$.

By evaluating the system at each of these grid points r_j , we obtain

$$\begin{aligned}\partial_t u(r_j, t) &= -u(r_j, t) + \int_{-L}^L w_p(r_j - r') f[u(r', t)] dr', \quad j \in \{1, \dots, n\}, \\ u(r_j, 0) &= u_0(r_j).\end{aligned}$$

Here, the integral term can be approximated with numerical integration using the composite trapezium rule. Generally, for a function g , this rule is given by [47]

$$\int_{-L}^L g(r') dr' \approx \frac{h}{2} (g(r_1) + g(r_{n+1})) + \sum_{j=2}^n g(r_j) h.$$

Now, if g is some $2L$ -periodic function then $g(r_{n+1}) = g(L) = g(-L) = g(r_1)$, and hence the formula simplified to

$$\int_{-L}^L g(r') dr' \approx \sum_{j=1}^n g(r_j) h.$$

Applying this to our integral, with $g(r') = w_p(r_j - r') f[u(r', t)]$, and letting $U_j(t)$ denote the approximation of $u(r_j, t)$, we arrive at the following system of ordinary differential equations:

$$\begin{aligned}U_j'(t) &= -U_j(t) + h \sum_{k=1}^n w_p(r_j - r_k) f[U_k(t)], \quad t \in [0, T], \\ U_j(0) &= u_0(r_j),\end{aligned}$$

which can be expressed more compactly in vector form. Let $U(t) = (U_1(t), \dots, U_n(t))^T \in \mathbb{R}^n$, with $t \in [0, T]$, be the vector of solutions at the grid points, which approximate $u(r_j, t)$. Also let $F : \mathbb{R}^n \rightarrow \mathbb{R}^n$, for which $(F(U))_i \mapsto f(u_i)$, i.e., $F(U)$ is obtained by applying the firing rate function f to each component of U . This allows us to find

$$\begin{aligned}U'(t) &= -U(t) + AMF(U), \quad t \in [0, T] \\ U(0) &= \Xi,\end{aligned}$$

where Ξ is the vector of initial conditions, so $\Xi_j = u_0(r_j)$. Moreover, the matrix $M \in \mathbb{R}^{n \times n}$ is the *circulant matrix*^(vii)

$$M = \begin{bmatrix} hW_{n/2+1} & hW_{n/2+2} & \cdots & hW_1 & \cdots & hW_{n/2} \\ hW_{n/2} & hW_{n/2+1} & \cdots & hW_n & \cdots & hW_{n/2-1} \\ \vdots & \vdots & \ddots & \vdots & \ddots & \vdots \\ hW_2 & hW_3 & \cdots & hW_{n/2+2} & \cdots & hW_1 \\ hW_1 & hW_2 & \cdots & hW_{n/2+1} & \cdots & hW_n \\ hW_n & hW_1 & \cdots & hW_{n/2+3} & \cdots & hW_{n-1} \\ \vdots & \vdots & \ddots & \vdots & \ddots & \vdots \\ hW_{n/2+2} & hW_{n/2+3} & \cdots & hW_2 & \cdots & hW_{n/2+1} \end{bmatrix}. \quad (2.2)$$

Here, $W_i = w(r_i)$, and it is seen that M has entries $M_{jk} = hw_p(r_j - r_k)$. Notice that we

^(vii)This means, in our context, that it contains the synaptic kernel w as its rows, where each subsequent row is a cyclic shift of the previous.

have a matrix-vector product in the system, which generally requires computational time $\mathcal{O}(n^2)$. However, when using fast Fourier transforms (FFTs), it is possible to perform the operation with $\mathcal{O}(n \log n)$ [47].

Consider below the implementation in Python. This code forms the basis of further simulations we shall do. Also see *Appendix A*.

```

import numpy as np
import matplotlib.pyplot as plt
from scipy.integrate import solve_ivp

# Functions and parameters
def wFun(x):
    return (1 - np.abs(x)) * np.exp(-np.abs(x))

def fFun(u):
    return 1 / (1 + np.exp(-mu * (u - theta)))

mu = 50
theta = 0.2

# Spatial grid and circulant matrix
n = 2*10
L = 10 * np.pi
h = 2 * L / n
x = -L + np.arange(n) * h

M = np.zeros((n, n))
y = wFun(x) * h
iShift = np.arange(-n // 2, n // 2)
for i in range(n):
    M[i, :] = np.roll(y, int(iShift[i]))

# Right-hand side of the dynamical system
def rhs(t, u):
    u = -u + np.dot(M, fFun(u))
    return u

# Solving
u0 = 0.01 * np.random.randn(n)
tspan = (0, 100)
sol = solve_ivp(rhs, tspan, u0, method='RK45',
    t_eval=np.linspace(tspan[0], tspan[1], 1000))

```

Chapter 3

Exploring the Neural Field Equation: Pattern Formation

As mentioned above, the neural field model helps to gain insight into how patterns of activity can emerge spontaneously. To explore this, we start by introducing a tuning parameter A in front of the integral term, thereby allowing us to tune how much non-linearity is important. We assume that there is no external input for now, i.e., $I(r, t) \equiv 0$. We then have

$$\partial_t u(r, t) = -u(r, t) + A \int_{\Omega} w(r, r') f[u(r', t)] dr'. \quad (3.1)$$

One might say that A represents scaling of the synaptic connections strength w , or equivalently, the scaling of the firing rate f . Moreover, we choose

$$w(z) = \frac{1}{\sqrt{\pi}} e^{-z^2} - \frac{1}{\sigma \sqrt{\pi}} e^{-(z/\sigma)^2}, \quad (3.2)$$

where $\sigma \in \mathbb{R}_{>1}$, and

$$f(u) = \frac{1}{1 + \exp(-\mu u + \theta)} - \frac{1}{1 + \exp(\theta)}, \quad (3.3)$$

where $\mu \in \mathbb{R}_{>0}$ and $\theta \in \mathbb{R}$. Also see [Figure 3.1](#). These might seem completely arbitrary at this point, but this choice will become clear later.

The initial state of the network is a spatially uniform (flat) zero state, where we apply some random Gaussian noise on top. In simulation, where we have n points, we therefore have that the initial condition vector u_0 consists of components

$$u_{0,i} = 0.01 X_i, \quad X_i \sim \mathcal{N}(0, 1), \quad i \in \{1, \dots, n\},$$

see [Figure 3.2\(a\)](#). Now, if we set A to be a small value, say, $A = 0.5$, the decay term dominates the recurrent input and the network is unable to sustain any activity on its own. That is to say, the system *remains* in a flat state showing no patterns.

However, if we now set $A = 1.5$, something more interesting happens, see [Figure 3.2\(b\)](#). Instead of a flat state, the system spontaneously (that is, without any external input) forms a spatially periodic pattern of activity. Apparently, by varying A , the horizontal state becomes unstable leading to the emergence of patterns. Immediately, one might ask why such a pattern is formed. This is largely the result of the synaptic kernel w . The chosen function follows a ‘Mexican hat’ profile, meaning it provides short-range excitation while long-range connections are inhibited. The system selectively amplifies spatial fluctuations of a certain wavelength which grow while other decay. Ultimately, the resulting state is a spatially non-uniform stable steady

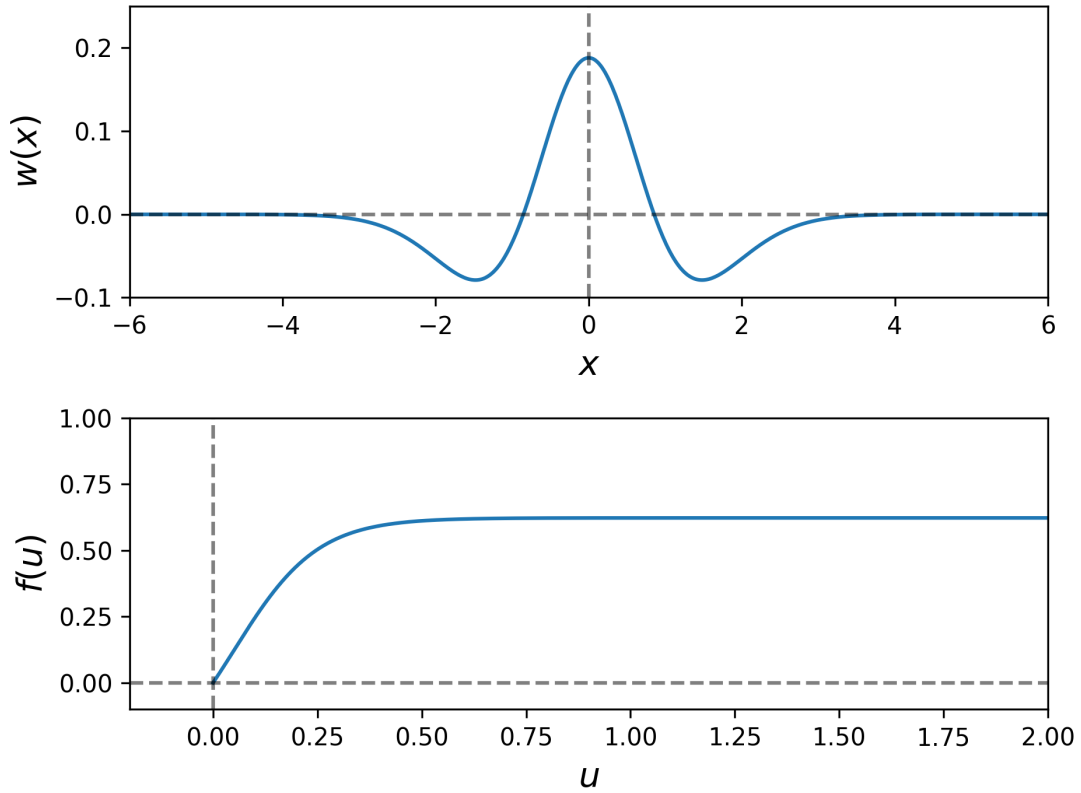


Figure 3.1: The chosen functions for the synaptic kernel w (top) and the firing rate f (bottom). Notice that w exhibits short-range excitation and long-range inhibition. Parameters were $\sigma = 1.5$, $\mu = 50$ and $\theta = 0.5$.

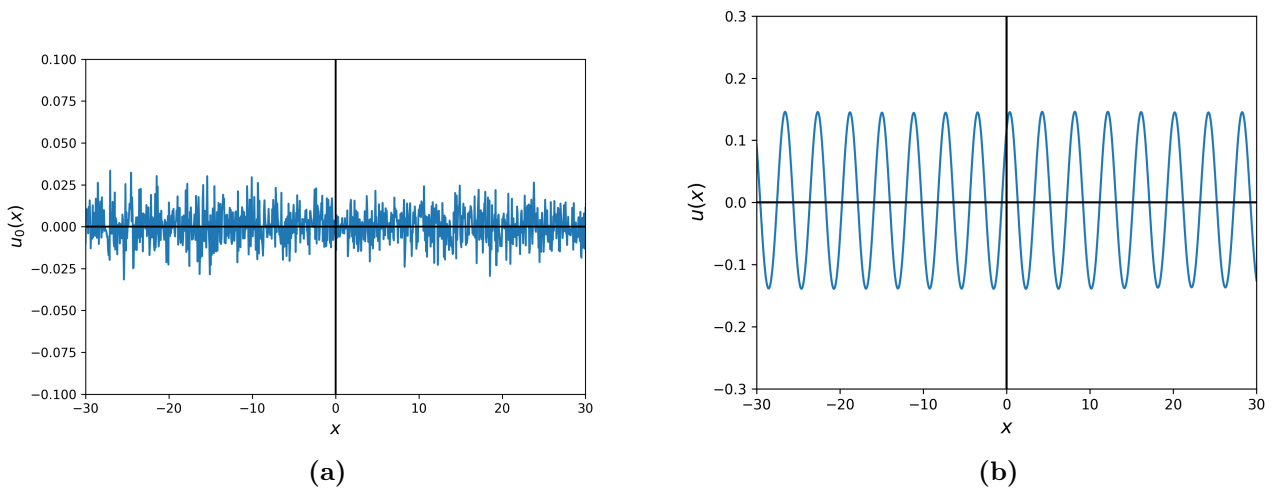


Figure 3.2: The formation of a pattern: (a) Some random Gaussian noise as initial condition. (b) Result of simulation with $A = 1.5$. After some time, a wave pattern starts to emerge. The snapshot is taken at $t = 100$.

state. We refer to this phenomenon as a *Turing bifurcation*^(viii). It is essential to distinguish this from a Hopf bifurcation, which refers to the emergence of oscillations in time, but that is not the case for us. A Turing bifurcation leads to oscillations in space.

3.1 Linear Stability of Homogeneous Equilibria

In order to better understand the Turing bifurcation which can yield spatial patterns, we return to the neural field equation, see *Equation (1.1)*. In particular, we start out by considering a spatially homogeneous equilibrium $U \in \Omega := \mathbb{R}$ ^(ix). Moreover, we have to assume that there is no external input, $I(r, t) \equiv 0$, as otherwise there is no spatially homogeneous equilibrium in general. Notice that U satisfies

$$\begin{aligned} 0 &= -U + \int_{\mathbb{R}} w(r - r') f(U) dr' \\ \implies U &= f(U) \int_{\mathbb{R}} w(r - r') dr' \\ &= f(U) \int_{\mathbb{R}} w(z) dz & (\star) \\ &= f(U) W_0. & (\star\star) \end{aligned}$$

To obtain $(\star)$, one must notice that w is even due to isotropy, and then apply a change of variables. Then $(\star\star)$ quickly follows from the assumption that w is integrable.

The solutions to the equilibrium condition $U = f(U)W_0$ are seen graphically in *Figure 3.3*. It is seen that the number of equilibria can change as parameters are varied. This gives rise to standard bifurcations. However, the equilibria continue to be flat, and are hence not of the desired type. Hence, the next step is to introduce a perturbation to the system.

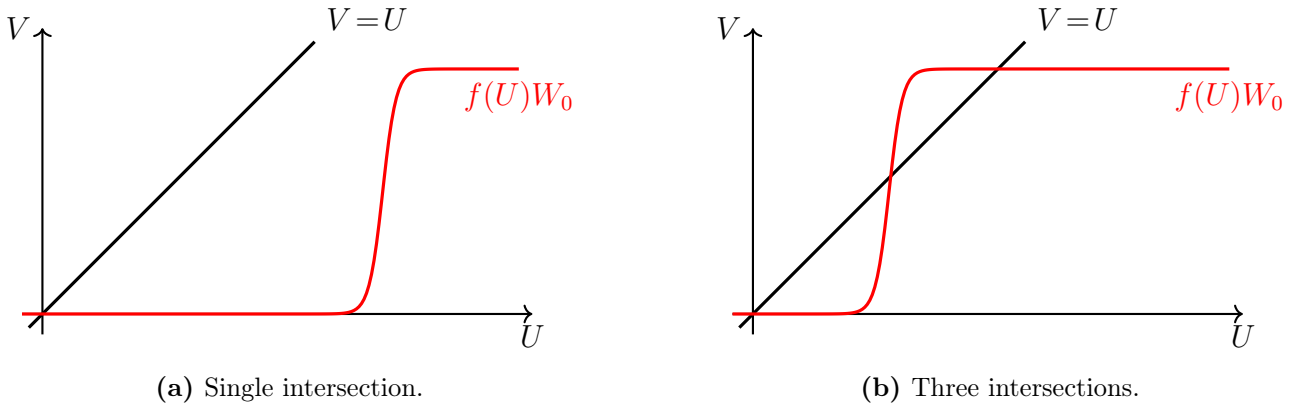


Figure 3.3: Illustration of equilibria and the effect of changing parameters giving rise to standard ODE bifurcations.

^(viii)For brevity, we shall call it a Turing bifurcation. However, originally such bifurcations only refer to the context of reaction-diffusion systems with local interactions. Yet, in our context of neural fields we are looking at non-locally coupled neurons without diffusion, where the instabilities show a similar bifurcation structure [18, 3]. Hence, to be more careful, one may refer to these as Turing-like bifurcations.

^(ix)Note that the subsequent analysis is readily generalizable to $\Omega = \mathbb{R}^3$. However, since $\Omega = \mathbb{R}$ is sufficient to reveal the mechanisms for pattern formation, we confine to it for clarity.

Let $\mathcal{F}$ be an operator such that $\partial_t u(r, t) = \mathcal{F}[u(\cdot, t)](r)$, and hence

$$\mathcal{F}[u(\cdot, t)](r) = -u(r, t) + \int_{\mathbb{R}} w(r - r') f[u(r', t)] dr'.$$

Moreover, we introduce a small perturbation $\xi(r, t)$ (satisfying $\|\xi(r, t)\| \ll 1$) by letting $u(r, t) = U + \xi(r, t)$. We shall consider the linear stability of the homogeneous equilibrium U . Plugging the previous into $\partial_t u(r, t) = \mathcal{F}[u(\cdot, t)](r)$, we have

$$\begin{aligned} \partial_t[U + \xi(r, t)] &= \mathcal{F}[U + \xi(\cdot, t)](r) \\ &= \mathcal{F}(U) + [D\mathcal{F}(U)\xi(\cdot, t)](r) + \mathcal{O}(\|\xi\|^2) \\ &= [\mathcal{L}_U \xi(\cdot, t)](r) + \mathcal{O}(\|\xi\|^2) \end{aligned}$$

by using that $\mathcal{F}(U) = 0$ since U is an equilibrium. We moreover applied a Taylor expansion [S], where $\mathcal{L}_U$ is the Fréchet derivative of $\mathcal{F}$ at U . Notice that we can write

$$\begin{aligned} \partial_t[U + \xi(r, t)] &= -U - \xi(r, t) + \int_{\Omega} w(r - r') f[U + \xi(r', t)] dr' \\ &= -U + \int_{\Omega} w(r - r') f(U) dr' \\ &\quad - \xi(r, t) + \int_{\Omega} w(r - r') f'(U) \xi(r', t) dr' \\ &\quad + \mathcal{O}(\|\xi\|^2), \end{aligned}$$

implying that $\mathcal{L}_U$ is explicitly given by

$$\mathcal{L}_U : v(r) \mapsto -v(r) + \int_{\mathbb{R}} w(r - r') f'[U(r')] v(r') dr'.$$

Disregarding second-order terms, we are left with the following:

$$\partial_t \xi(r, t) = -\xi(r, t) + f'(U) \int_{\Omega} w(r - r') \xi(r', t) dr' \quad (3.4)$$

for which $\xi(r, 0) = \xi_0(r)$. The goal is to construct solution of the linear problem, as this allows for insights to the non-linear case.

In order to investigate how solutions to Equation (3.4) behave as $t \rightarrow \infty$, we are obstructed by the convolution term. A way to resolve this is to use the *convolution theorem*, which, loosely speaking, states that taking the Fourier transform of the convolution leads to a product in Fourier space [43]. This allows for solving the problem, after which the inverse Fourier transform can be applied to return back to original coordinates. Consider the (forward) Fourier transform (FT):

$$\mathcal{F} : \xi(r) \mapsto \hat{\xi}(k) = \int_{\mathbb{R}} \xi(x) \exp(-ikr) dr,$$

as well as the inverse Fourier transform (IFT):

$$\mathcal{F}^{-1} : \hat{\xi}(k) \mapsto \xi(r) = \frac{1}{2\pi} \int_{\mathbb{R}} \hat{\xi}(k) \exp(ikr) dk,$$

where $k \in \mathbb{R}$ is called the wavelength, and $r \in \mathbb{R}$ [43]. Performing a FT in the spatial variable r on Equation (3.4) with use of the convolution theorem gives us, as outlined earlier, a product:

$$\mathcal{F} \left[\int_{\Omega} w(r - r') \xi(r', t) \, dr' \right] = \hat{w}(k) \hat{\xi}(k, t).$$

Hence, we find

$$\begin{aligned} \mathcal{F} [\partial_t \xi(r, t)] &= \mathcal{F} \left[-\xi(r, t) + f(U) \int_{\Omega} w(r - r') \xi(r', t) \, dr' \right] \\ \implies \partial_t \hat{\xi}(k, t) &= -\hat{\xi}(k, t) + f'(U) \hat{w}(k) \hat{\xi}(k, t) \\ &= \underbrace{[-1 + f'(U) \hat{w}(k)]}_{=: \lambda(k)} \hat{\xi}(k, t), \end{aligned}$$

with $\hat{\xi}(k, 0) = \hat{\xi}_0(k)$. It is readily seen that the solution of this system is $\hat{\xi}(k, t) = \hat{\xi}_0(k) \exp[\lambda(k)t]$, and therefore, by the IFT we conclude that

$$\xi(r, t) = \frac{1}{2\pi} \int_{\mathbb{R}} \hat{\xi}_0(k) \exp(ikr) \exp[\lambda(k)t] \, dk.$$

Returning to Figure 3.2(a), we now have a better understanding of the initial condition, which consisted of Gaussian noise on the zero state. For clarity with respect to notation, one might write $u_0(r) = U + \xi(r, 0) =: \xi_0(r)$, where we let $U = 0$. The important insight as a result of Fourier theory, is that $\xi_0(k)$ can be decomposed:

$$\xi_0(r) = \frac{1}{2\pi} \int_{\mathbb{R}} \hat{\xi}_0(k) \exp(ikr) \, dk.$$

In other words, $\xi_0(r)$ decomposes into a constant part, and a family of sinusoidal parts with increasing wavenumber k . Such parts are referred to as modes, and a particular mode corresponding to wavenumber k is weighted by $\hat{\xi}_0(k)$, and has wavelength $2\pi/k$.

The long-term behaviour of the perturbation $\xi(r, t)$ is determined by the term $\exp[\lambda(k)t]$. Notice that $\lambda(k)$ was defined as

$$\lambda(k) := -1 + f'(U) \hat{w}(k), \quad \text{dispersion relation.}$$

In particular, $\text{Re } \lambda(k)$ determines the growth of mode k . If $\text{Re } \lambda(k) < 0$, the corresponding mode decays and is dampened. Conversely, if $\text{Re } \lambda(k) > 0$, mode k is amplified. With regards to stability, we can say that if *all* modes decay, then the spatially homogeneous steady state U must be stable. That is, we have a (linear) stability criterion for the spatially homogeneous steady state U : if $\text{Re } \lambda(k) < 0$ for all $k \in \mathbb{R}$, then U is linearly stable. Again, $\lambda(k) = -1 + f'(U) \hat{w}(k)$, where $\hat{w}$ is the Fourier transform of w , the latter being an even and real-valued function. This means that $\hat{w}(k)$ is also even and real. Therefore, the stability criterion can be simplified by omitting $\text{Re}(\cdot)$. If we furthermore substitute the dispersion relation, and rearrange the terms, we find that

$$\text{if } \hat{w}(k) < \frac{1}{f'(U)} \text{ for all } k \in \mathbb{R}, \text{ then } U \text{ is linearly stable.} \quad (3.5)$$

Let's return to our previous example, *Equations (3.1)-(3.3)* where we had introduced a parameter A in front of the convolution. We repeat the equations from the example here for ease:

$$\partial_t u(r, t) = -u(r, t) + A \int_{\Omega} w(r, r') f[u(r', t)] dr', \quad (3.6)$$

and the functions w and f (synaptic kernel and firing rate, respectively) were

$$w(z) = \frac{1}{\sqrt{\pi}} e^{-z^2} - \frac{1}{\sigma\sqrt{\pi}} e^{-(z/\sigma)^2}, \quad f(u) = \frac{1}{1 + \exp(-\mu u + \theta)} - \frac{1}{1 + \exp(\theta)}.$$

Of course, the parameter A changes the dispersion relation. But by linearity of the Fourier transform, this remains only a slight alteration whereby A is simply pulled in front. We have

$$\lambda(k) = -1 + A f'(U) \hat{w}(k)$$

as the dispersion relation. Subsequently, if

$$A < \frac{1}{\hat{w}(k) f'(U)} \text{ for all } k \in \mathbb{R},$$

then U is linearly stable. Let $U = 0$ be the trivial steady state of *Equation (3.6)*. In order to specify its stability, the Fourier transform of w and the derivative of f should be found. It can be verified that [43]

$$\hat{w}(k) = \exp\left(-\frac{k^2}{4}\right) - \exp\left(-\frac{\sigma^2 k^2}{4}\right),$$

is the Fourier transform of $w(r)$. Moreover, the derivative of f is

$$f'(u) = \frac{\mu \exp(\theta - \mu u)}{[1 + \exp(\theta - \mu u)]^2}.$$

If A is sufficiently large, according to the stability criterion, then the steady state $U = 0$ is linearly unstable to perturbations $\exp(ikr)$ for all k satisfying

$$\hat{w}(k) > \frac{1}{A f'(0)}.$$

The values k for which $\hat{w}(k)$ is maximal are called critical wavenumbers, which are denoted by $\pm k_c$ [21]. The system undergoes a Turing-like bifurcation at the critical value of A given by

$$A_c := \frac{1}{\hat{w}(k_c) f'(0)} \approx 1.465, \quad (3.7)$$

where $k_c \approx 1.610$, which is obtained by solving $\hat{w}'(k_c) = 0$.

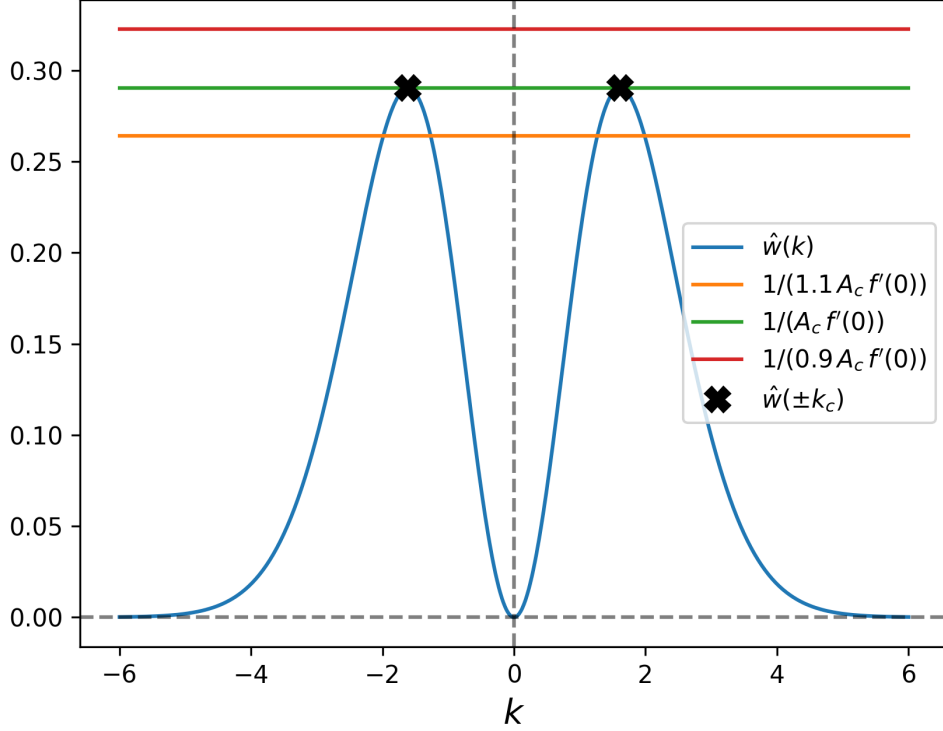


Figure 3.4: The function $\hat{W}(\xi)$ with maxima at $\pm\xi_c$.

In view of the stability criterion, we plot $\hat{w}(k)$ together with the ‘threshold lines’ corresponding to A_c and some sub- and super-critical values, see *Figure 3.4*. When the parameter A is sub-critical, say $A = 0.9A_c$, the stability criterion is satisfied. As is depicted in *Figure 3.4*, this graphically corresponds to the entire curve of $\hat{w}(k)$ lying below the horizontal line $1/[1.1A'f(0)]$. In this regime, the dispersion relation $\lambda(k)$ is negative for all wavenumbers $k \in \mathbb{R}$ which implies that all small perturbations are absorbed, thereby returning the system to the state $U = 0$.

As A is increased to the critical value $A = A_c$, the line $1/[A_c f'(0)]$ becomes tangent to $\hat{w}(k)$, the latter occurs precisely at $k := \pm k_c$. As A is further increased to become $1.1A_c$, the threshold line $1/[1.1A_c f'(0)]$ drops further below the maxima of $\hat{w}(k)$. This creates a band of unstable wavenumbers around $\pm k_c$, as seen in the figure (the values k for which $\hat{w}(k)$ is above the yellow line). Modes corresponding to these wavenumbers will be amplified rather than dampened. In particular, the modes that correspond to $\pm k_c$, expand maximally. In passing, note that at the points $\pm k_c$ the curve $\hat{w}(k)$ experiences its maxima. These modes exhibiting the greatest amplification will dominate, in the sense that other modes, although possibly also growing quite quickly, become negligible in comparison. The emerging initial spatial structure will therefore be characterized by a periodic pattern with a wavelength of approximately $2\pi/k_c$. The exact result, and particularly the final stable pattern, lies beyond the scope of this linear approach as it is the consequence of non-linear interactions.

However, we can verify by running a simulation that the wavelength approximation is not entirely inaccurate. We set $A = 1.1A_c$, and choose the following parameters:

$$\mu = 10, \quad \theta = 0.5, \quad \sigma = 1.5.$$

Next to that, the initial condition u_0 consists of Gaussian noise as before:

```
u0 = 0.01 * np.random.randn(n)
```

Running the simulation leads to *Figure 3.5*.

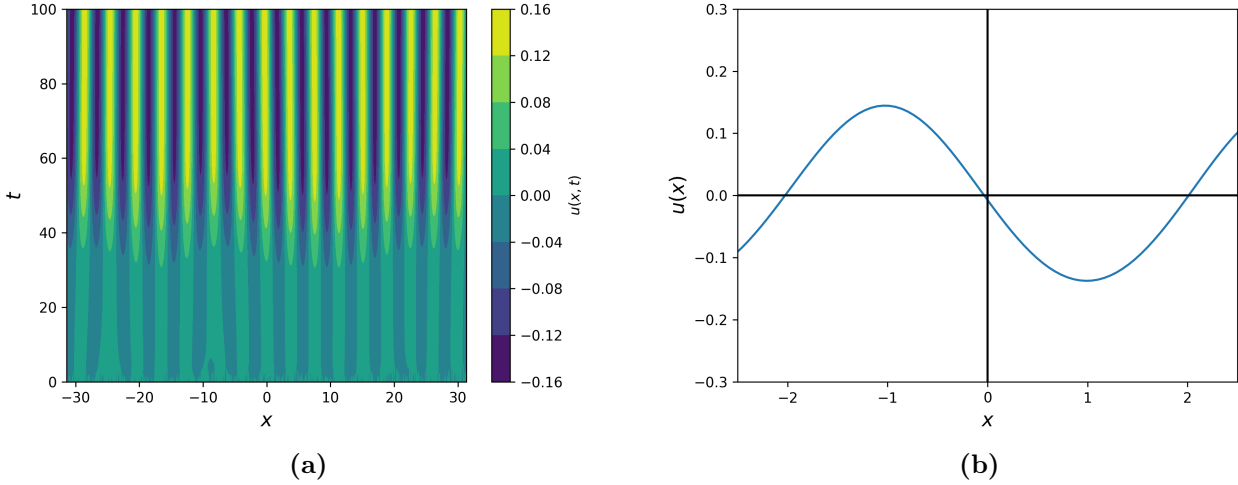


Figure 3.5: Simulation of neural field equation with supercritical parameter $A = 1.1A_c$: **(a)** Contour plot showing the pattern formation across time. **(b)** Snapshot at $t = 100$, zooming into the pattern near zero. Notice a wavelength of approximately 4, hence agreeing with the theoretical value.

We have answered the natural question at which value of A the change of behaviour occurs, i.e., finding the critical value. As a next step, we now let A vary with time:

$$A(t) = \varepsilon t, \quad (3.8)$$

so A increases linearly in time at rate ε . We impose $A(0) = 0$ as initial condition. The system of equations is adopted to:

$$\begin{aligned} \partial_t u(r, t) &= -u(r, t) + A(t) \int_D w(r - r') f[u(r', t)] dr', \\ \partial_t A(t) &= \varepsilon. \end{aligned}$$

which in code becomes^(x):

```
def rhs(t, u):
    u = u[:-1]
    A = u[-1]

    # M is circulant matrix, fFun is firing rate function
    du = -u + np.dot(A * M, fFun(u))
    dA = epsilon * t
    return np.append(du, dA)
```

^(x)This method might seem cumbersome but this allows for further extensions of the model later on.

That is, the first n (spatial grid) values of u are used for the neural field equation, and at the last index of u we have the dynamics of A , which is captured in the aforementioned equations.

From our earlier results, we know that the system does not display patterns until $A(t)$ exceeds A_c . Associated with that is the critical time:

$$t_c = \frac{A_c}{\varepsilon}, \quad (3.9)$$

When $t = t_c$ the system undergoes a Turing-like bifurcation, and patterns similar to *Fig. 3.5* start to emerge (see *Fig. 3.6*). However, as the value of A keeps increasing, so does the amplitude of the waves associated with $u(r, t)$. The reason for this is apparent when looking at *Eq. (??)*. Note that merely the amplitude is increasing, while the wavelength remains fixed, as the bifurcation leads to a mode with dominant wavelength $2\pi/\xi_c$ that persists for $A(t) > A_c$ and there are no other bifurcations occurring.

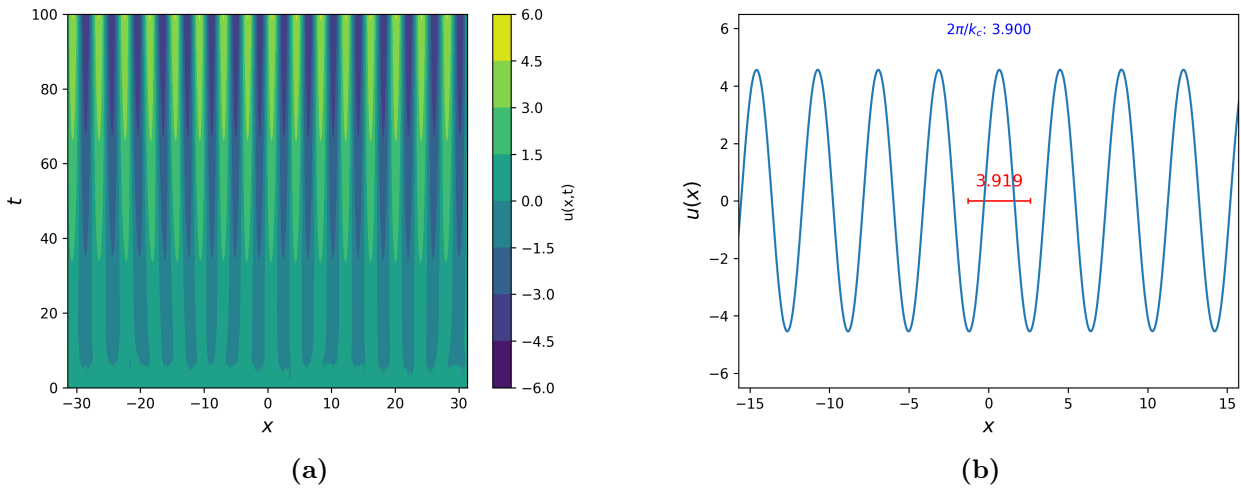


Figure 3.6: Solution $u(x, t)$ at $t \approx 100$ with $A(t) = \varepsilon t$, and $\varepsilon = 0.1$. Notice a wavelength of approximately 3.9. This wavelength was computed using the SciPy Signal library.

The aforementioned pattern formation can be cross-checked with the diagram in *Figure 3.7*, which was produced with numerical simulations. Indeed, when $A(t)$ crosses the critical value ($A(t) > A_c$), the spatial pattern starts to emerge, which is reflected by an increase in $\|u(t)\|$. The periodic pattern has a wavelength of $2\pi/k$ that is temporally constant. Only the amplitude is increased whenever $A(t)$ is increased.

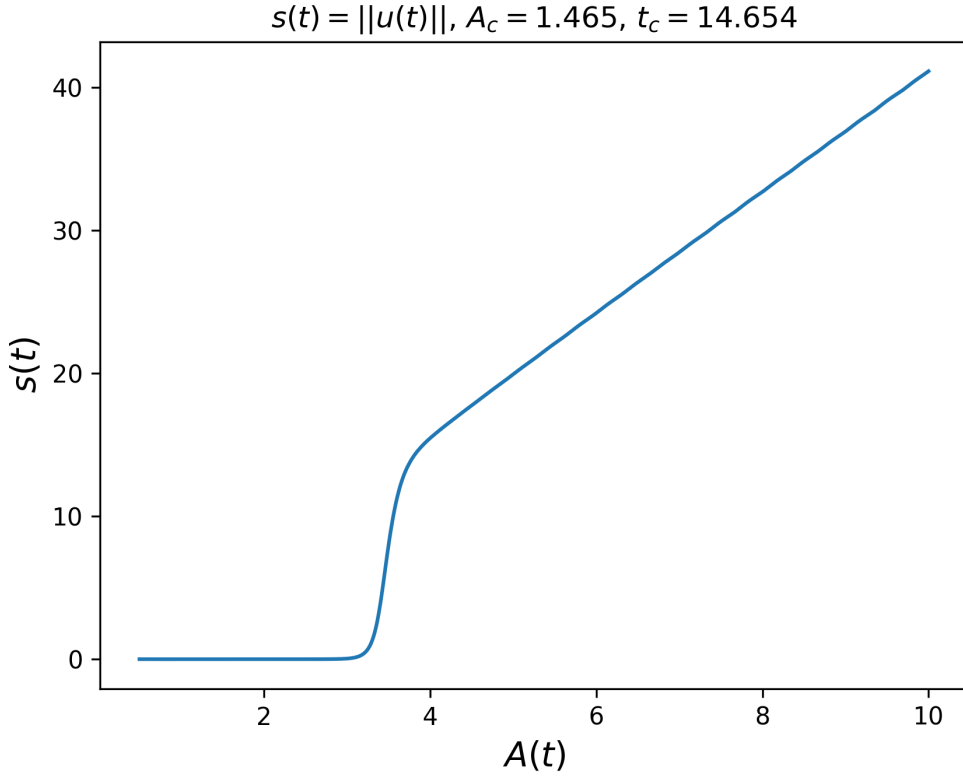


Figure 3.7: Bifurcation diagram with $s(t) = ||u(t)||$, where $\epsilon = 0.1$. Note a critical time point of approximately 15.

3.2 Bump Solutions

We have just seen that the neural field equation is able to support a flat steady state. Moreover, upon achieving a critical parameter value, wave patterns begin to occur. We now switch away from a *flat* state, and instead consider solutions that correspond to *persistent* and *localized* neural activity, where a spatially localized group of neurons show spiking behavior [14, 33]. This is also referred to as a *bump* of activity.

The neural field equation is capable of supporting these localized bumps of activity. By studying bumps in this framework, we can investigate the stability of bumps and how they are affected by various factors. In the analysis below, we closely follow [19, Sections 1 and 3], to which we also refer for further details.

As an example of bumps, see Figures 3.8 and 3.9. We say that the steady state $U(r)$ is a bump solution to the neural field equation if it satisfies [19]

- (i) $\lim_{z \rightarrow \pm\infty} w(z) = 0$;
- (ii) $U(r) \geq \theta$ for all $x \in [r_1, r_2]$, where $r_1, r_2 \in \Omega$ satisfying $r_1 < r_2$.

Recall that

$$\mathcal{F}[u(\cdot, t)](r) = -u(r, t) + \int_{\mathbb{R}} w(r - r') f[u(r', t)] dr'.$$

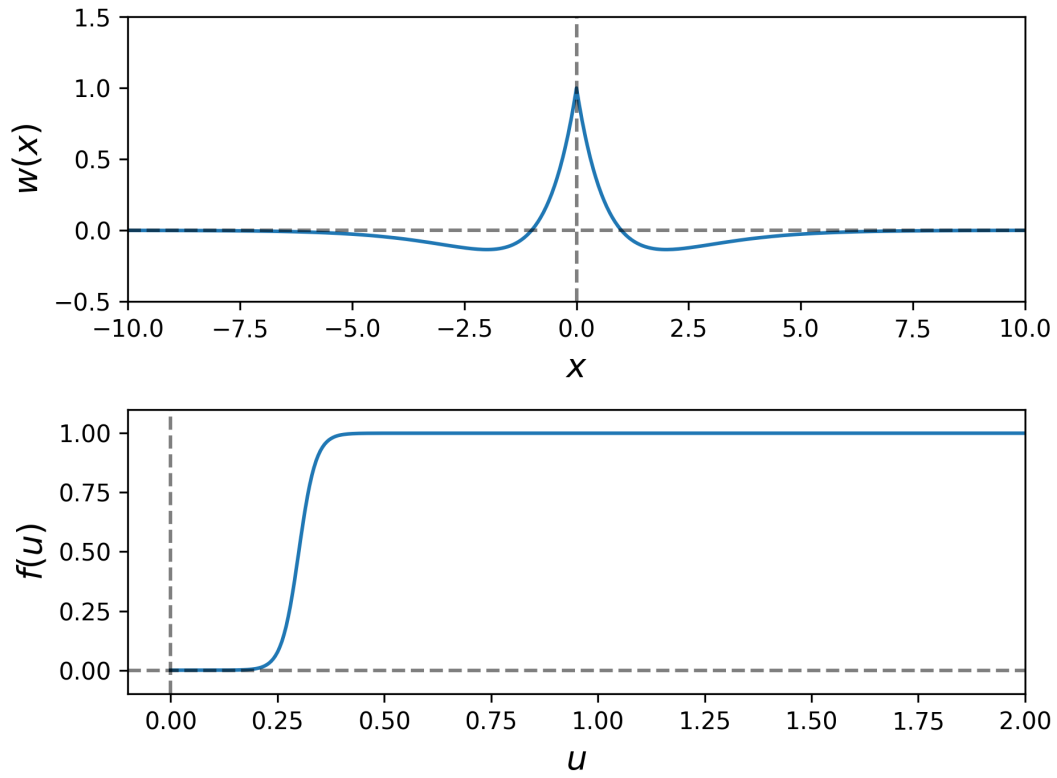


Figure 3.8: The synaptic kernel and firing rate function: $w(z) = (1 - |z|) \exp(-|z|)$ and $f(u) = \frac{1}{1 + \exp[-\mu(u - \theta)]}$. Parameters were $\mu = 50, \theta = 0.3$.

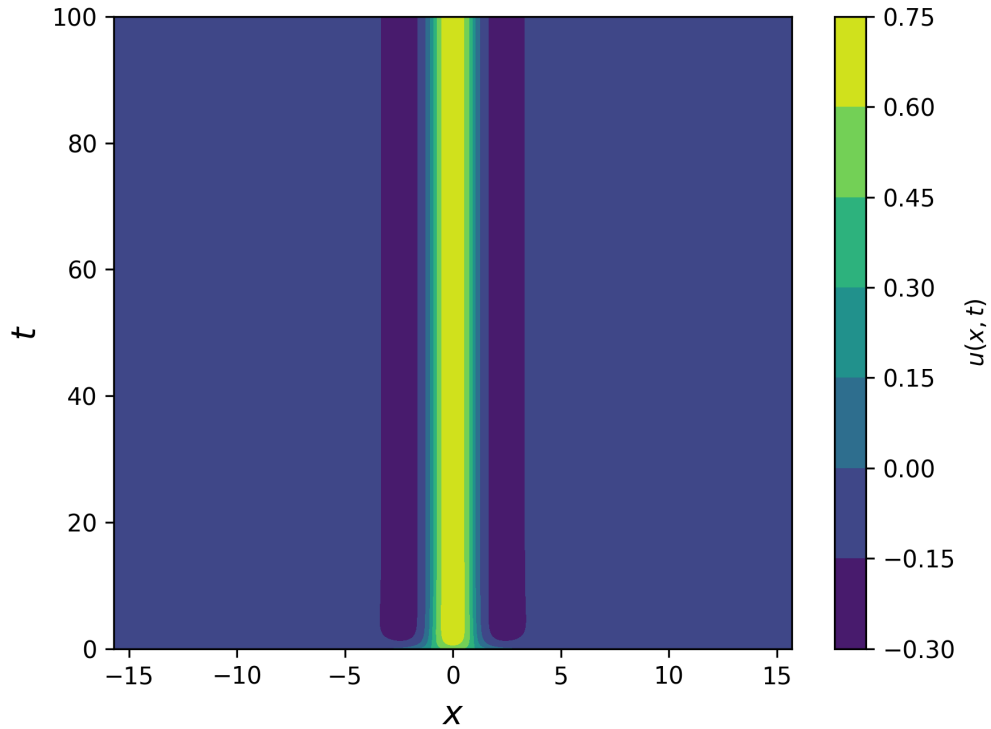


Figure 3.9: Contour plot showing persistent localized activity.

In order to determine the stability of $U(r)$ to small perturbations of the type $\xi(r, t) = \exp(\lambda t)v(r)$, we are concerned with eigenpairs $(\lambda, v(r))$ to the eigenvalue problem $\lambda v = \mathcal{L}_U v$ where

$$\mathcal{L}_U : v(r) \mapsto -v(r) + \int_{\mathbb{R}} w(r - r') f'[U(r')] v(r') \, dr'.$$

So, the stability problem can be summarized as finding eigenpairs $(\lambda, v(r))$ such that

$$\lambda v(r) = -v(r) + \int_{\mathbb{R}} w(r - r') f'[U(r')] v(r') \, dr'.$$

Within the Heaviside world with $f(v) := \mathcal{H}(v - \theta)$ as the firing rate function, we find that $f'[U(r')] = \delta[U(r') - \theta]$. Here, δ is the Dirac delta function. Following [27], we can say that the Dirac delta function of a function $f \in \mathcal{C}^1(\mathbb{R})$ with simple roots $r_1, \dots, r_n$, where $n \in \mathbb{Z}_{>0}$, is given by

$$\delta[f(r)] = \sum_{i=1}^n \frac{\delta(r - r_i)}{|f'(r_i)|}.$$

Hence, we find

$$\lambda v(r) = -v(r) + \int_{\mathbb{R}} w(r - r') \sum_{i=1}^2 \frac{\delta(r' - r_i)}{|U'(r_i)|} v(r') \, dr'$$

which can be rearranged as follows

$$\begin{aligned} (\lambda + 1)v(r) &= \int_{\mathbb{R}} w(r - r') \sum_{i=1}^2 \left[\frac{\delta(r' - r_i)}{|U'(r_i)|} \right] v(r') \, dr' \\ &= \sum_{i=1}^2 \frac{1}{|U'(r_i)|} \underbrace{\int_{\mathbb{R}} w(r - r') \delta(r' - r_i) v(r') \, dr'}_{=w(r-r_i)v(r_i) \text{ sifting property of } \delta} \\ &= \sum_{i=1}^2 \frac{w(r - r_i)}{|U'(r_i)|} v(r_i). \end{aligned} \tag{3.10}$$

We would like to further explore this, especially unpacking the $U'(r_i)$ part.

Definition 3.1 (Active region). The active region of $\Omega := \mathbb{R}$ to be given by

$$\mathcal{R}_\Omega(t) = \{r \mid u(r, t) > 0\}.$$

Often it is assumed that it follows the form $\mathcal{R}_\Omega(t) = [r_1(t), r_2(t)]$, meaning u is positive within a bounded interval [19]. Indeed, when we impose the Heaviside world and a bump solution $U(r)$, then

$$\begin{aligned} U(r) &= \int_{\Omega} w(r - r') \mathcal{H}[U(r') - h] \, dr' \\ &= \int_{\mathcal{R}_\Omega} w(r - r') \, dr' \\ &= \int_{r_1}^{r_2} w(r - r') \, dr'. \end{aligned}$$

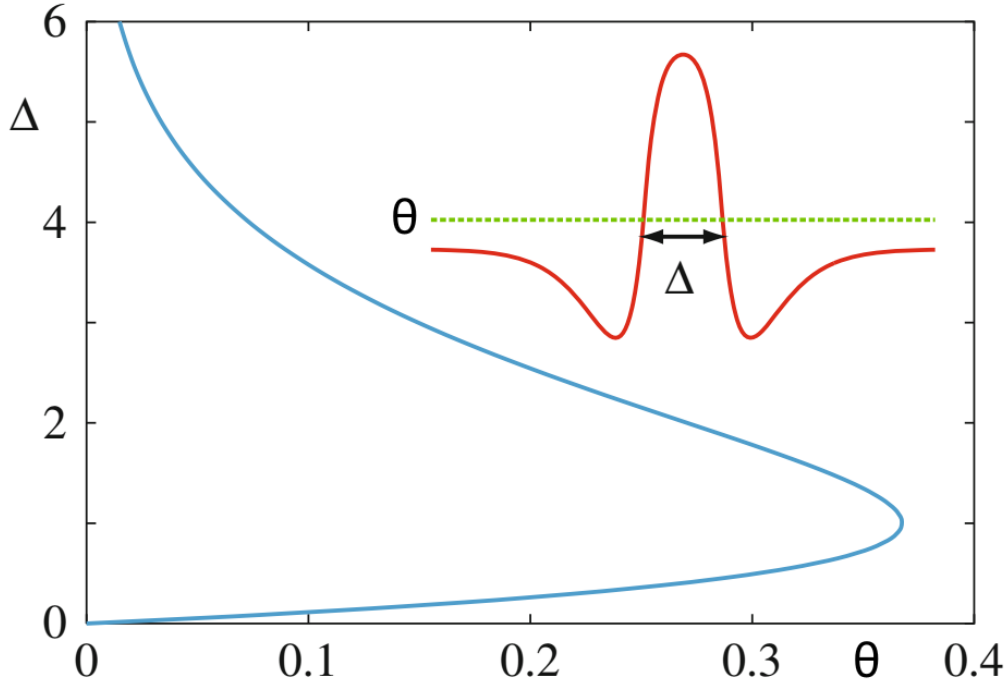


Figure 3.10: Bifurcation diagram: upper branch is stable, lower branch is unstable.
Taken and amended from [19].

This means that in order to find $U(r)$, we must find $r_1, r_2 \in \mathbb{R}$. On the one hand, we have $U(r_1) = U(r_2) = \theta$ (threshold) by construction, and on the other hand, from the above, we can obtain

$$\begin{aligned}
 U(r_1) &= \int_{r_1}^{r_2} w(r_1 - r') dr' \\
 &= \int_{r_1}^{r_2} w(r' - r_1) dr' && w \text{ is even} \\
 &= \int_0^{r_2 - r_1} w(z) dz && \text{substitution.}
 \end{aligned}$$

Let $\Delta := r_2 - r_1$, which represents the width of $\mathcal{R}_\Omega(t)$. Combining results, we conclude that

$$\theta = \int_0^\Delta w(z) dz. \quad (3.11)$$

Also see the solution branch in *Figure 3.10*.

The above forms a way to construct bumps. Namely, for some synaptic kernel w one has to find h and Δ satisfying *Equation (3.11)*. As a next step, to actually find $U(r)$, we note due to homogeneity of w , we can set, without loss of generality, $r_1 = 0$ and $r_2 = \Delta$. Then

$$U(r) = \int_{r_1}^{r_2} w(r - r') dr' = \int_0^\Delta w(r - r') dr',$$

is the bump solution.

We can now return to *Equation (3.10)*. In particular, we know what U' looks like now:

$$U(r) = \int_0^\Delta w(r - r') dr' \implies U'(r) = w(r) - w(r - \Delta).$$

In rewriting the earlier result, we use a vector $\mathbf{v}$ with components $\mathbf{v}_i = v(x_i)$ and a matrix M defined below. If we evaluate at $r = x_1$ and $r = x_2$, the roots of $U(r) - h$, we find the following:

$$\lambda \mathbf{v} = \begin{bmatrix} \frac{w(0)}{|w(0) - w(\Delta)|} - 1 & \frac{w(\Delta)}{|w(0) - w(\Delta)|} \\ \frac{w(\Delta)}{|w(0) - w(\Delta)|} & \frac{w(0)}{|w(0) - w(\Delta)|} - 1 \end{bmatrix} \mathbf{v} =: M \mathbf{v}. \quad (3.12)$$

We can now conclude on bump stability [19]. Consider the eigenpairs $(\lambda_i, \mathbf{v}_i)$ for $M \in \mathbb{R}^{2 \times 2}$. There always exists one solution with eigenvalue zero with eigenmode $U'(r)$. This zero eigenvalue does not influence stability. The remaining eigenvalue is purely real, and let us denote it simply by λ . The solution is stable if $\lambda < 0$, and unstable when $\lambda > 0$. Also see *Figure 3.10*, from which we can further conclude that the branch of *Equation (3.11)* corresponding to the largest Δ (i.e., the broadest bump) is stable. The narrower bump is unstable.

Now, from *Equation (3.10)*, we can obtain the eigenmode $v(r)$ associated with λ :

$$v(r) = \frac{1}{\lambda + 1} \sum_{i=1}^2 \frac{w(r - r_i)}{|w(r) - w(r - \Delta)|} v(r_i),$$

using that $U'(r) = w(r) - w(r - \Delta)$. This eigenmode is helpful because it describes the dynamics near the steady state, and thereby hints what shape to expect upon instability.

Before concluding this part, we mention a result that will be referred to later.

Claim 3.2. The evolution equation is invariant to spatial translations.

Proof. Let $u(r, t)$ be a solution. We have that $\partial_t u(x, t) = \mathcal{F}[u(\cdot, t)](r)$. Now, let $r_0 \in \mathbb{R}$. Then,

$$\begin{aligned} \partial_t u(r + r_0, t) &= \underbrace{-u(r + r_0, t) + \int_{\mathbb{R}} w(r + r_0 - r') f[u(r', t)] dr'}_{=\mathcal{F}[u(\cdot, t)](r + r_0)} \\ &= \underbrace{-u(r + r_0, t) + \int_{\mathbb{R}} w(r - z) f[u(z + r_0, t)] dz}_{=\mathcal{F}[u(\cdot + r_0, t)](r)} \quad r' = z + r_0 \end{aligned}$$

Therefore, $u(r + r_0, t)$ is also a solution. This demonstrates that the translated function obeys the same dynamics. Hence the equation is translationally invariant.

Hence, solutions are unique up to spatial translations, meaning that, without loss of generality, one may fix $r_0 = 0$. Note that the zero eigenvalue from earlier is a direct manifestation of translation symmetry, without effect on stability. $\square$

Chapter 4

Introducing Short-Term Synaptic Plasticity

Previously, we have examined bumps of neural activity and their significance for working memory. However, an essential aspect we must consider if we choose to look at timescales on which working memory works, is the dynamic nature of synaptic connection strengths. As we have seen, activity-dependent changes cause synaptic connections to strengthen or weaken. In the current model, this is an important missing component, as it is currently assumed that all connections weights are static with time. Therefore, we now extend the earlier model by incorporating short-term synaptic plasticity (STSP).

The goal is to translate the neuroscientific description of STSP into the appropriate mathematical framework. In doing so, we mainly follow a paper by Cihak and Kilpatrick [14]. We have seen that there are two prominent mechanisms underlying STSP: depression and facilitation. We briefly review those from the perspective of laying out their dynamics through a differential equation. To that end, let d and q represent the depression and facilitation dynamics, respectively. In particular, d corresponds to the fraction of available presynaptic resources^(xi) and q represents the increase in synaptic efficacy due to facilitation dynamics [14, 33, 54].

If multiple action potentials arrive in quick succession, the initial response is often an enhancement of vesicle release (facilitation). We refer to this as strengthening of the synapse, which we say occurs at some rate β . However, this facilitation is inherently limited due to the finite availability of synaptic vesicle resources, characterized by the parameter q_0 . On the other hand, the mechanisms of depression are also at play. The action potentials invading the synapse cause the synaptic vesicles to be progressively depleted. For instance, experimentally it has been shown that 20 Hz low-frequency stimulation in particular synapses can reduce the readily releasable pool (RRP) to about 40% of control conditions, and moreover release vesicles about two times slower [38]. That is, a synapse can exhaust its resources, leading to a decrease in release probability, possibly below the original baseline value. We now say that the synapse is weakened at some rate α . The idea is that whenever the synapse has released a considerable amount of synapses, it is short on resources and therefore reducing its vesicle outputs thereby allowing for replenishing of the vesicles.

It must be noted that neural activity acts on the order of 10 ms, while short-term synaptic plasticity has a slower timescale of seconds [14]. Introducing time constants τ_u , τ_d , and τ_q for the respective variables in the system, we set the membrane time constant to $\tau_u = 1$ and say that $u(r, t)$ has time units of 10 ms. Moreover, τ_d and τ_q are then $\mathcal{O}(10^2)$. Note that in this text, we often take $\tau_d = \tau_q = 1$ unless otherwise specified meaning plasticity effects occur on the same timescale as the neural activity. In this case, d and q more rapidly approach their steady states. However, there are appropriate instances where we allow them to be $\tau_d, \tau_q \gg 1$, for instance when considering high frequency stimuli or traveling waves.

^(xi)We talk here about ‘presynaptic resources’ as one could measure, say, the amount of vesicles, but also the amount of neurotransmitters (which are in the vesicles).

From this discussion, it is seen that an adequate model for d and q is [14]

$$\begin{aligned}\tau_d \partial_t d &= 1 - d - \alpha d(1 + q)f, & \text{'weaker synapses due to fatigue'}, \\ \tau_q \partial_t q &= -q + \beta(q_0 - q)f, & \text{'stronger synapse after rapid activity'}.\end{aligned}$$

Recall that for neural activity we had $\partial_t u = -u + w \otimes f^{(\text{xii})}$. Now, STSP is incorporated by including d and q as scaling factors in the nonlocal term [14, 54, 32]

$$\partial_t u = -u + w \otimes d(1 + q)f,$$

For the dynamics of d , we confirm that when there is no firing ($f = 0$), we have $\partial d / \partial t = 1 - d$ and so $d \rightarrow 1$ in absence of activity. That is, we can call this the recovery term (or, ‘replenishing term’ as we are refilling the RRP’s). Indeed, when action potentials fire there is an added term $-\alpha d(1 + q)$ that decreases d in a manner where there is progressive depletion of the synaptic vesicles pool, but the rate of depletion decreases with time^(xiii). On the other hand, when facilitation is increased ($q > 0$), there are relatively more transmitters being released which is represented by an increase in $\alpha d(1 + q)$ yielding a decrease in d . On the other hand, for the dynamics of q , when $f = 0$ we see that $\partial q / \partial t = -q$ meaning when there is no firing, the system relaxes back to a state where there is no facilitation happening. In case of spiking activity, q increases at a rate β and is limited by q_0 .

Before continuing, we say that $d(1 + q)f$ is the *effective firing rate*. The function f is simply an indicator function, which is scaled by plasticity influences. We have seen that d (depression) has its baseline at 1 and decreases upon depression, so with increasing synaptic depression the effective firing rate is scaled to a lower value. Similarly, q (facilitation) has its baseline at 0 and increases upon facilitation, subsequently having an increasing effect on the effective firing rate.

Writing the model out in full, we have [14]

$$\begin{aligned}\partial_t u(r, t) &= -u(r, t) + w(r) \otimes [d(r, t)(1 + q(r, t))f(u(r, t))] \\ \tau_d \partial_t d(r, t) &= 1 - d(r, t) - \alpha d(r, t)(1 + q(r, t))f(u(r, t)) \\ \tau_q \partial_t q(r, t) &= -q(r, t) + \beta(q_0 - q(r, t))f(u(r, t)),\end{aligned}\tag{4.1}$$

where

- u : neural activity; d : depression dynamics; q : facilitation dynamics;
- τ_d, τ_q : synaptic time constants^(xiv);
- w : the synaptic kernel defined by

$$w(z) = (1 - |z|) \exp(-|z|);$$

^(xii)Here, $w \otimes f$ is shorthand for the spatial convolution. In full, it is given by

$$w(r) \otimes f[u(r, t)] = \int_{\mathbb{R}} w(r - r') f[u(r', t)] dr',$$

cf. Equation (1.1).

^(xiii)If $d = 0.1$, then the RRP’s are almost fully depleted (so there is not much more to deplete), and any continued depletion occurs at a lower rate than when, say, $d = 0.9$.

^(xiv)As these time constants are usually set to one, they will often not be omitted for brevity.

- f : the firing rate often given by the Heaviside function

$$f(u) = \mathcal{H}(u - \theta) = 1_{u \geq \theta},$$

in analytical context. For numerical simulations, we use a sigmoid

$$f(u) = \frac{1}{1 + \exp[-\mu(u - \theta)]},$$

with high gain μ and some threshold θ , cf. *Section 1.3*;

- α : ‘rate of depression’, i.e., the rate at which connections are made weaker due to synaptic vesicle depletion;
- β : ‘rate of facilitation’, i.e., the rate at which facilitation strengthens synapses as a result of activity;
- q_0 : the amount of available presynaptic resources.

Before delving into some analysis on bumps and their stability, let us first consider a comparison between bumps with and without STSP effects. This is seen in *Figure 4.1*. In (a) the STSP dynamics have been shut off, whereas in (b) they are enabled with an emphasis on depression by choosing $\alpha = 0.5$, $\beta = 0.01$, and $q_0 = 1$, also see (c) and (d). The results show a peak that is much more dampened in amplitude during the pronounced depression.

Equilibria Solutions

In what follows, we analyze steady states within the Heaviside world where $f(u) = \mathcal{H}(u - \theta)$. Let $(U(r), D(r), Q(r))$ be a steady state of the system in *Equation (4.1)*, which must satisfy the following:

$$\begin{aligned} 0 &= -U(r) + w(r) \otimes [D(r) (1 + Q(r)) \mathcal{H}(U(r) - \theta)] \\ 0 &= 1 - D(r) - \alpha D(r) (1 + Q(r)) \mathcal{H}(U(r) - \theta) \\ 0 &= -Q(r) + \beta (q_0 - Q(r)) \mathcal{H}(U(r) - \theta). \end{aligned}$$

Recall we defined the active region of a bump to be the region where $u(r, t) \geq \theta$, see *Definition 3.1*. We made the assumption that the active region is of the form $[r_1(t), r_2(t)]$. Since we are considering equilibria, the time dependence drops. Moreover, due to translational invariance, following from *Claim 3.2*, and assuming U is even, we can pick the center of mass to always be at $r = 0$ around which the bump is symmetric [14]. To that end, we say $[-a, a]$ is the active region. Notice that outside of the active region ($U < \theta$) the firing rate is zero, while inside ($U \geq \theta$) it is one.

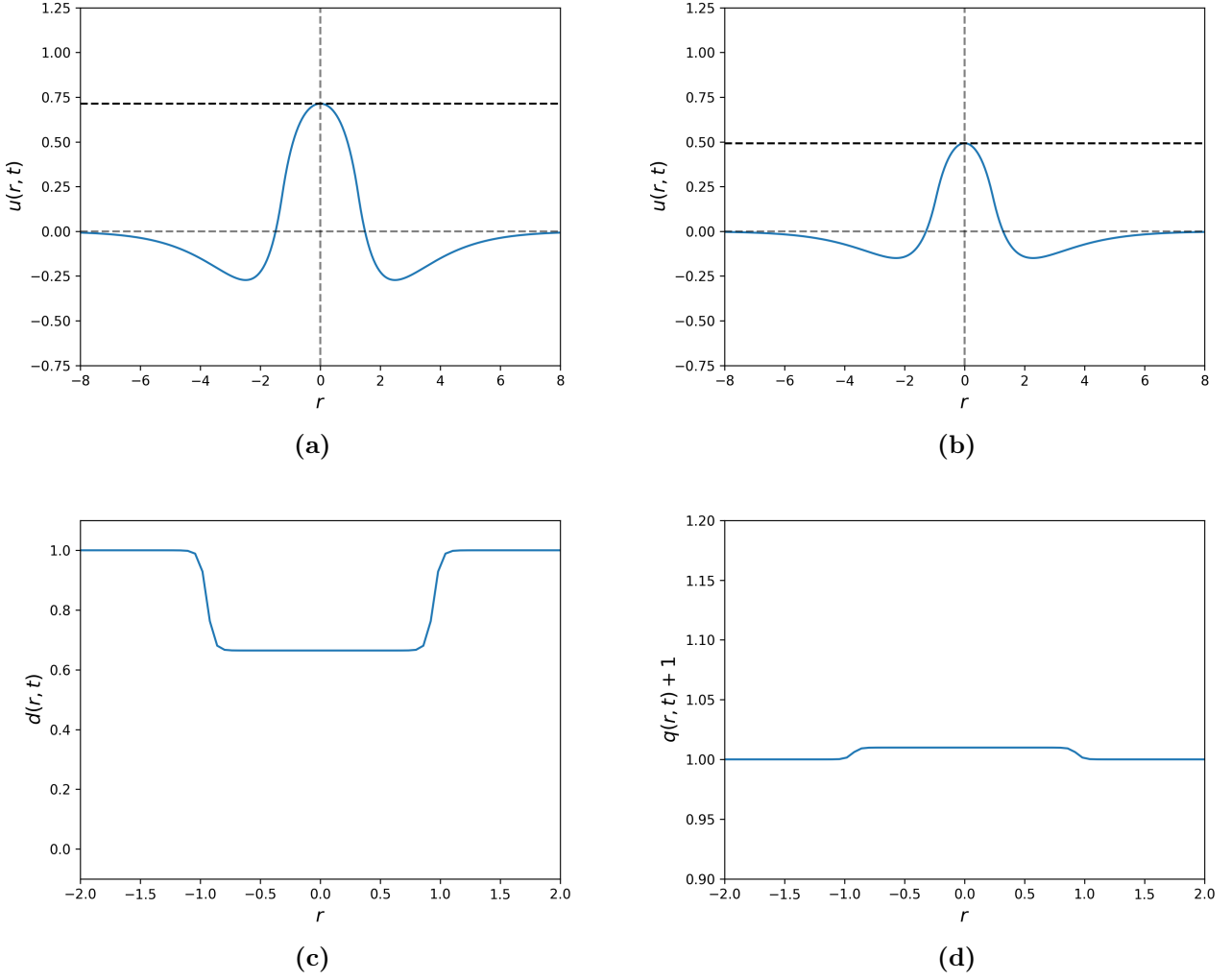


Figure 4.1: The effects of STSP: all functions, parameters, and initial conditions were the same across simulations, where $w(z) = (1 - |z|) \exp(-|z|)$, f a sigmoid with $\mu = 50$ and $\theta = 0.2$, and $u_0(r) = 0.5 \exp(-r^2/2)$. Moreover, all plots are shown at $t = 100$. **(a)** A bump without STSP dynamics. Notice the bump has its peak just above 0.70. **(b)** A bump with STSP dynamics with corresponding parameters $\alpha = 0.5$, $\beta = 0.01$, and $q_0 = 1$. This time, it is seen that the peak is lowered to around 0.5, due to the relatively prominent depression rate. **(c)** Depression dynamics d . **(d)** Facilitation dynamics, presented as $q + 1$.

The following can now be obtained.

- **Case I** (inside the active region). We have $\mathcal{H}(U(r) - \theta) = 1$ for $r \in [-a, a]$, and hence

$$\begin{aligned} 0 &= -U(r) + w(r) \otimes [D(r) (1 + Q(r))] \\ 0 &= 1 - D(r) - \alpha D(r) (1 + Q(r)) \\ 0 &= -Q(r) + \beta (q_0 - Q(r)). \end{aligned}$$

Here, $Q(x)$ can be readily found and substituted into the equation above in order to obtain $D(r)$, after which $U(r)$ can be obtained by substituting both $Q(r)$ and $D(r)$. This respectively yields

$$\begin{aligned} Q^* &= \frac{q_0 \beta}{1 + \beta} \\ D^* &= \frac{1}{1 + \alpha(1 + Q^*)} \\ &= \frac{1}{1 + \alpha \left(1 + \frac{q_0 \beta}{1 + \beta}\right)} \\ U(r) &= \int_{-a}^a w(r - r') [D(r') (1 + Q(r'))] dr' \\ &= \int_{-a}^a w(r - r') \frac{1 + Q^*}{1 + \alpha(1 + Q^*)} dr' \\ &= \frac{1 + \frac{q_0 \beta}{1 + \beta}}{1 + \alpha \left(1 + \frac{q_0 \beta}{1 + \beta}\right)} \int_{-a}^a w(r - r') dr'. \end{aligned}$$

- **Case II** (outside the active region). Here, $\mathcal{H}(U(r) - \theta) = 0$ for $r \in (-\infty, -a) \cup (a, \infty)$. Hence we find that

$$\begin{aligned} Q^* &= 0 \\ D^* &= 1 \\ U(r) &= \int_{-\infty}^{-a} w(r - r') \cdot 0 dr' \\ &\quad + \int_{-a}^a w(r - r') [D(r') (1 + Q(r')) \underbrace{\mathcal{H}(U(r') - \theta)}_{=1 \text{ for } r' \in [-a, a]}] dr' \\ &\quad + \int_a^{\infty} w(r - r') \cdot 0 dr' \\ &= \frac{1 + \frac{q_0 \beta}{1 + \beta}}{1 + \alpha \left(1 + \frac{q_0 \beta}{1 + \beta}\right)} \int_{-a}^a w(r - r') dr'. \end{aligned}$$

So, the steady states $D(r)$ and $Q(r)$ of the synaptic plasticity dynamics are piecewise constant functions with jumps at the boundaries of the active region, $r = \pm a$:

$$\begin{aligned}
 U(r) &= \frac{1 + \frac{q_0\beta}{1+\beta}}{1 + \alpha \left(1 + \frac{q_0\beta}{1+\beta}\right)} \int_{-a}^a w(r - r') \, dr' \\
 D(r) &= \begin{cases} \frac{1}{1 + \alpha \left(1 + \frac{q_0\beta}{1+\beta}\right)} & \text{if } r \in [-a, a] \\ 1 & \text{if } r \in (-\infty, -a) \cup (a, \infty) \end{cases} \\
 Q(r) &= \begin{cases} \frac{q_0\beta}{1 + \beta} & \text{if } r \in [-a, a] \\ 1 & \text{if } r \in (-\infty, -a) \cup (a, \infty). \end{cases}
 \end{aligned}$$

This is reflected in the simulations shown in *Figure 4.1*, although one must note that there a sigmoid for the firing rate function was used instead of the idealized Heaviside. Exploiting $U(\pm a) = \theta$, we can find that

$$\begin{aligned}
 \theta = U(a) &= \underbrace{\frac{1 + \frac{q_0\beta}{1+\beta}}{1 + \alpha \left(1 + \frac{q_0\beta}{1+\beta}\right)}}_{=:S} \underbrace{\int_{-a}^a w(a - r') \, dr'}_{=:I(a)} \\
 &= SI(a).
 \end{aligned}$$

Notice that in this equation, we relate the bump geometry (characterized by a) to the synaptic kernel w and the model parameters: the threshold θ (threshold) and the plasticity parameters α , β , and q_0 .

To obtain an explicit result, we choose the Mexican hat synaptic kernel given by

$$w(z) = (1 - |z|) \exp(-|z|).$$

It can be obtained that

$$I(a) = \int_{-a}^a w(a - r') \, dy = 2a \exp(-2a),$$

meaning we have found explicit criteria for threshold crossing:

$$\theta = SI(a) = 2a \frac{1 + \frac{q_0\beta}{1+\beta}}{1 + \alpha \left(1 + \frac{q_0\beta}{1+\beta}\right)} \exp(-2a). \quad (4.2)$$

If we pick $\theta = 0.2$, $q_0 = 1$, $\beta = 0.01$ and $\alpha = 0.05$ and solve numerically for a then one obtains

$$a_1 \approx 0.14 \quad \text{and} \quad a_2 \approx 1.24.$$

Two remarks are worth making at this point. First, generally speaking, satisfaction of the above condition is necessary but not sufficient for the existence of a bump which would require that $U(r) < \theta$ for $(-\infty, a) \cup (a, \infty)$ and $U(r) > \theta$ for $r \in (-a, a)$ [33, 46]. That is, a solution to *Equation (4.2)* is a candidate for a bump with the corresponding half-width, but its existence is not generally not guaranteed. Under certain regulatory assumptions applied to the model without

synaptic dynamics, conditions for the existence of such a bump have been established by Amari [1], where he refers to the existence of an a -solution. It must be noted that it can readily be checked whether the obtained half-width values lead to a bump solution. For instance, in Figure 4.2 we confirm that the aforementioned parameters produce a bump that has half-width $a \approx 1.24$.

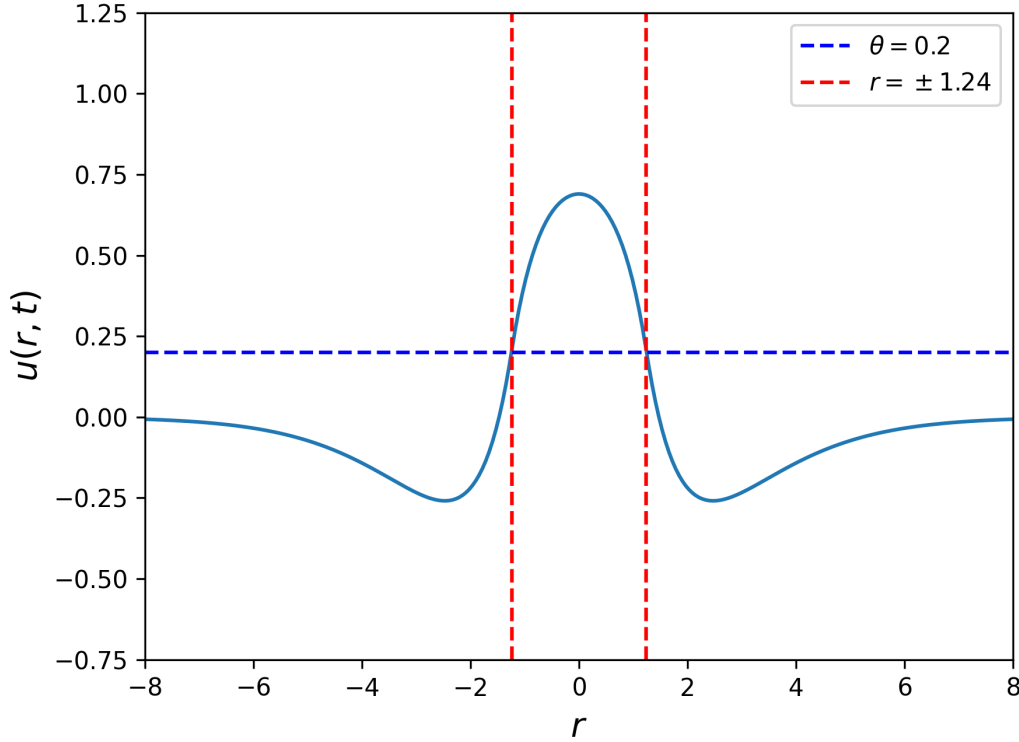


Figure 4.2: Plot of u at $t \approx 100$, also plotting $y = 0.2$ (for θ) and $x = \pm 1.24$. Parameters are $\mu = 50$ and $\theta = 0.2$, and the synaptic kernel is $w(z) = (1 - |z|) \exp(-|z|)$.

The second remark is with regard to the other acquired value, a_1 . Indeed, obtaining two distinct solutions a_1, a_2 suggests that it is possible for two bump solutions to exist (one narrow and one broad). The latter has initially been investigated by Amari within the Heaviside world in [45, 1], but later also in models with synaptic plasticity effects, as seen in [46, 32, 14]. Following these papers, it is seen that the narrow bump solutions are unstable. The significance of this bump is that it defines the critical threshold for bump persistence. An initial activity that is wider than the narrow bump will be attracted to the stable broad bump solution, while an initial profile that is even narrower will ultimately decay to the rest state. We will look in more detail to stability in the next section.

4.1 Stability of Spatially Homogeneous Version

The stability analysis of steady states in the full model is a complex problem. Therefore, as an intermezzo we choose to first consider a spatially homogeneous version of the model, which greatly simplifies analysis and allows for some insight at this reduced level.

In the spatially homogeneous case, the spatial convolution clearly disappears. The system becomes^(xv)

$$\begin{aligned}\partial_t u(t) &= -u(t) + \left(\int_{\mathbb{R}} w(z) dz \right) d(t)(1 + q(t))\mathcal{H}(u(t) - \theta) \\ \partial_t d(t) &= 1 - d(t) - \alpha d(t)(1 + q(t))\mathcal{H}(u(t) - \theta) \\ \partial_t q(t) &= -q(t) + \beta(q_0 - q(t))\mathcal{H}(u(t) - \theta)\end{aligned}\tag{4.3}$$

Let us define $W_0 := \int_{\mathbb{R}} w(z) dz$. If we take, say, $w(x) = (1 - |x|)\exp(-|x|)$, then $W_0 = 0$. Moreover, let (U^*, D^*, Q^*) denote the steady state in the *spatially homogeneous* case. We look at U^* satisfying either $U^* < \theta$ (inactive) or $U^* > \theta$ (active)^(xvi). To that end, when relevant, we denote to which region a component belongs, e.g., U_I^* and U_A^* for sub-threshold (inactive) and super-threshold (active), respectively. Similarly as before, we can now explicitly find the steady states when distinguishing between the two cases.

If $U^* < \theta$, then we have the system

$$\begin{aligned}\partial_t u(t) &= -u(t) \\ \partial_t d(t) &= 1 - d(t) \\ \partial_t q(t) &= -q(t)\end{aligned}\tag{4.4}$$

from which we obtain

$$(U_I^*, D_I^*, Q_I^*) = (0, 1, 0) =: \mathbf{x}_I^*.\tag{4.5}$$

If, on the other hand, $U^* \geq \theta$, then the system becomes

$$\begin{aligned}\partial_t u(t) &= -u(t) + W_0 d(t)(1 + q(t)) \\ \partial_t d(t) &= 1 - d(t) - \alpha d(t)(1 + q(t)) \\ \partial_t q(t) &= -q(t) + \beta(q_0 - q(t))\end{aligned}\tag{4.6}$$

which can be solved by subsequent substitution as before, which yields

$$(U_A^*, D_A^*, Q_A^*) = \left(\frac{1 + \frac{q_0\beta}{1+\beta}}{1 + \alpha \left(1 + \frac{q_0\beta}{1+\beta}\right)} W_0, \frac{1}{1 + \alpha \left(1 + \frac{q_0\beta}{1+\beta}\right)}, \frac{q_0\beta}{1 + \beta} \right) =: \mathbf{x}_A^*.\tag{4.7}$$

^(xv)Here q is the easiest to solve, as it is not dependent on u or r . It is seen that

$$\begin{aligned}\dot{q}(t) &= -q(t) + \beta[q_0 - q(t)] = -(1 + \beta)q(t) + \beta q_0 \\ \implies q(t) &= \frac{\beta q_0}{1 + \beta} + C \exp(-(1 + \beta)t) \\ q(0) &= \frac{\beta q_0}{1 + \beta} + C \exp(0) \implies C = q(0) - \frac{\beta q_0}{1 + \beta} \\ \implies q(t) &= \frac{\beta q_0}{1 + \beta} + \left(q(0) - \frac{\beta q_0}{1 + \beta} \right) \exp(-(1 + \beta)t).\end{aligned}$$

Now that q is known, we could tackle r . The equations will not be friendly however when we plug the obtained solution for q into r .

^(xvi)To determine stability of these steady states, we assume that the equilibrium U^* is not *on* the threshold, but strictly away from it, so $U^* \neq \theta$.

Suppose we say the system in Equation (4.3) is given by $\partial_t \mathbf{x} = F(\mathbf{x})$ with $\mathbf{x} = (u, d, q)$, then the Jacobian matrix is

$$J_F = \begin{bmatrix} \frac{\partial F_u}{\partial u} & \frac{\partial F_u}{\partial d} & \frac{\partial F_u}{\partial q} \\ \frac{\partial F_d}{\partial u} & \frac{\partial F_d}{\partial d} & \frac{\partial F_d}{\partial q} \\ \frac{\partial F_q}{\partial u} & \frac{\partial F_q}{\partial d} & \frac{\partial F_q}{\partial q} \end{bmatrix},$$

and therefore, from Equations (4.4) and (4.6), we have

$$J_F(\mathbf{x}_I^*) = \begin{bmatrix} -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{bmatrix} \quad \text{and} \quad J_F(\mathbf{x}_A^*) = \begin{bmatrix} -1 & (1 + Q_A^*)W_0 & D_A^*W_0 \\ 0 & -1 - \alpha(1 + Q_A^*) & -\alpha D_A^* \\ 0 & 0 & -1 - \beta \end{bmatrix}.$$

From this, we immediately conclude local stability for $\mathbf{x}_I^*$, since the eigenvalues of a diagonal matrix are on its main diagonal, which are all negative: $\lambda_1 = \lambda_2 = \lambda_3 = -1$. Moreover, finding the eigenvalues of $J_F(\mathbf{x}_A^*)$ is not a demanding endeavor either, as it is an upper triangular matrix. The eigenvalues are the diagonal elements, so

$$\begin{aligned} \lambda_1 &= -1 < 0 \\ \lambda_2 &= -(1 + \alpha(1 + Q_A^*)) \\ \lambda_3 &= -(1 + \beta) < 0 \quad \text{as } \beta > 0. \end{aligned}$$

For $\text{sgn}(\lambda_2)$, notice that $Q_A^* = q_0\beta/(1 + \beta) > 0$ for $\beta, q_0 > 0$, and moreover that $\alpha > 0$. Hence $\lambda_2 < 0$. Therefore, all eigenvalues are negative, which means the active region exhibits local stability as well.

4.2 Stability of Full Model

Recall that analyzing the stability of steady states of the full model in Equation (4.1) is complex. This is due to the discontinuity of $D(r)$ and $Q(r)$, which is in turn a result of choosing the Heaviside as a firing rate function. The mentioned synaptic functions are piecewise constant, which leads to the system being *piecewise* smooth. Therefore, the usual approach of linearizing about a specific solution to conclude on stability cannot be used [33]. Therefore, we have to turn to other means. A natural thought at this point is to use a sigmoid function instead of a Heaviside, and ultimately take a high gain limit ($\mu \rightarrow \infty$). This has been explored in [33, 17] through construction of the Evans function. However, this approach leads to inaccurate descriptions of stability. In particular, in [33] the authors look at a model with synaptic depression, where this method of stability analysis failed to capture certain instabilities^(xvii). In literature, an alternative approach is described in [14, 33, 32, 8], which we look at below. The full details can be found in the mentioned references, we highlight a sketch by closely following the highlights of their work.

^(xvii)The authors then discuss another approach, which we shall follow.

Let ψ, φ, ξ be smooth perturbations and $\varepsilon \ll 1$. We consider

$$\begin{aligned} u(r, t) &= U(r) + \varepsilon\psi(r, t) \\ d(r, t) &= D(r) + \varepsilon\varphi(r, t) \\ q(r, t) &= Q(r) + \varepsilon\xi(r, t), \end{aligned}$$

and substitute these into our model with $f(u) = \mathcal{H}(u - \theta)$:

$$\begin{aligned} \partial_t u(r, t) &= -u(r, t) + w(r) \otimes [d(r, t)(1 + q(r, t))\mathcal{H}(u(r, t) - \theta)] \\ \partial_t d(r, t) &= 1 - d(r, t) - \alpha d(r, t)(1 + q(r, t))\mathcal{H}(u(r, t) - \theta) \\ \partial_t q(r, t) &= -q(r, t) + \beta(q_0 - q(r, t))\mathcal{H}(u(r, t) - \theta), \end{aligned} \tag{4.8}$$

In the following, note that $\partial_t[U(r) + \varepsilon\psi(r, t)] = \varepsilon\partial_t\psi(r, t)$ and moreover notice we immediately discard $\mathcal{O}(\varepsilon^2)$. The substitution of $u(r, t) = U(r) + \varepsilon\psi(r, t)$ into the first line of Equation (4.8) yields

$$\begin{aligned} \varepsilon\partial_t\psi(r, t) &= -\varepsilon\psi(r, t) \\ &\quad - U(r) \\ &\quad + \int_{\mathbb{R}} w(r - r')[D(r') + \varepsilon\varphi(r', t)][1 + Q(r') + \varepsilon\xi(r', t)]\mathcal{H}[U(r') + \varepsilon\psi(r', t) - \theta] dr' \\ &= -\varepsilon\psi(r, t) \\ &\quad - \int_{\mathbb{R}} w(r - r')[D(r') + D(r')Q(r')]\mathcal{H}[U(r') - \theta] dr' \\ &\quad + \int_{\mathbb{R}} w(r - r')[D(r') + D(r')Q(r')]\mathcal{H}[U(r') + \varepsilon\psi(r', t) - \theta] dr' \\ &\quad + \varepsilon \int_{\mathbb{R}} w(r - r')[D(r')\xi(r', t) + Q(r')\varphi(r', t) + \varphi(r', t)]\mathcal{H}[U(r') + \varepsilon\psi(r', t) - \theta] dr' \\ &= -\varepsilon\psi(r, t) \\ &\quad + \int_{\mathbb{R}} w(r - r')D(r')[1 + Q(r')][\mathcal{H}(U(r') + \varepsilon\psi(r', t) - \theta) - \mathcal{H}(U(r') - \theta)] dr' \\ &\quad + \varepsilon \int_{\mathbb{R}} w(r - r')[D(r')\xi(r', t) + \varphi(r', t)(1 + Q(r'))]\mathcal{H}[U(r') + \varepsilon\psi(r', t) - \theta] dr' \end{aligned}$$

In a manner, expressions for $\varepsilon\partial_t\varphi(r, t)$ and $\varepsilon\partial_t\xi(r, t)$ can be found. They also contain a difference of Heaviside functions of the form $\mathcal{H}(U(r) + \varepsilon\psi(r, t) - \theta) - \mathcal{H}(U(r) - \theta)$, which causes jump discontinuities.

The perturbations $\varepsilon\psi$ cause the point at which the bump crosses the threshold θ to shift. Recall that these points were denoted by $\pm a$. The amount by which the intersection point shifts we shall denote by $\varepsilon\Delta_{\pm}(t)$. The new intersection points are then located where the perturbed activity $u(x, t)$ crosses the threshold θ , that is, the following is satisfied:

$$u(\pm a + \varepsilon\Delta_{\pm}(t), t) = \theta.$$

Applying a Taylor expansion, we obtain

$$\begin{aligned}\theta &= u(\pm a + \varepsilon \Delta_{\pm}(t), t) = U(\pm a + \varepsilon \Delta_{\pm}(t), t) + \varepsilon \psi(\pm a + \varepsilon \Delta_{\pm}(t), t) \\ &= U(\pm a) + \varepsilon \Delta_{\pm}(t) U'(\pm a) + \varepsilon \psi(\pm a, t) + \mathcal{O}(\varepsilon^2).\end{aligned}$$

The stationary bump $U(r)$ by construction satisfies $U(\pm a) = \theta$, implying that in $\mathcal{O}(\varepsilon)$ we have

$$\begin{aligned}0 = \varepsilon \Delta_{\pm}(t) U'(\pm a) + \varepsilon \psi(\pm a, t) &\implies \Delta_{\pm} = -\frac{\psi(\pm a, t)}{U'(\pm a)} \\ &= \pm \frac{\psi(\pm a, t)}{|U'(a)|}\end{aligned}$$

using that $U'(a) < 0$ (decreasing) and $U'(-a) > 0$ (increasing).

The jump discontinuities alluded to above occur near the bump boundary. Dynamically, as the bump shifts its bump boundary, depression and facilitation change steady state. Either from (D^*, Q^*) in Equation (4.5) to the one in Equation (4.7) or the other way around. Now, φ and ξ could undergo $\mathcal{O}(1/\varepsilon)$ changes. The *auxiliary fields*

$$\begin{aligned}\Phi(r, t) &= \int_{-a+\varepsilon\Delta_{-}(t)}^{a+\varepsilon\Delta_{+}(t)} w(r-r') \varphi(r', t) dr' \\ \Xi(r, t) &= \int_{-a+\varepsilon\Delta_{-}(t)}^{a+\varepsilon\Delta_{+}(t)} w(r-r') \xi(r', t) dr'\end{aligned}$$

are introduced to shield against large changes. These equations can be differentiated and combined with expressions for $\partial_t \psi$, $\partial_t \varphi$, and $\partial_t \xi$ to obtain in $\mathcal{O}(1)$:

$$\begin{aligned}\int_a^{a+\varepsilon\Delta_{+}(t)} w(r-r') D(r') (1 + Q(r')) dr' &= \varepsilon \Delta_{+}(t) w(r-a) G(\Delta_{+}(t)) \\ \int_{-a+\varepsilon\Delta_{-}(t)}^{-a} w(r-r') D(r') (1 + Q(r')) dr' &= -\varepsilon \Delta_{-}(t) w(r+a) G(-\Delta_{-}(t)),\end{aligned}$$

where

$$G(\Delta) = \begin{cases} 1 & \text{if } \Delta > 0 \\ \frac{1 + \frac{q_0 \beta}{1+\beta}}{1 + \alpha \left(1 + \frac{q_0 \beta}{1+\beta}\right)} & \text{if } \Delta < 0, \end{cases}$$

see Equation (4.7). So, the local dynamics depends on whether the boundary moves inwards or outwards which is indeed captured by the sign of $\Delta_{\pm}$. For an inward moving boundary, $\pm \Delta_{\pm} < 0$, the local dynamics are the result of synaptic efficacy governed by plasticity effects. On the other hand, with an outward moving boundary, $\pm \Delta_{\pm} > 0$, the local dynamics are not influenced by plasticity effects.

Ultimately, a linearized expressions for $\partial_t \psi$, $\partial_t \Phi$, and $\partial_t \Xi$ can be found by combining results and only taking $\mathcal{O}(1)$ terms. A problem is then that it is not possible to find all solutions of the system $\mathbf{F} = (\partial_t \psi, \partial_t \Phi, \partial_t \Xi)^T$. To make further progress, one may assume that $\Delta_{\pm}(t)$ never switches sign, which means the system has *seperable* solutions:

$$(\psi(r, t), \Phi(r, t), \Xi(r, t))^T = \exp(\lambda t) (\psi(r), \Phi(r), \Xi(r))^T, \quad \lambda \in \mathbb{R}.$$

These solutions then determine stability. To that end, eigenpairs $(\lambda, \psi(r))$ of the eigenvalue problem resulting from the above must be considered. Due to the jump discontinuities focusing the stability criterion to the bump boundary, we only need to look at $r = \pm a$. In [14], it is concluded that indeed narrow bumps (originating from the solution with small bump width) are unstable, and moreover that synaptic plasticity can generate traveling waves due to drift instabilities. Finally, oscillatory instabilities can arise, although in such a case the assumption that $\Delta_{\pm}(t)$ never changes sign does not hold. The authors in [14] speculate that instability persists provided a complex eigenvalue with a positive real part belongs to every combination of perturbation signs. Again, we refer to [14, 33, 32, 8] for a more extensive treatment of the work above.

Chapter 5

Next-Generation Neural Model

At this point, we have thoroughly examined the model by Wilson and Cowan, together with synaptic depression and facilitation. The main motivation of adding synaptic dynamics was to improve accuracy of how neural activity evolves by capturing a process that is crucial at timescales at which working memory operates. It must be said that the population model developed in the 70s by Wilson and Cowan allowed for many insights through mathematical analysis, but also due to using the model in computational neuroscience [10]. However, there are some limitations we discuss hereafter. First is its phenomenological basis. The firing rate function is experimentally known to be a sigmoid (and, as we know, for mathematical ease often a Heaviside is adapted) [59]. That is, the model is not obtained bottom-up from biophysical descriptions of neuron action potentials [10]. Instead, we are neural fields where the notion of individual neurons are lost. This seemed necessary when we addressed about the intricacies Huxley-Hodgkin model, making it simply too complicated to arrive at large-scale dynamics from that starting point. Another limitation of the Wilson and Cowan model is that there is no consideration of intrinsic noise and correlations [13]. It assumes that there is an asynchronicity of neuron spiking. However, it is known in neuroscience that one of the things that real brain activity pattern reveals is that neurons synchronize and desynchronize their activity, see for instance [55].

In this section, we explore how those limitations can be dealt with, arriving at a model is derived from the fundamental biophysics of neural spiking. The idea is to consider a model describing microscopic dynamics, and from this derive the macroscopic behavior by using statistical methods on a large number of neurons^(xviii). We follow a paper by Montbrió, Pazó, and Roxin, in which the dynamics of spiking neurons is linked to the firing rate [36]. There is other literature that, for instance, puts more focus on the evolution of synchronicity, as further outlined in [10].

As suggested, to establish macroscopic properties from the microscopic world, we first require a suitable model that describes dynamics at the level of individual neurons, after which statistical methods are to be applied. While the Hodgkin-Huxley model, mentioned earlier, provides a detailed description at the microscopic level, its treatment of ion flow leading to action potentials exceeds our intended scope and would significantly complicate the process. Instead, a candidate which is simpler yet sufficiently represents dynamics near the spiking threshold is the quadratic

(xviii) To better understand what this means, let us consider a well-known analogue for those with some background in physics. We turn to the theory of thermodynamics, using [7] as a reference. Consider a collection of atoms. Describing their precise motions is extremely complex, yet these microscopic motions are exactly what underlies macroscopic properties such as temperature and pressure. Initially, these properties were characterized from a macroscopic viewpoint through classical thermodynamics, a theory that deliberately neglects detailed information about individual particles. Notice we find ourselves in a similar predicament in the context of the neural field model, where individual spikes of neurons are out of scope. Later, the physicists developed statistical thermodynamics. The idea is that macroscopic properties are derived by applying statistical methods to the behavior of individual atoms and molecules. Also see [7]. A similar approach can be adopted in neural modeling.

integrate-and-fire (QIF) model [36]. Below, we shortly introduce this model to subsequently derive a macroscopic observables.

In the QIF model, a single neuron is described by its membrane potential V satisfying [18, 29]

$$\begin{aligned} \partial_t V &= V^2 + I \\ \text{if } V &\geq V_{\text{peak}} \text{ then } V \leftarrow V_{\text{reset}}. \end{aligned}$$

Here, I is the input current, and the second line is referred to as the ‘resetting rule’. As soon as the membrane potential of the neuron reaches a threshold or peak value V_{peak} , the neuron is said to emit a spike and its potential is reset to V_{reset} , also see *Figure 5.1*.

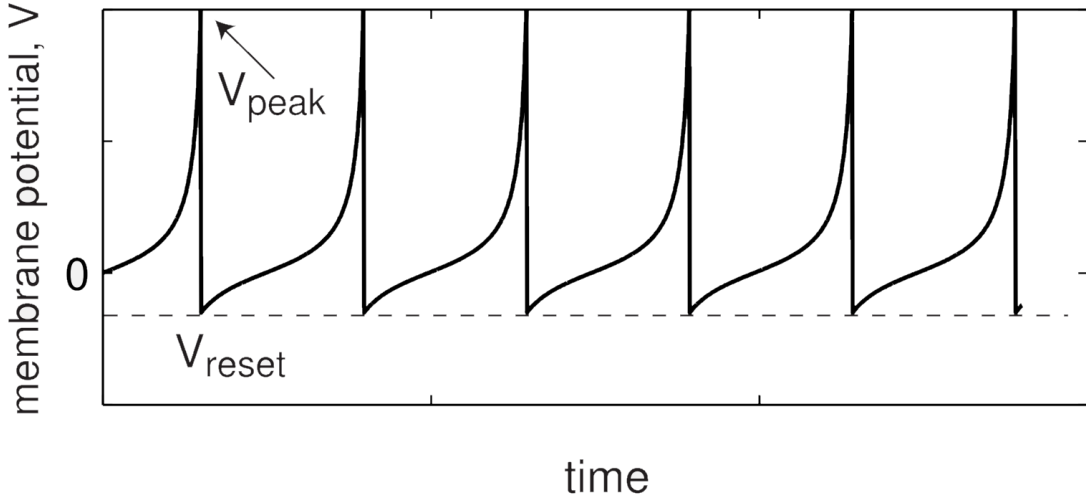


Figure 5.1: Dynamics of a QIF neuron where $I > 0$. Amended from [29].

Let us consider a network of $N \in \mathbb{Z}_{>0}$ neurons, assumed to be a heterogeneous all-to-all coupled population of QIF neurons [36]. We then say that the microscopic state of these neurons are captured by $\{V_j\}_{j=1,\dots,N}$. Applying indexing to the single neuron QIF model, we say the membrane potential of neuron j satisfies

$$\begin{aligned} \partial_t V_j &= V_j^2 + I_j \\ \text{if } V_j &\geq V_{\text{peak}} \text{ then } V_j \leftarrow V_{\text{reset}}, \end{aligned}$$

where $j \in \{1, \dots, N\}$. The input current I_j is given by $I_j = \eta_j + Js(t) + I(t)$, where η_j and $I(t)$ are external components, the first one being fixed for a neuron j while the second one is allowed to vary with time. Moreover, $Js(t)$ forms the recurrent input component consisting of the synaptic weight J , and the mean synaptic activation $s(t)$, given by

$$s(t) = \frac{1}{N} \sum_{j=1}^N \sum_{k: t_j^k < t} \int_{-\infty}^t a_\tau(t - t') \delta(t' - t_j^k) dt'.$$

A few more symbols need explaining, namely, t_j^k is the time at which the k th spike of the j th neuron is emitted, $a_\tau(t)$ is the normalized synaptic activation upon an incoming spike with synaptic time scale τ , and $\delta(t)$ is the Dirac delta function.

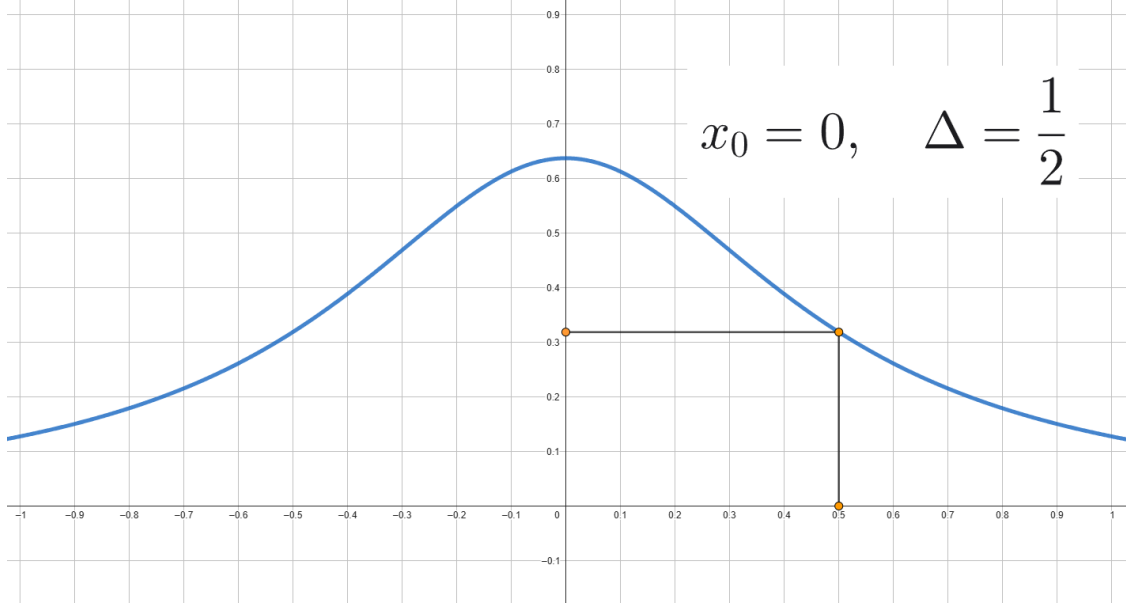


Figure 5.2: Probability density function of the Lorentzian distribution with parameters $x_0 = 0$ and $\Delta = \frac{1}{2}$.

Now, we can take $N \rightarrow \infty$. The goal is being able to describe the macroscopic dynamics of firing rate and membrane potential. We highlight the most important ingredients that allow for this, following the work of [36], to which we refer for the full treatment. In the limit, we have

$$\dot{V} = V^2 + \eta + Js(t) + I(t)$$

and we let

$$\rho(V | \eta, t) dV$$

be the fraction of neurons with membrane potential between V and $V + dV$. Notice that η is now a continuous random variable $\eta \sim g(\eta)$, with g being some probability density function. Moreover, we can say that

$$\int_{-\infty}^{\infty} \rho(V | \eta, t) g(\eta) d\eta,$$

is the total voltage density at time t . In order to have the number of neurons as a conserved quantity we find

$$\partial_t \rho + \partial_V [(V^2 + \eta Js + I)\rho] = 0.$$

To make progress, the authors introduce an ansatz known as the ‘Lorentzian ansatz’ (LA).

Definition 5.1 (Lorentzian distribution). The Lorentzian distribution (also called Cauchy distribution) has probability density function [11]

$$f(x | x_0, \Delta) = \frac{1}{\pi \Delta} \frac{1}{1 + \left(\frac{x-x_0}{\Delta}\right)^2}, \quad x \in \mathbb{R}.$$

Here, $x_0 \in \mathbb{R}$ represents the location of the peak around which the distribution is centered, and $\Delta \in \mathbb{R}_{>0}$ is the half-width of the distribution. Also see Figure 5.2.

Lorentzian ansatz. The density of membrane potentials are distributed according to the Lorentzian distribution:

$$\rho(V \mid \eta, t) = \frac{1}{\pi} \frac{x(\eta, t)}{[V - y(\eta, t)]^2 + x(\eta, t)^2}.$$

It is then concluded that this equation fully characterizes the macroscopic dynamics of the QIF neuron network [36]. The authors are subsequently able to derive the total firing rate $r(t)$ and mean membrane potential $v(t)$. Further progress is made when assuming $\eta \sim \text{Lor}(x_0 =: \bar{\eta}, \Delta)$, that is

$$g(\eta) = \frac{1}{\pi\Delta} \frac{1}{1 + \left(\frac{\eta - \bar{\eta}}{\Delta}\right)^2}.$$

This allows for the following firing-rate equations (FREs):

$$\dot{r} = \frac{\Delta}{\pi} + 2rv \tag{5.1}$$

$$\dot{v} = v^2 + \bar{\eta} + Jr + I(t) - \pi^2 r^2 \tag{5.2}$$

describing the population firing rate and mean membrane potential, respectively. This system of equations provides a macroscopic description of the spiking network. Notice the similarities between the QIF model and the FREs. Another insight, which cannot be stressed enough, is that the firing rate is now an *emergent property* rather than some supplied function, the latter being the case before. The firing rate is a dynamic variable, coupled to the average potentials.

It is worth observing how the FREs behave, see *Figure 5.3*. The plots are analogous to and agree with [36, *Figure 2*]. The chosen initial conditions were $r_i = 0$ (no firing rate) and $v_i = -2$ (negative mean membrane potential). As the plot suggests, the system settles into a stable state *near* the initial condition. The stability shall be confirmed later. In (a), the following external input stimulus is applied:

$$I(t) = \begin{cases} 3, & \text{if } 0 \leq t \leq 30, \\ 0, & \text{otherwise,} \end{cases}$$

and in (b), a sinusoidal stimulus $I(t) = 3 \sin[(\pi/20)t]$ is used. As soon as the external stimulus is activated, there is a response where r and v grow, and then show oscillatory behavior, reflecting the synchronicity of neurons. In (a), the system reflexes to a particular state, approximately $(r, v) = (1.5, 0)$. When the stimulus turns off at $t = 30$, both r and v do not return to the low activity steady state from before, but instead stay elevated. Again, as will become clear this also is a steady state. That is, whenever there is no stimulus, it is possible for a high activity stable state to coexist with a low activity stable state. Such stability properties will be further addressed shortly.

First, however, we cannot overlook the spike rasters. These plots show the spiking behavior of 300 randomly selected neurons according to the QIF model. Each dot represents an individual spike. Notice the advancement with the next-generation neural model in the sense that we can capture both the macroscopic field dynamics and the corresponding microscopic spiking activity, within a single overview. We refer to *Appendix A* on the simulation code involved for such spiking plots.

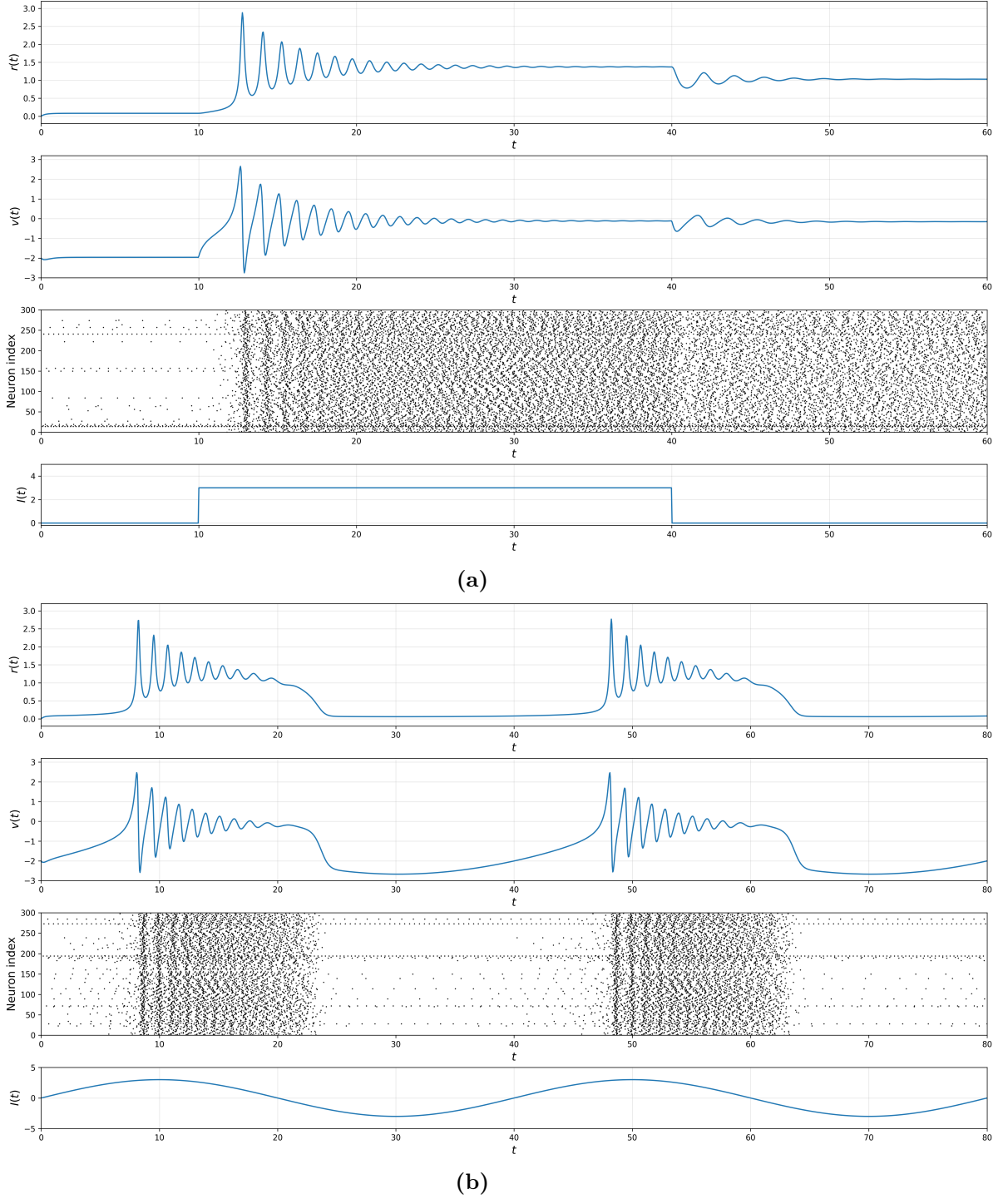


Figure 5.3: Simulations following and agreeing with [36]. The parameters in both simulations were $\Delta = 1$, $J = 15$, and $\bar{\eta} = -5$. **(a)** Uniform external current $I = 3$ between $t = 10$ and $t = 40$, otherwise zero. **(b)** Sinusoidal external current $I(t) = 3 \sin[(\pi/20)t]$.

Before, it was suggested that multiple stable states can coexist. To formalize this, we glance at stability analysis on the FREs. In doing so, we assume that the time-varying external stimulus

is always zero, i.e., $I \equiv 0^{(\text{xix})}$. In Figure 5.4(a) we find the phase diagram as provided in [36, Figure 1]. The system exhibits three regions of interest. Namely, a region of parameter values that lead to a stable node, which corresponds to a state of low activity. The region exhibiting a stable focus *generally* corresponds to a state of high activity. Then lastly, there is a region of bistability. Here, it is possible for low and high activity states to coexist. Note that the point marked with $\blacktriangle$ in the diagram corresponds to $\bar{\eta} = -5$ and $J = 15$ when taking $\Delta = 1$, which are exactly the values of the parameters in Figure 5.3.

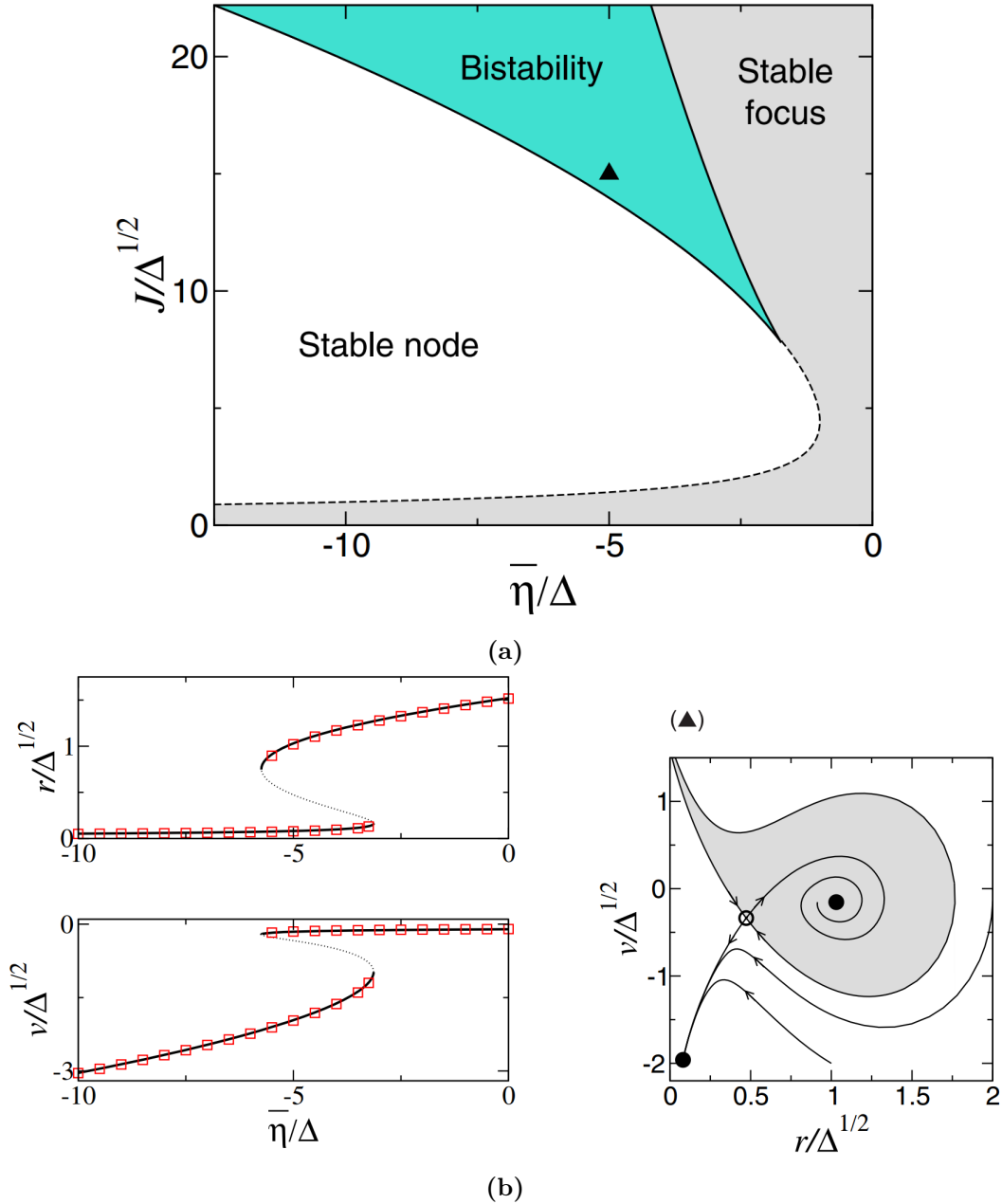


Figure 5.4: Taken from [36]: (a) Phase diagram showing regions where there is a stable node, bistability, or a stable focus. (b) Bifurcation diagram and phase portrait for the point $\blacktriangle$ ($J/\sqrt{\Delta} = 15$ and $\bar{\eta}/\Delta = -5$).

^(xix)Notice that in Figure 5.3(a) both the low activity state and the high activity state are achieved while $I(t) = 0$. The applied stimulus had the effect of bringing the system from the low to the high activity state.

Chapter 6

Combining Short-Term Synaptic Plasticity with Next-Generation Model

As a next step, we reintroduce the STSP dynamics into the spatially extended next-generation neural model. To the best of our knowledge, this has not been described in literature yet. The goal is to reconcile the FREs in *Equations* (6.11) and (6.12) with the earlier discussed synaptic dynamics:

$$\begin{aligned}\partial_t u &= -u + w \otimes d(1+q)f + I \\ \partial_t d &= 1 - d - \alpha d(1+q)f \\ \partial_t q &= -q + \beta(q_0 - q)f.\end{aligned}$$

To that end, let us review what we know. First, in the next-generation neural model, the description of neural activity follows from the coupling of rate and membrane potential. That is, there is no firing rate function that needs to be imposed on the model but the firing rate is instead an emerging property. Moreover, when synaptic depression (d) and facilitation (q) were introduced into the Wilson and Cowan model, they were implemented as prefactors in the nonlocal term, thereby yielding a scaled effective synaptic strength. Combining these matters, we propose the following model:

$$\partial_t r = \frac{\Delta}{\pi} + 2rv \tag{6.1}$$

$$\partial_t v = v^2 + \bar{\eta} - \pi^2 r^2 + I(t) + Jd(1+q)r \tag{6.2}$$

$$\partial_t d = 1 - d - \alpha d(1+q)r \tag{6.3}$$

$$\partial_t q = -q + \beta(q_0 - q)r \tag{6.4}$$

6.1 Equilibria Solutions

Assume $r(t) > 0$ and $I(t) = 0$ for all t . Notice that the steady states satisfy

$$0 = \frac{\Delta}{\pi} + 2R^*V^* \tag{6.5}$$

$$0 = v^2 + \bar{\eta} - \pi^2 r^2 + I(t) + Jd(1+q)r \tag{6.6}$$

$$0 = 1 - d - \alpha d(1+q)r \tag{6.7}$$

$$0 = -q + \beta(q_0 - q)r. \tag{6.8}$$

From *Equations* (6.1)-(6.4), it is generally not algebraically possible to find the steady states, but we can obtain an equation exclusively in r which can then be numerically solved. Let us furthermore assume boundedness of solutions. The above equations can be rewritten to become

$$\begin{aligned} V^* &= -\frac{\Delta}{2\pi R^*} \\ 0 &= (V^*)^2 + \bar{\eta} - \pi^2 r^2 + JD^*(1 + Q^*)R^* \\ D^* &= \frac{1}{1 + \alpha R^*(1 + Q^*)} = \frac{1 + \beta R^*}{1 + (\alpha + \beta)R^* + \alpha\beta(1 + q_0)(R^*)^2} \\ Q^* &= \frac{q_0\beta R^*}{1 + \beta R^*}. \end{aligned}$$

Notice how V^* , D^* , and Q^* ultimately depend only on R^* . Therefore, the second line can be rewritten as

$$0 = \frac{\Delta^2}{4\pi^2(R^*)^2} + \bar{\eta} - \pi^2(R^*)^2 + J \frac{R^* + \beta(1 + q_0)(R^*)^2}{1 + (\alpha + \beta)R^* + \alpha\beta(1 + q_0)(R^*)^2} =: F(R^*). \quad (6.9)$$

Any bounded R^* must satisfy *Equation* (6.9), i.e., $F(R^*) = 0$, after which V^* , D^* , and Q^* can be found. Since $R^* > 0$, and considering the allowed values of the parameters, a valid value for the other components of the steady states always exists.

6.2 Stability Analysis

To determine stability of a given steady state $\mathbf{x}^* = (R^*, V^*, D^*, Q^*)^T$, we consider the Jacobian matrix J . Let the system in *Equations* (6.1)-(6.4) be given by $\partial_t \mathbf{x} = G(\mathbf{x})$, where $\mathbf{x} = (r, v, d, q)^T$. The Jacobian is then

$$J_G = \begin{bmatrix} \frac{\partial G_r}{\partial r} & \frac{\partial G_r}{\partial v} & \frac{\partial G_r}{\partial d} & \frac{\partial G_r}{\partial q} \\ \frac{\partial G_v}{\partial r} & \frac{\partial G_v}{\partial v} & \frac{\partial G_v}{\partial d} & \frac{\partial G_v}{\partial q} \\ \frac{\partial G_d}{\partial r} & \frac{\partial G_d}{\partial v} & \frac{\partial G_d}{\partial d} & \frac{\partial G_d}{\partial q} \\ \frac{\partial G_q}{\partial r} & \frac{\partial G_q}{\partial v} & \frac{\partial G_q}{\partial d} & \frac{\partial G_q}{\partial q} \end{bmatrix}.$$

It can be verified that

$$J_G(\mathbf{x}) = \begin{bmatrix} 2v & 2r & 0 & 0 \\ -2\pi^2 r + Jd(1 + q) & 2v & J(1 + q)r & Jdr \\ -\alpha d(1 + q) & 0 & -1 - \alpha(1 + q)r & -\alpha dr \\ \beta(q_0 - q) & 0 & 0 & -1 - \beta r \end{bmatrix}. \quad (6.10)$$

The stability of a steady state $\mathbf{x}^*$ is then determined by evaluating $J_G(\mathbf{x}^*)$ and determining its eigenvalues. Recall that the eigenvalues $\{\lambda_i\}_{i=1,\dots,4}$ dictate the (relevant) stability types as follows [37]:

- Case I: *unstable*. If any eigenvalue λ_i has a positive real part, $\text{Re}(\lambda_i) > 0$ for some i , the steady state is said to be unstable.
- Case II: *stable node*. If all eigenvalues are purely real and negative, $\text{Re}(\lambda_i) < 0$ and $\text{Im}(\lambda_i) = 0$ for all i , the steady state is a stable node.

- Case III: *stable focus*. If all eigenvalues have negative real parts, $\text{Re}(\lambda_i) < 0$ for all i , and there is a complex conjugate pair among the eigenvalues. That is, the Jacobian matrix has an eigenvalue pair $\lambda_{\pm} = -a \pm bi$, where $a, b > 0$. The solutions show spiraling behavior, and the corresponding steady state is called a stable focus.

To begin with, we pick a set of parameters to make some initial investigations. To make comparisons with Montbrió *et al.*, we choose $J = 15$ and $\bar{\eta} = -5$ (which corresponds to their ‘point $\blacktriangle$ ’ as seen in Figure 5.4). Furthermore, we let $\Delta = 1$, $\alpha = 0.05$, $\beta = 0.01$, $q_0 = 2$. The function $F(R^*)$ from Equation (6.9) and its roots are seen in Figure 6.1. As there are three roots, it is clear that $F(R^*) = 0$ leads to three solutions: R_1^*, R_2^*, R_3^* , and they turn out to be given explicitly by

$$R_1^* = 0.0810 \dots$$

$$R_2^* = 0.4949 \dots$$

$$R_3^* = 0.9431 \dots$$

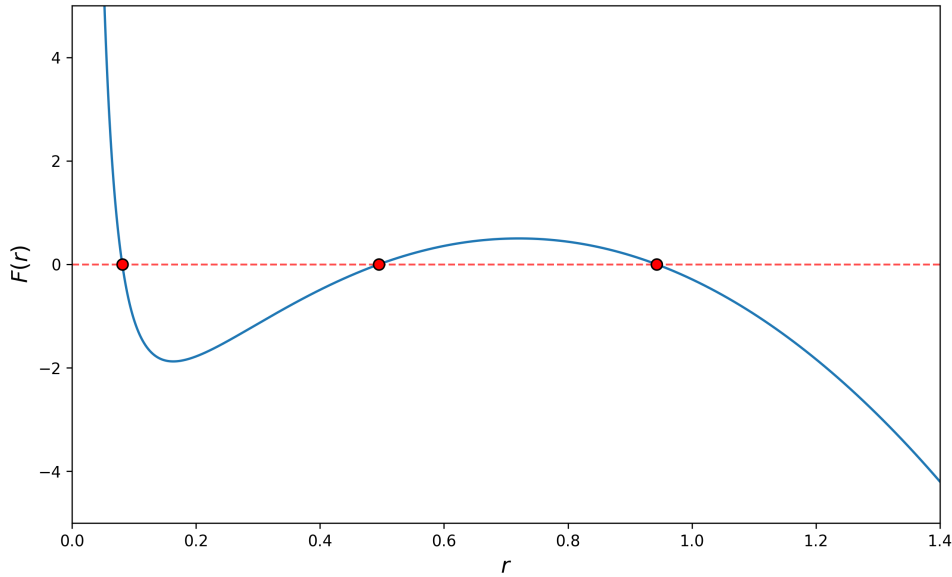


Figure 6.1: Plotting the function F from Equation (6.9) with the parameters $\Delta = 1$, $J = 15$, $\bar{\eta} = -5$, $\alpha = 0.05$, $\beta = 0.01$, and $q_0 = 2$. Choosing these parameters leads to F having three roots, implying that there exist three steady states.

From these solutions, all steady states for the entire system with chosen parameters can be provided using Equations (6.5)-(6.8). We find

$$\mathbf{x}_1^* \approx (0.0810, -1.9624, 0.9959, 0.0016)^T$$

$$\mathbf{x}_2^* \approx (0.4949, -0.3215, 0.9756, 0.0098)^T$$

$$\mathbf{x}_3^* \approx (0.9431, -0.1687, 0.9541, 0.0186)^T.$$

Let Σ_i be the spectrum of $J_G(\mathbf{x}_i^*)$. That is, Σ_i is the set of eigenvalues of the Jacobian corresponding to the steady state $\mathbf{x}_i^*$, where $i \in \{1, 2, 3\}$. We have

$$\begin{aligned}\Sigma_1 &\approx \{-5.3977, -2.4514 - 1.0049i, -1.0009\} \\ \Sigma_2 &\approx \{1.5641, -2.8964, -0.9743, -1.0097\} \\ \Sigma_3 &\approx \{-0.2881 + 2.7738i, -0.2881 - 2.7738i, -1.1425, -1.0138\},\end{aligned}$$

from which it is immediately concluded that $\mathbf{x}_1^*$ is a stable node as all its eigenvalues are purely real and negative. Furthermore, considering this equilibrium has the lowest value for R^* (near zero), we refer to the state as a low activity stable state. Continuing, we find that $\mathbf{x}_2^*$ is an unstable steady state as the first eigenvalue is real but positive. Moreover, $\mathbf{x}_3^*$ is a stable focus (corresponding to a high activity state) due to the complex conjugate pair of eigenvalues in the spectrum.

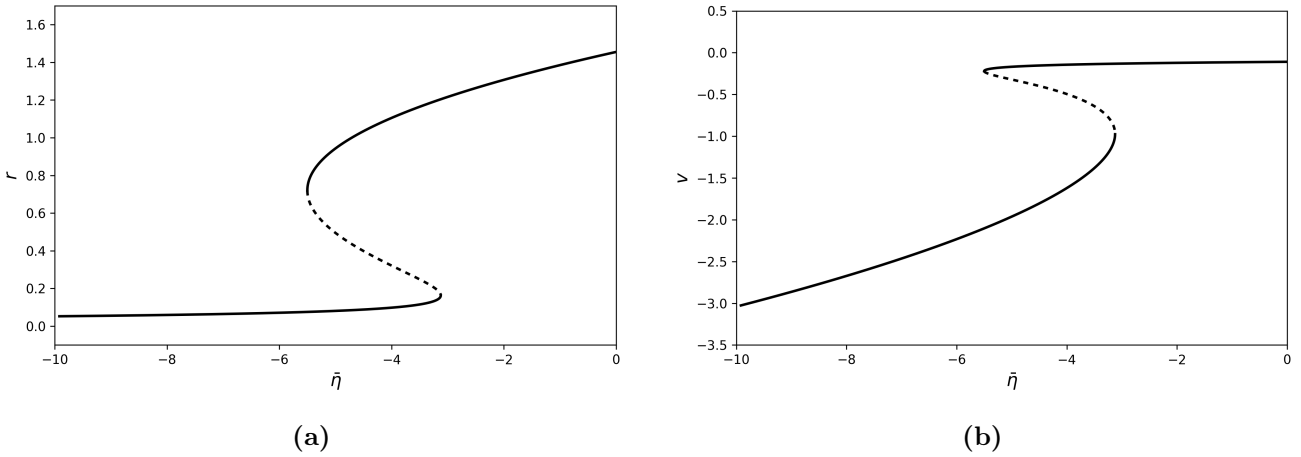


Figure 6.2: Bifurcation diagrams for $J = 15$, dashed line signifies instability: **(a)** Plotting population firing rate r versus $\bar{\eta}$. **(b)** Plotting mean membrane potential v versus $\bar{\eta}$.

A bifurcation diagram corresponding to the parameters picked earlier is provided in *Figure 6.2*. This diagram can be compared with the next-generation model without synaptic effects in *Figure 6.3(b)*. These diagrams exhibit great similarity, implying that the stability behavior is not influenced radically by the inclusion of synaptic dynamics for the current parameter choice. However, since $\alpha > \beta$ (higher depression rate compared to facilitation rate), the negative feedback resulting from the synaptic depression suggests that the firing rate r for stable states is lowered. Making quantitative comparisons are obstructed by the otherwise similarity between the diagrams. However, we can write *Equation (6.9)* as

$$-\frac{\Delta^2}{4\pi^2(R^*)^2} - \bar{\eta} + \pi^2(R^*)^2 = JD^*(1 + Q^*)R^*,$$

and recognize that synaptic depression hence moves the graph in *Figure 6.1* down, resulting in the outer roots (corresponding to stable states) moving to the left (that is, lower r). Upon re-doing

the calculations but now with $\alpha = \beta = 0$, we find

$$R_1^* \approx 0.0811 \dots$$

$$R_2^* \approx 0.4729 \dots$$

$$R_3^* \approx 1.0305 \dots$$

The stable states R_1^* and R_3^* have indeed a higher firing rate upon the removal of depression dominated synaptic plasticity, although the former (low state) only marginally. Note that R_2^* is lower but also serves a different role, as this state is unstable [37, 57].

Numerically, we use Python to find the steady states. This is done using, for instance, the `fsolve` function from the `SciPy` library which finds the root of a function. This allows for solving the system in *Equations* (6.5)-(6.8). An initial guess needs to be supplied from which the solver starts finding a steady state. In general, any guesses too far from the a particular steady state can lead to missing it, as the solver might converge to another state. This is a risk that is in particular relevant when there are three steady states, relatively close to each other. Then, the middle one can easily be missed. However, on our case, the middle exists whenever the high activity state exists and it is generally unstable. That is, whenever it exists it cannot be sustained, which is akin to the earlier concept of always having an unstable narrow bump. Therefore, we choose to supply initial guesses that are sufficiently low or high such that the unstable middle state is not captured. Then `fsolve` was used to find the steady states, supplying it with the Jacobian for better convergence. We refer to *Figures* 6.1 and 6.2 for this idea. Other, more cumbersome, approaches include supplying sufficiently many initial guesses and using `fsolve`. Also `brentq` can be employed, for which an interval needs to be provided. Using these with the entire system, or simply with *Equation* (6.9), gets the desired steady states.

The stability of the next-generation neural model with synaptic plasticity is outlined in *Figure* 6.3. How it is constructed is outlined in *Appendix A*^(xx).

Upon a first glance, we can make some initial observations. Notice the case $\alpha = \beta = 0$, implying that the steady states are $D^* = 1$ and $Q^* = 0$. We indeed recover the original results where there were no synaptic effects, see *Figure* 5.4. Arguably, the most notable change when varying the values of α and β is that the regions change in their size. Especially the region for stable focus (green) has enlarged substantially. On the diagonal plots (where $\alpha = \beta$), the stable focus area expanded such that there is only a small loop where the system is in a stable node. For the ‘off-diagonal’ plots where $\alpha \neq 0 \neq \beta$, notice that the stable focus is larger when $\beta > \alpha$. The result suggests that situations where $\alpha = \beta$ generally lead to situations where the system spirals towards the steady state likely due to the continuous interplay between synaptic depression and facilitation mechanisms. Whenever $\alpha > \beta$, the dynamics of the dominant depression would increase the damping, ultimately showing a stable node more often. For cases $\beta > \alpha$ (but both not zero), the plots show that the stable focus regions are larger compared to any counterpart mirrored in the diagonal. Furthermore, the plots show that the bistability regions shrinks with increasing α and enlarges with increasing β . Lastly, we mention that no additional instability was introduced beyond the ‘intermediate unstable states’ discussed earlier.

^(xx)We point out that we confirmed the stability plot in *Figure* 6.3. This agreement with existent literature suggests that the corresponding code is successful in correctly classifying stability of steady states of the model by Montbrió *et al.*, to which end we extend it to the new, expanded model.

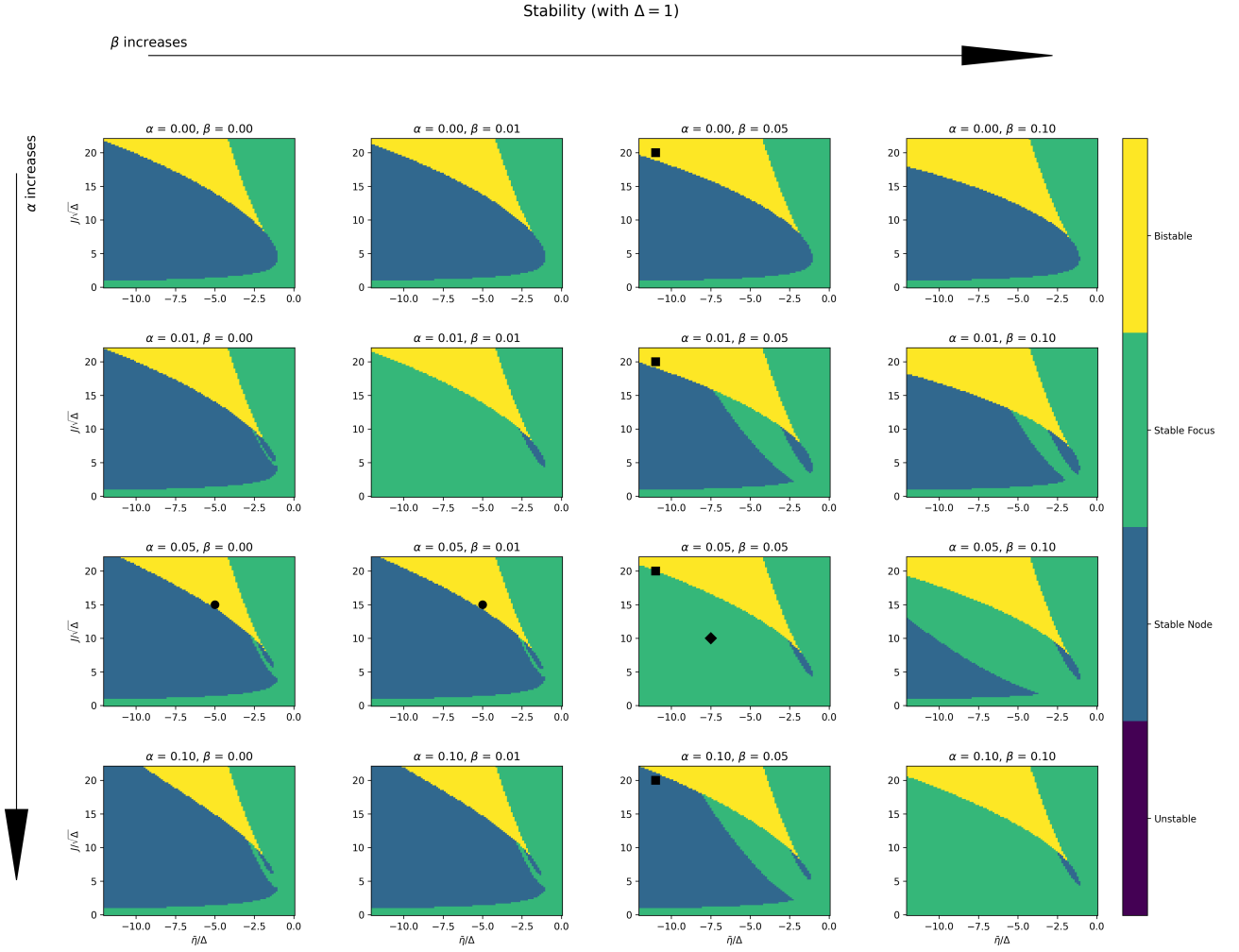


Figure 6.3: MPCK model stability, varying α (depression) and β (facilitation). We always have $\Delta = 1$ and $q_0 = 2$. Some points are marked as they will be referred to later: ■ is on $J = 20, \bar{\eta} = -11, \alpha \in \{0, 0.01, 0.05, 0.1\}, \beta = 0.05$; ◆ is on $J = 10, \bar{\eta} = -7.5, \alpha = \beta = 0.05$; ● is on $J = 15, \bar{\eta} = -5, \alpha = 0.05, \beta \in \{0, 0.01\}$.

Let us investigate this further. We pick:

$$J = 10, \quad \bar{\eta} = -7.5, \quad \alpha = \beta = 0.05,$$

associated with the point ◆. In *Figure 6.4(a)* the function $F(r)$ is plotted from which it is seen that there is one steady state, given by:

$$\mathbf{x}^* \approx (0.0604, -2.6324, 0.9969, 0.0060)^T$$

with corresponding eigenvalues

$$\Sigma \approx \{-6.2982, -4.2317, -1.0029 + 0.0011i, -1.0029 - 0.0011i\}.$$

This confirms that the steady state is a stable focus. However, two remarks can be made here.

First, in the case of Montbrió *et al.*, it was seen that a stable focus generally corresponds to a high state of activity. Here, it is instead a low state. Furthermore, although there is a complex conjugate pair, its complex part is quite small. This means any spiraling behavior in the phase portraits will be rather weak and less obvious, see Figure 6.4(b)-(d).

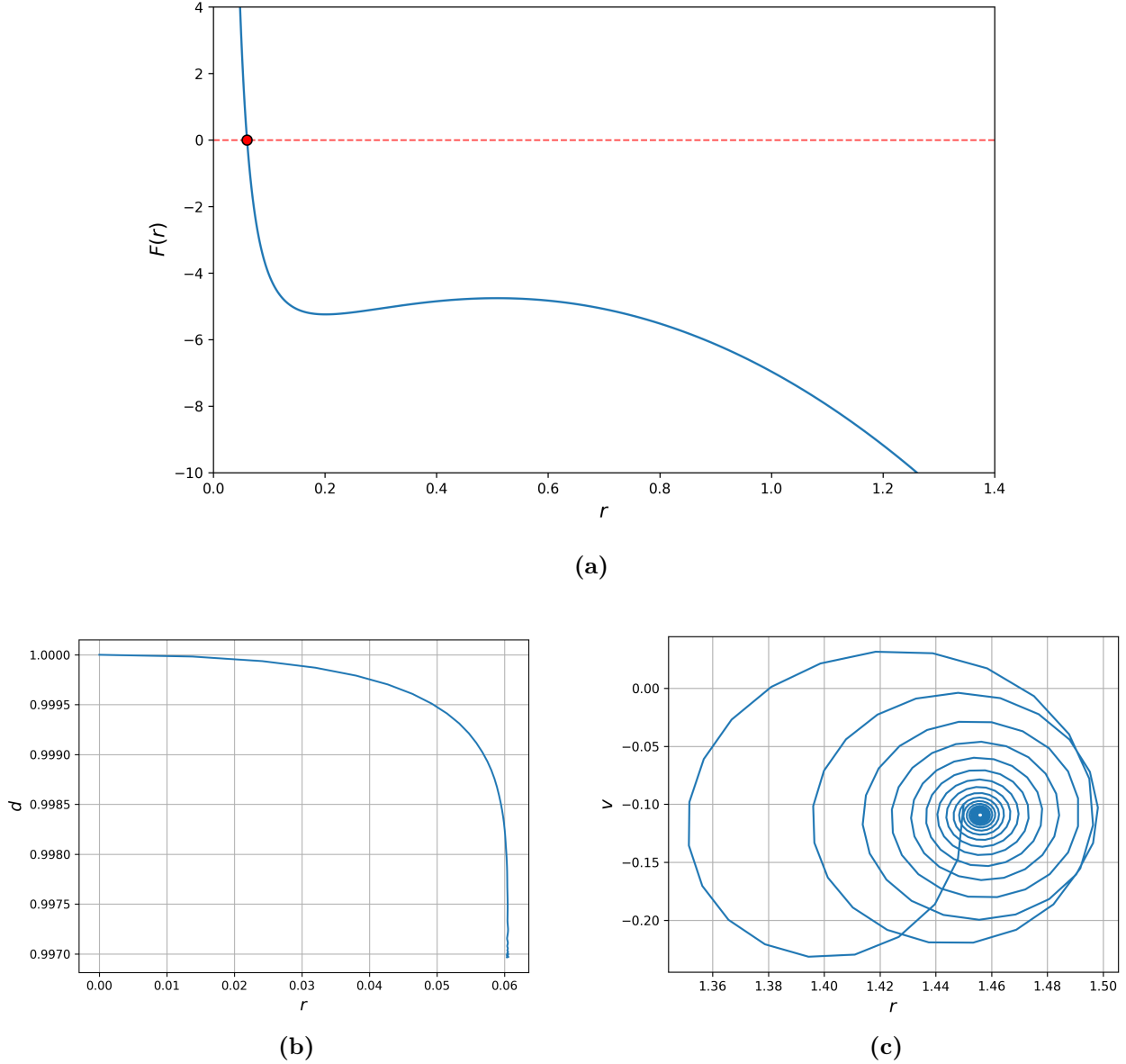


Figure 6.4: Analysis of point ♦: (a) The equation $F(\mathbf{x}^*) = 0$ yields one solution. (b) Phase portrait of r versus d for parameters $J = 10$, $\bar{\eta} = -7.5$, and $\alpha = \beta = 0.05$ (stable focus). (Out of all possible phase portrait combinations, we picked one best representing the presence of a complex conjugate pair.) (c) Phase portrait of r versus v for parameters $J = 20$, $\bar{\eta} = -2.5$, and $\alpha = \beta = 0.05$ (stable focus). This point shows more pronounced spiraling behavior, indicating a difference between the original stable focus region, and the expanded regions seen in cases with synaptic plasticity.

We now turn to ■ in *Figure 6.3*. These are peculiar points in the sense that the stability type changes considerably with varying α . Recall that the parameters in ■ are

$$J = 20, \quad \bar{\eta} = -11, \quad \beta = 0.05.$$

Algorithm 1 Plotting norm and eigenvalues

- 1: Set parameters $\Delta = 1, J = 20, \bar{\eta} = -11, \beta = 0.05, q_0 = 2$
 - 2: Let `alpha_vals` be 100 points between 0 and 0.1
 - 3:
 - 4: *Step 1.* Solving $F(r) = 0$ with $\alpha = 0$ provides first solutions $\mathbf{x}_i^*$ where $i \in \{1, 2, 3\}$
 - 5:
 - 6: *Step 2.* For each initial guess $\mathbf{x}_i^*$:
 - 7: `previous_solution =  $\mathbf{x}_i^*$`
 - 8:
 - 9: **for** α in `alpha_vals`, starting with second index **do**
 - 10: Find solution using `fsolve`, supplying `previous_solution`, Jacobian, and α
 - 11: `previous_solution = solution`
 - 12:
 - 13: *Step 3.* Plot results by showing each branch in a single plot
-

To investigate, we start by plotting the changing α values against r thereby obtaining *Figure 6.5*. The points were found by first solving $F(r) = 0$ to obtain initial $\mathbf{x}^*$'s for current fixed parameters and $\alpha = 0$ (beginning value). For each of these solutions we run through a for loop, increasing α each time, and taking the solution of the previous run as the initial guess of the next. See *Algorithm 1* above.

Looking at the various branches, it is immediately seen that the system is *bistable* for $\alpha \in [0, 0.5]$ (approximately). Furthermore, it is confirmed that there are generally two types of stable steady states, as in Montbrió *et al.* First, a quiescent state (low activity) which is stable for all $\alpha \in [0, 0.1]$. But for $\alpha \in [0, 0.5]$ there is also a high-activity state. A critical transition occurs at $\alpha_{\text{crit}} \approx 0.5$. Here, the ‘upper’ and ‘lower’ high-activity branches meet, yielding a saddle-node bifurcation. Also see *Figure 6.6*, where the eigenvalues are plotted for each branch.

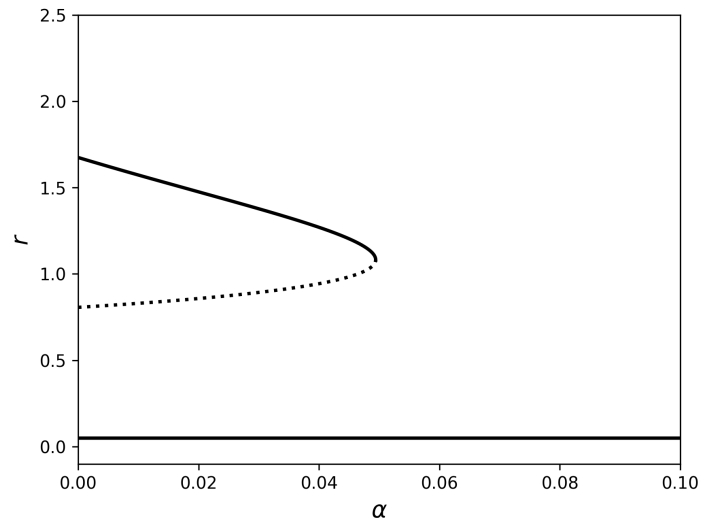
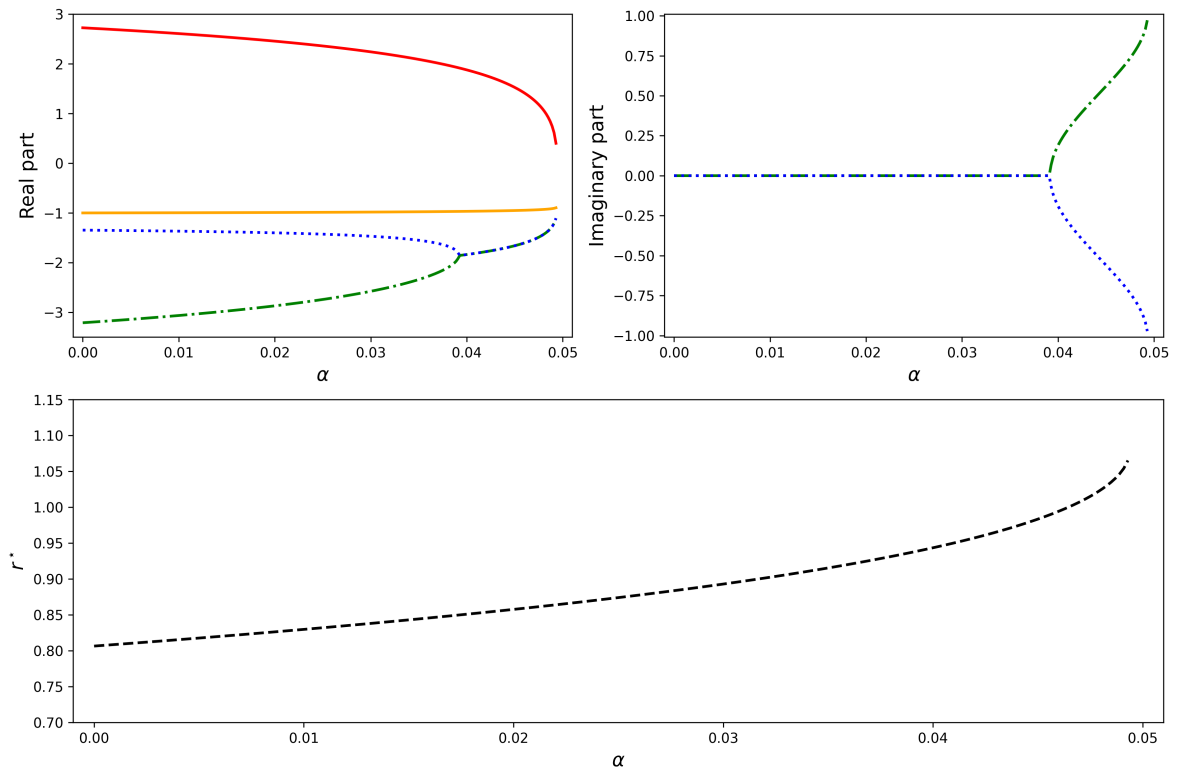
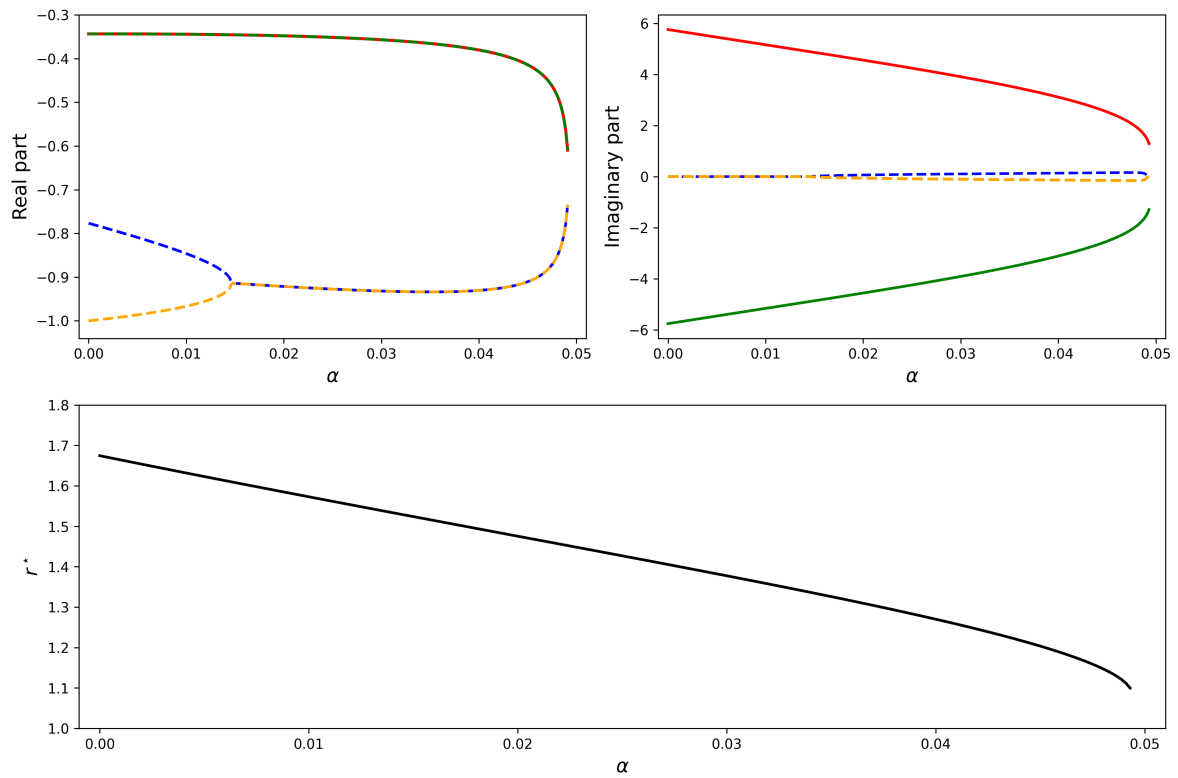


Figure 6.5: Bifurcation diagram of population firing rate r versus α , see *Algorithm 1*.



(a)



(b)

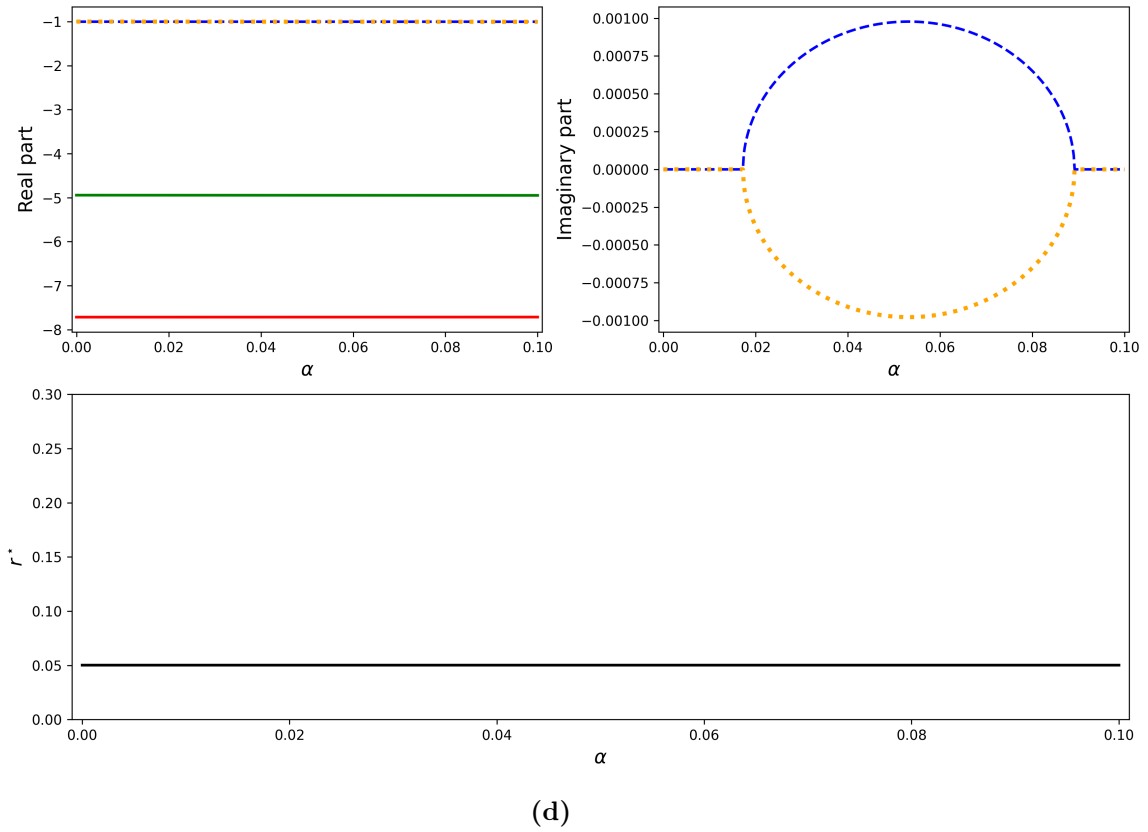
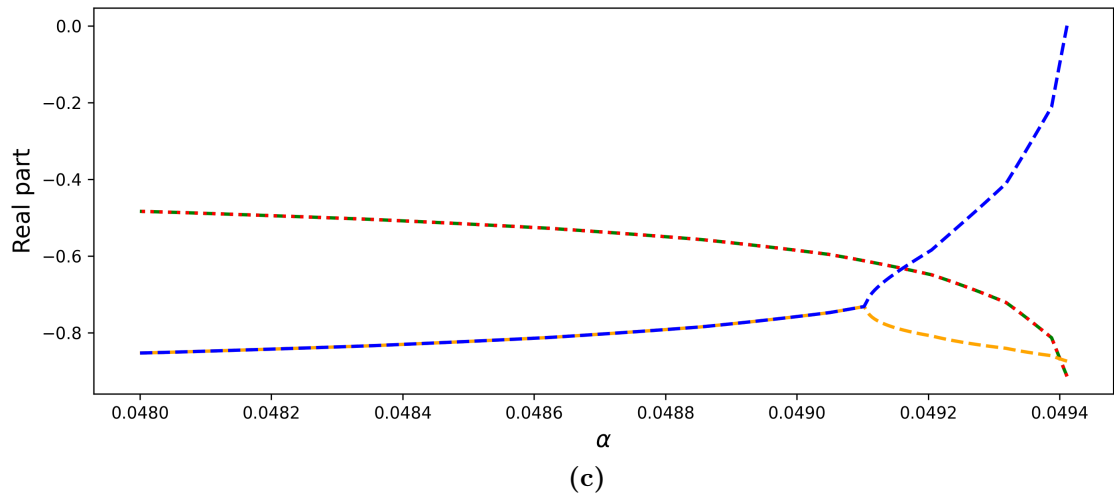


Figure 6.6: Plotting eigenvalues for (a) middle unstable branch, (b) upper stable branch (high activity), (d) lower stable branch (low activity). In (c) we zoom into the region $\alpha \approx 0.5$ of plot (b). A saddle-node bifurcation can be seen.

We now perform some further simulations on the new combined model. Recall *Figure 5.3* where we looked at the model by Montbrió *et al.* We revisit this arrangement, now of course for the expanded model in *Equations (6.1)-(6.4)*. The same uniform stimulus is applied:

$$I(t) = \begin{cases} 3, & \text{if } 10 \leq t \leq 40, \\ 0, & \text{otherwise.} \end{cases}$$

Furthermore, we let $\Delta = 1$, $J = -15$ and $\bar{\eta} = -5$. The idea is to run simulations for:

$$\boxed{\text{(i) } \alpha = 0.05, \beta = 0.01} \quad \text{and} \quad \boxed{\text{(ii) } \alpha = 0.05, \beta = 0}.$$

It is seen from *Figure 6.3* that in both cases we are in a region of bistability.

- (i) $\alpha = 0.05, \beta = 0.01$. See *Figure 6.8*. This represents a familiar result, in the sense that it overall follows the earlier results in *Figure 5.3*. The main difference is that the plasticity effects have an effect on the values of the steady states. In the present case, firing rate will be lower due to dominant depression. Upon application of the stimulus, the system abandons the low state and becomes elevated. When the current is removed at $t = 40$, the system converges to a high activity steady state.
- (ii) $\alpha = 0.05, \beta = 0$. See *Figure 6.7*. Now something remarkable is happening. Whereas before we had seen situations of low activity pre-stimulus and high activity post-stimulus, we now see that the system returns to its low activity state after the stimulus shuts off. The parameters, consisting of depression with rate $\alpha = 0.05$ and no facilitation $\beta = 0$, make it impossible for the system to. That is, . It is important to be reminded of the fact that with the current parameters, we are in the bistable region, thus a high activity state certainly exists, however, it simply cannot be reached.

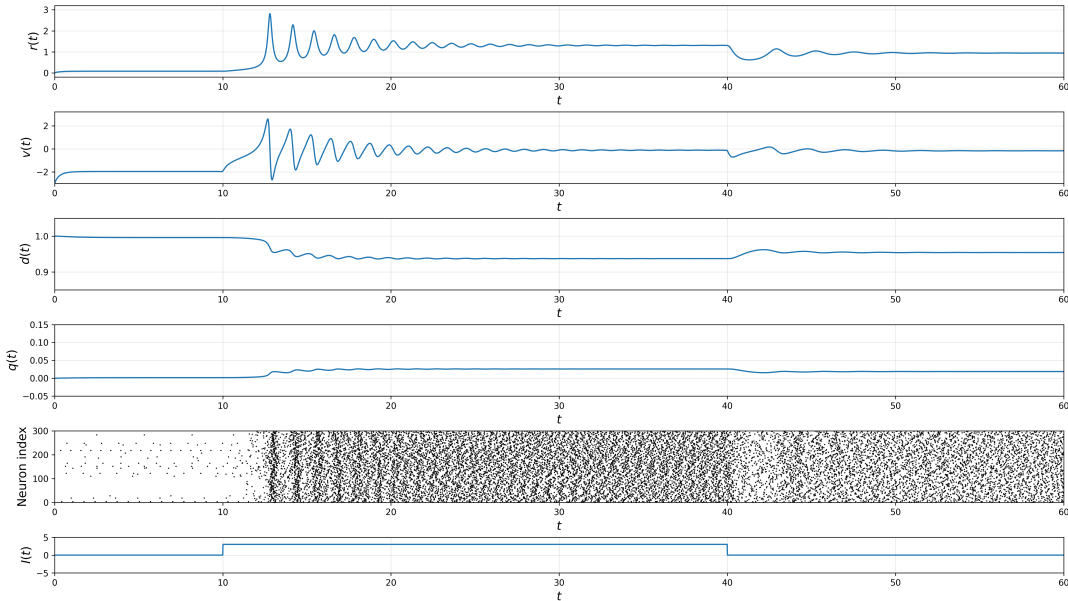


Figure 6.7: Parameters: $\Delta = 1$, $J = 15$, $\bar{\eta} = -5$, $\alpha = 0.05$, $\beta = 0.01$, $q_0 = 2$. Initial value: $(r_i, v_i) = (0.01, -3)$. Spiking neuron plot is according to the QIF model, using 10^4 neurons and randomly plotting 300.

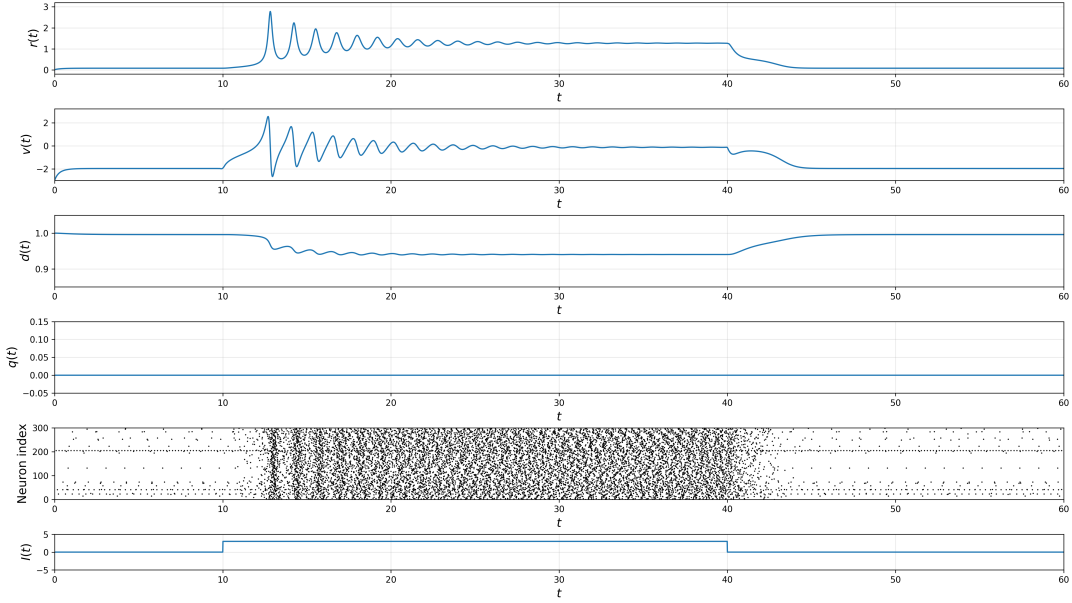


Figure 6.8: Simulation with $\beta = 0$ and otherwise the same as above. Notice how the system is not able to reach the other stable state this time.

We conclude this section by considering a synaptic timescales $\tau_d = 150$ and $\tau_q = 250$, and a stimulus consisting of pulses and low-frequency waves. The former is to further explore the behavior observed directly above, where the system can no longer achieve an existent high state when depression is sufficiently dominant. The latter serves a suggestion made earlier that a stimulus with frequency of 20 Hz is able to induce synaptic depression [38]. Consider

$$I(t) = \begin{cases} 5 & \text{if } t \in [40, 60] \cup [100, 120] \cup [160, 180] \cup [220, 240] \\ 2.5 + A \sin(2\pi f t_s) & \text{if } t \in [350, 750] \\ 0 & \text{otherwise} \end{cases}$$

where $A = 2.5$ is the amplitude of the wave, $f = 20$ Hz is the frequency, and $t_s = 0.01t$ to account for the membrane time constant. The results are presented in the figures below. The square external stimuli are applied at various times, they stay on for 20 time units, and in between pulses there is a 40 time unit off period.

Consider the parameters $\alpha = 0.05$ and $\beta = 0.01$ in *Figure 6.9*. When the stimulus first turns on, depression and facilitation are moving away from their baseline at 1 and 0, respectively. Due to the synaptic timescales $\tau_d, \tau_q \gg 1$, the system has not fully restored when the second pulse arrives. After four pulses, the system is sufficiently depressed such that it cannot maintain the high activity state upon stimulus removal, and instead the rate r drops to the low activity state. In case where $\alpha = 0.05$ and $\beta = 0$, depression is even more dominant. The result of this in simulation is that now only three pulses are required to reach a point where the high activity state is not maintained in the idle period of the stimulus, see *Figure 6.10*. In both cases, we see that a ‘low’-frequency stimulation of 20 Hz yields clear depression of the system. As soon as the stimulus is turned off, d and q slowly relax back to their baseline steady state.

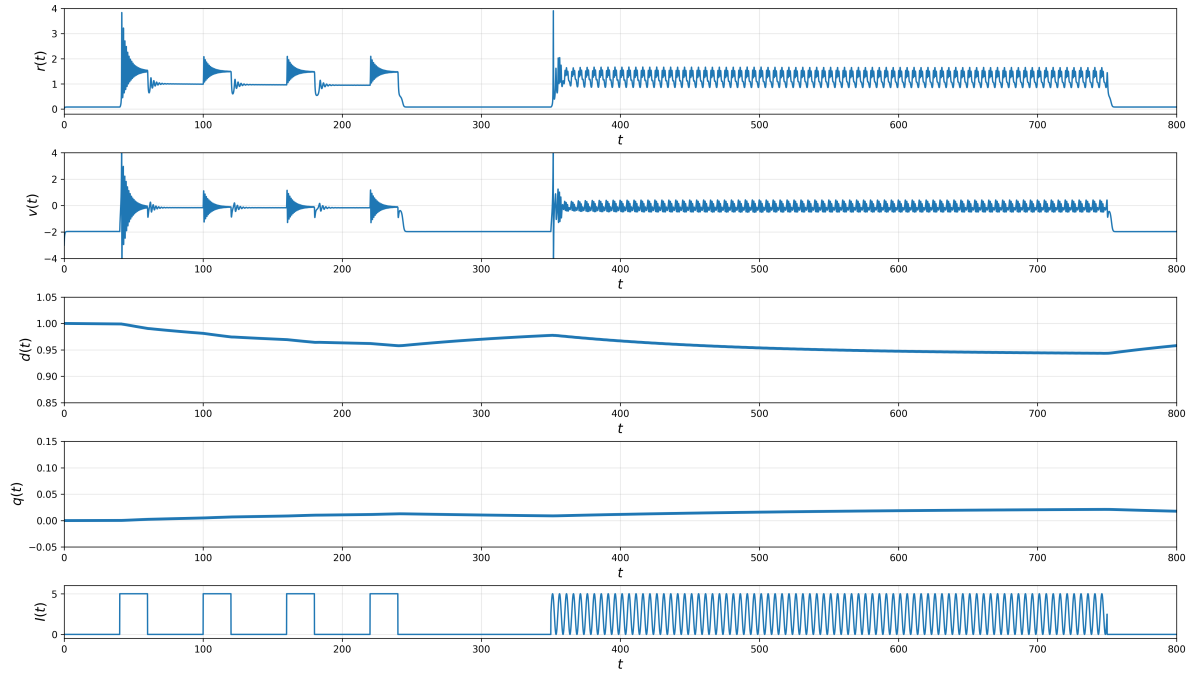


Figure 6.9: Parameters were $\Delta = 1, J = 15, \bar{\eta} = -5, \alpha = 0.05, \beta = 0.01, q_0 = 2, \tau_d = 150$, and $\tau_q = 250$.

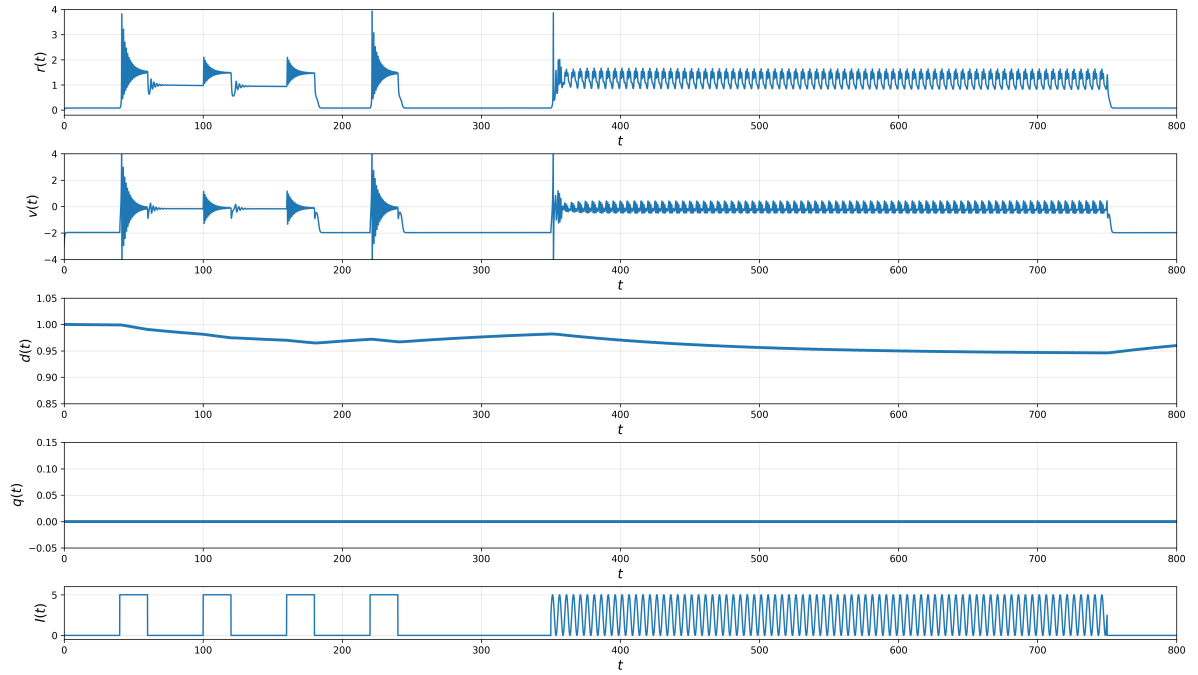


Figure 6.10: Parameters were $\Delta = 1, J = 15, \bar{\eta} = -5, \alpha = 0.05, \beta = 0, q_0 = 2, \tau_d = 150$, and $\tau_q = 250$.

6.3 Spatial Extension and Spiking Network Simulation

Up till now, we have not yet considered the spatially extended version of the next-generation neural model. After the work by Montbrió *et al.* [36], continued progress was made to expand their work to a spatially extended system. For detailed insight how this is done, we refer to [9]^(xxi). The spatially extended next-generation model is given as follows [48]. We still consider N QIF neurons, which satisfy

$$\begin{aligned}\partial_t V_j &= V_j^2 + \eta_j + J s_j + I \\ \text{if } V_j &\geq V_{\text{peak}} \text{ then } V_j \leftarrow V_{\text{reset}},\end{aligned}$$

where^(xxii)

$$s_j(t) = \frac{1}{N} \sum_{k=1}^N w(x_j, x_k) \sum_{k: t_j^k < t} \int_{-\infty}^t a_\tau(t - t') \delta(t' - t_j^k) dt'.$$

Here, w is the synaptic kernel which has been extensively discussed. We consider a similar spatial layout as in Section 2. The cortex is modeled as the interval $\Omega = [-L, L)$ containing N evenly spaced grid points $x_j = -L + (j - 1)h$, where $L \gg 1$, $h = 2L/N$, and $j \in \{1, \dots, N\}$. At each grid point there is a neuron. Moreover, in line with earlier, the fixed external components η_j are distributed using the Lorentzian distribution [48]:

$$\eta_j = \bar{\eta} + \Delta \tan \left(\frac{\pi}{2} \frac{2j - N - 1}{N + 1} \right).$$

All external components $\{\eta_j\}_{j=1, \dots, N}$ are associated, without repetition, with the grid points $\{x_j\}_{j=1, \dots, N}$.

The FREs need to accommodate a spatial structure, which means that a convolution is introduced [9, 48]:

$$\partial_t r = \frac{\Delta}{\pi} + 2rv \tag{6.11}$$

$$\partial_t v = v^2 + \bar{\eta} + Jw \otimes r + I - \pi^2 r^2. \tag{6.12}$$

It is clear from Equations (6.1)-(6.4) that the spatially extended next-generation neural model with synaptic plasticity is then given by introducing the synaptic kernel w through a convolution:

$$\partial_t r = \frac{\Delta}{\pi} + 2rv \tag{6.13}$$

$$\partial_t v = v^2 + \bar{\eta} - \pi^2 r^2 + I(t) + Jw \otimes d(1 + q)r \tag{6.14}$$

$$\partial_t d = 1 - d - \alpha d(1 + q)r \tag{6.15}$$

$$\partial_t q = -q + \beta(q_0 - q)r \tag{6.16}$$

^(xxi)In the mentioned paper ‘ θ -neurons’ are extensively discussed. The authors note however that the QIF model is equivalent to the θ -neuron model through transformation.

^(xxii)Previously, we reserved r to denote the spatial variable. However, as that symbol is now occupied by the firing rate, we switch to using x instead.

We followed [48] and hence for consistency and to allow for comparisons, we choose the same synaptic kernel as the authors picked, that is

$$w(z) = \exp(-|z|) - \frac{1}{4} \exp\left(-\frac{|z|}{2}\right),$$

see Figure 6.11(a). Notice that $W_0 = \int_{\Omega} w(z) dz = 1$.

To study localized bump solutions again, we choose $\Delta = 2$, $J = 15\sqrt{\Delta} = 15\sqrt{2}$, and $\bar{\eta} = -10$, also consistent with the author's choice the mentioned paper. To see the effect due to synaptic plasticity dynamics, we first run a simulation without any STSP, and thereafter a simulation with STSP using $\alpha = 0.05$, $\beta = 0.01$, and $q_0 = 2$. In order to produce a bump, we use a transient localized external current:

$$I(x, t) = \begin{cases} 5 & \text{if } (x, t) \in [-2.5, 2.5] \times [0, 5] \\ 0 & \text{otherwise} \end{cases}$$

Performing solutions yield the results seen in Figures 6.11(b)-(d). It is apparent that the bump with STSP is lower compared to the one without STSP, consistent with earlier results, cf. Figures 4.1(a)-(b). The contour plots in (c)-(d) give an overview of the temporal evolution of the bump, where it is seen that (d), where STSP is enabled, is less bright (i.e., lower r) and narrower compared to (c). Moreover, both contour plots reveal the synchronizing behavior of the firing rate during bump formation and while it is approaching the steady state.

Recall that the (modified) FREs are derived from the microscopic model of QIF neurons. This means we can also plot the corresponding spiking behavior seen macroscopically in Figure 6.11 above. We review this process extensively in Appendix A.

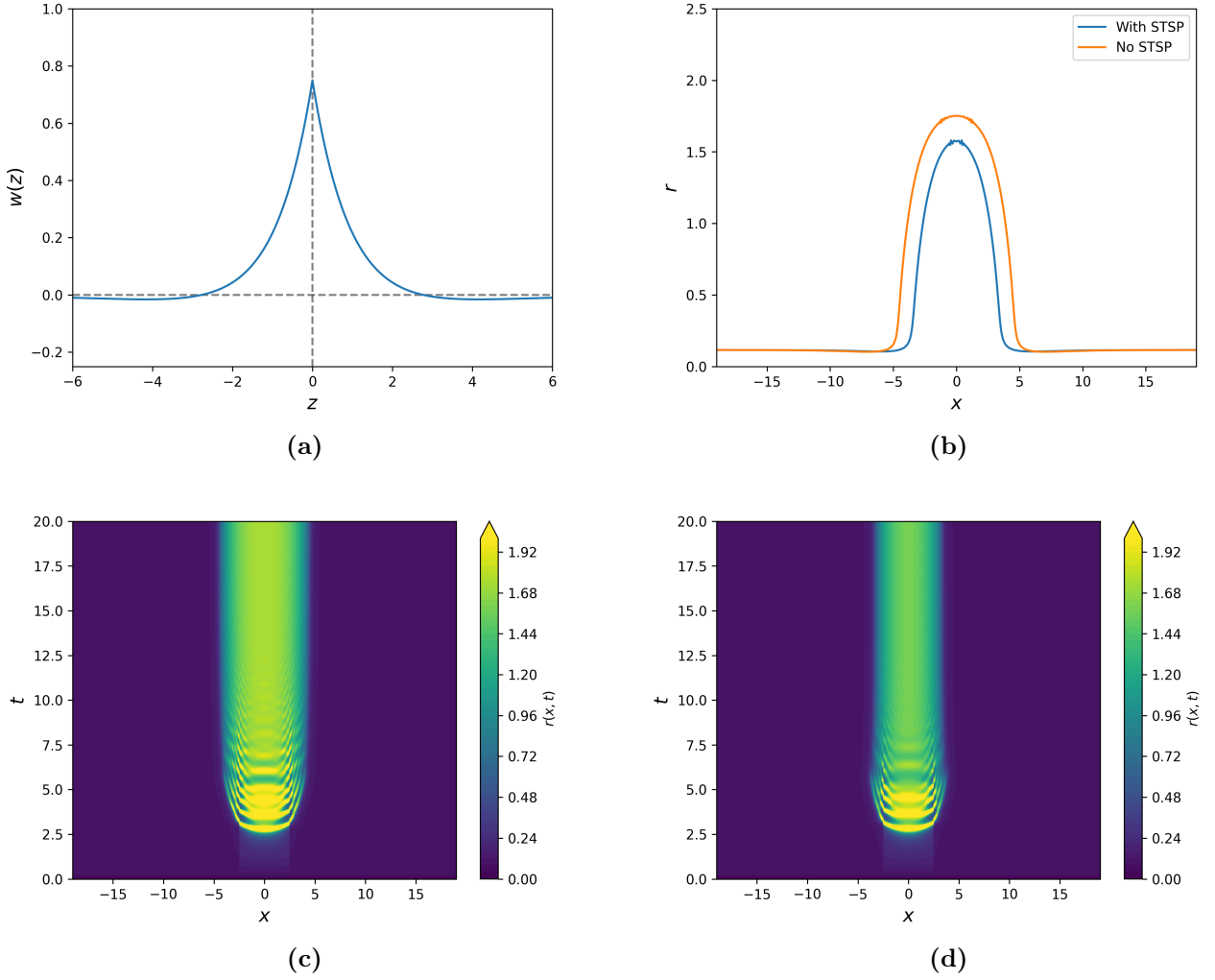


Figure 6.11: The effects of STSP. Parameters were $\Delta = 2, J = 15\sqrt{2}, \bar{\eta} = -10, \alpha = 0.05, \beta = 0.01$, and $q_0 = 2$. **(a)** Chosen kernel function $w(z) = \exp(-|z|) - \frac{1}{4} \exp(-|z|/2)$. **(b)** Activity bump of the next-generation neural model with (blue) and without (orange) STSP effects. **(c)** The corresponding contour plot of *disabled* STSP. **(d)** The corresponding contour plot of *enabled* STSP.

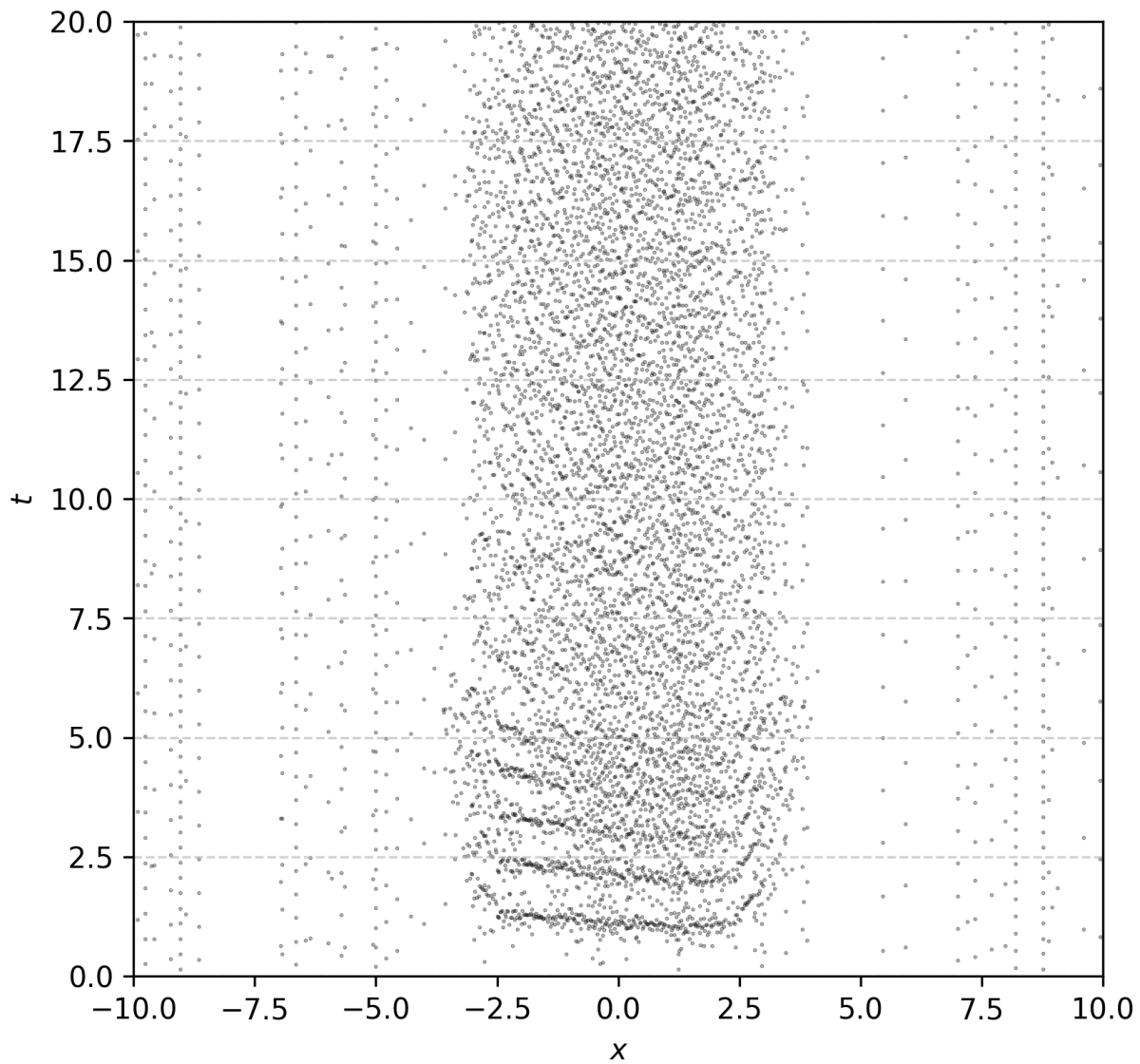


Figure 6.12: Spiking neuron raster. The neurons follow the QIF model. Parameters were $\Delta = 2$, $J = 15\sqrt{2}$, $\bar{\eta} = -10$, $\alpha = 0.05$, $\beta = 0.01$, and $q_0 = 2$

Chapter 7

Wave Entrainment to Apparently Moving Stimuli

We have seen how the next-generation neural model with STSP behaves. In this section, the goal is to advance the analysis even further and see how the model responds to *moving stimuli*. For many animals, the ability to perceive and predict the motion of objects is crucial: from a predator tracking its prey to a primate navigating its environment. The importance of prediction is reflected in the fact that it takes time for sensory information from the eyes to be processed as such. Therefore, any visual information is inherently delayed. It has been found that the visual system can generate a representation of the future position of a moving object before sensory information has arrived [6]. This also serves a role in observing objects intermittently, where estimates are needed while the object is temporarily out of sight based on obtained spatial or temporal-spatial information [40]. This brings us to the idea of *apparent motion*, where a sequence of transient and localized stimuli is perceived as a single, continuously moving object [51].

An underlying component in the reacting and predicting mechanisms of the visual system are coherent neural activity patterns. Traveling waves are instances of such patterns. They occur within the cortex in response to stimuli, but can also arise spontaneously. It has been established that neural activity waves can interact with external stimuli to aid in generating representations of a moving object [12]. Furthermore, such waves are involved in the actual motor output of the eyes. We refer to [51], where the above is discussed in more detail.

In this section, we study how traveling pulses respond when perturbed by external inputs that model a moving or apparently moving visual stimulus. We investigate the phenomenon of entrainment, which refers to the idea that a neural activity wave can alter its speed to synchronize with an external stimulus. By modeling stimuli both as continuously moving and as a series of intermittent flashes (apparent motion), we can study conditions under which entrainment occurs.

To begin, we revisit the model in *Equation 4.1*. In order to allow for diffusive-like displacement of the bump, we change to longer timescales by adding synaptic timescale variables $\tau_d, \tau_q \gg 1$ [32, 51, 14], resulting in

$$\begin{aligned}\partial_t u &= -u + w \otimes d(1 + q)f + \varepsilon I \\ \tau_d \partial_t d &= 1 - d - \alpha d(1 + q)f \\ \tau_q \partial_t q &= -q + \beta(q_0 - q)f.\end{aligned}$$

For tractability, we disable facilitation dynamics ($\beta = 0$) in the initial simulations. Moreover, notice that ε scales the external inputs, where we assume $\varepsilon \ll 1$ to ensure that the visual stimuli presented are weak compared to neural activity [51]. As described in [32, 51], a traveling wave can be subdivided in *pulses* and *fronts*. Examples of both are highlighted in *Figure 7.1*. The key component in whether a moving bump is a pulse or a front is the synaptic depression. Notice

how in *Figure 7.1(b)* the neural activity u is elevated at a certain value when a bump has moved by. However, upon sufficiently strong depression, such activity can no longer be sustained and therefore drops below the threshold, as seen in (a). (Indeed, the bump in (a) is also lower in amplitude due to the depression.)

Non-localized solutions to the above system are $u(x, t) = U(\xi)$ and $d(x, t) = D(\xi)$ (notice we turned off facilitation for now), where $\xi := x - ct$ is the traveling wave coordinate, with c being the wave speed. Notice that c is the speed at which the wave naturally travels, that is, without any external input. For traveling fronts explicit expressions for U, D , and c can be derived by exploiting the fact that a front crosses the threshold θ once, which can be assumed to occur at $\xi = 0$. This implies that the active region is given by $\mathcal{R}_{\mathbb{R}} = \{\xi \in \mathbb{R} \mid U(\xi) > \theta\} = (-\infty, 0)$. The aforementioned expressions are quite lengthy, and hence we do not mention them here. Similarly, expressions can be found for traveling pulses, this time exploiting the fact that the threshold is crossed twice and then considering the active region. We refer to [51] for details.

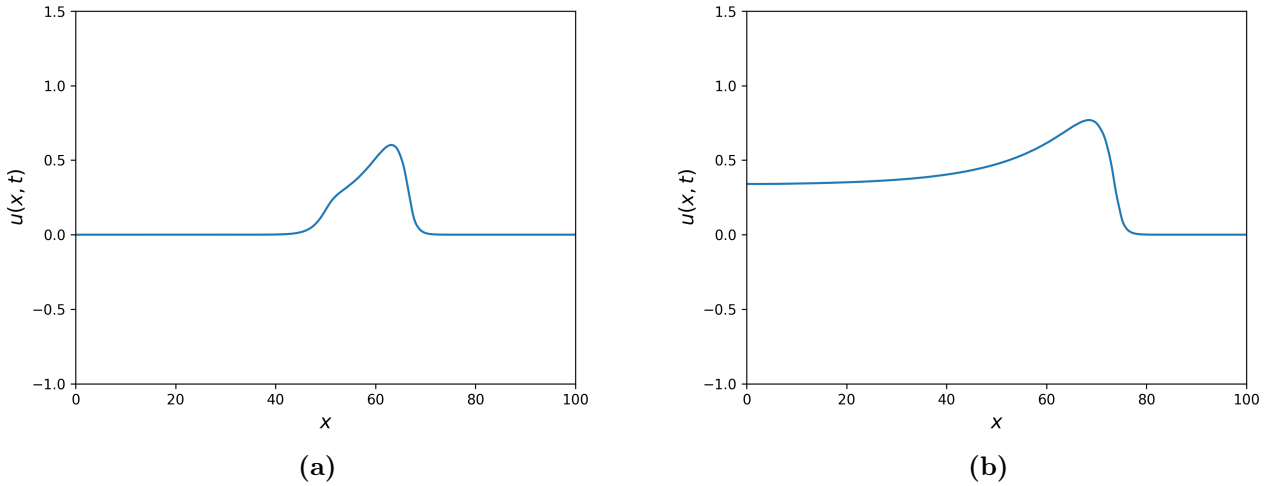


Figure 7.1: Moving bump: (a) Pulse. (b) Front. Both taken at $t \approx 33$.

We now consider an external stimulus representing apparent motion. The stimulus has certain time periods T_{on} , where it provides input, and T_{off} , where there is no input. A time frame in which the stimulus fully cycles between on and off once lasts for $T = T_{\text{on}} + T_{\text{off}}$. We say that the stimulus has width Δ_x and will flash for time T_{on} at some initial position, after which it turns off for a time $T_{\text{off}} = T - T_{\text{on}}$. Lastly, we call Δ_c the speed offset, as $(c + \Delta_c)T$ corresponds to the spatial advancement between the stimulus being switched on. With the notation in place, we can give the apparent motion stimulus:

$$I(x, t) = \mathbb{1}_{(x^* - \Delta_x, x^*)} \left(x - \left\lfloor \frac{t}{T} \right\rfloor (c + \Delta_c)T \right) \mathbb{1}_{(0, T_{\text{on}})} \left(t - \left\lfloor \frac{t}{T} \right\rfloor T \right), \quad (7.1)$$

where $\mathbb{1}_A$ denotes the indicator function defined by $\mathbb{1}_A(x) = 1$ if $x \in A$ and $\mathbb{1}_A(x) = 0$ otherwise. Moreover, $\lfloor \cdot \rfloor$ denotes the floor function given by $\lfloor x \rfloor = \max\{n \in \mathbb{Z} \mid n \leq x\}$.

Now, in *Figure 7.2*, we have a moving pulse. It is apparent that it is a pulse and not a front due to the fact that the posterior of the bump peak ultimately relaxes down to zero, see (a). In (b), we follow the leading edge of the moving pulse. We simplify the notion of the leading edge compared to the original work in [51] by saying it is the x -coordinate at which the moving

pulse first crosses the threshold. For the type of analysis we are aiming at, we deem this to be a sufficient notion. In parts (c) and (d) the apparent motion stimulus from Equation (7.1) is implemented. The orange rectangles represent the spatiotemporal parts where the stimulus is switched on. It is clear from the results that the wave is successfully entrained. Here, $\Delta_c = 0.5$ and $\varepsilon = 0.2$ (other parameters are described in the caption of Figure 7.2). This means that the bump alters its inherent speed in order to advance with the speed of the stimulus [51]. In (d), an example is shown where the entrainment fails. Here, the stimulus is made weaker and also moves faster: $\Delta_c = 1$ and $\varepsilon = 0.1$. These both have an effect where it is more difficult for the bump to entrain the wave. Notice that after entrainment failure, the pulse retreats to its own natural wave speed c . Another influence on whether a stimulus can successfully entrain the wave is the duration of T_{on} versus T_{off} . If a stimulus is only briefly switched on and then stays off for a relatively long time period, then it is clear that entrainment becomes impossible. From this it can be concluded that there is a *window* of entrainment; there is a set of parameters that lead to successful entrainment, outside of which entrainment will fail. It turns out that entrainment is successful when [51]

$$\Delta_c < \varepsilon \frac{T_{\text{on}}}{T} \frac{c}{K},$$

where K is a scaling factor, its definition is outside the scope of our treatment^(xxiii).

We now consider the next-generation neural model, with the same alterations as above, that is, synaptic weights $\tau_d, \tau_q \gg 1$ and scaling parameter ε .

$$\begin{aligned} \partial_t r &= \frac{\Delta}{\pi} + 2rv \\ \partial_t v &= v^2 + \bar{\eta} - \pi^2 r^2 + \varepsilon I + Jw \otimes d(1+q)r \\ \tau_d \partial_t d &= 1 - d - \alpha d(1+q)r \\ \tau_q \partial_t q &= -q + \beta(q_0 - q)r \end{aligned}$$

Several remarks are in order. Due to the model difference, some parameters needed to be re-tuned. A value of $\varepsilon = 5$ was chosen as it produced a weak enough stimulus that had an effect but was unable to transform the low activity into high activity by itself. Also, we set $\Delta = 1$, $J = 20$, $\bar{\eta} = -5$, $\alpha = 0.9$, $\beta = 0.2$, and $q_0 = 2$. The synaptic timescales are still $\tau_d = 20$ and $\tau_q = 30$. Another concept that needs to be re-considered is the leading edge. Before, we said this was the point at which the threshold $\theta = 0.2$ is crossed from the right, as we are considering right-moving bumps. We keep this simplified approach, only slightly increase the ‘threshold’ to 0.5 as values for r are generally higher than values for neural input u . Notice that there is no true threshold parameter θ in this model, as it was part of the firing rate, but that is now an emergent property in the system. As a last remark, note that the stimulus εI is applied to v , but since r is coupled to v , we plot with r .

In Figure 7.3 we present the results of the simulations using the apparent motion stimulus in Equation (7.1). Plots (a) and (b) show successful and failed entrainment, respectively. Parameters $\Delta_c = 0.7$ and $\varepsilon = 5$ ensured entrainment, while increasing the wave speed by setting $\Delta_c = 1.1$ and decreasing the stimulus strength, $\varepsilon = 4$, and moreover tuning $T_{\text{on}} = 0.4$ and T_{off} entrainment clearly failed. It was checked that they are both pulses, moving with a natural wave speed $c \approx 3.32$. This speed is noticeably higher than in the previous model. Also note that in the plots, the x -axis is extended further. The results show that the advanced model we can also model wave

^(xxiii)The scaling factor K is given as $K := \langle U', v \rangle + \tau_d \langle D', p \rangle$, which is further outlined in [51, Sections 4 and 5].

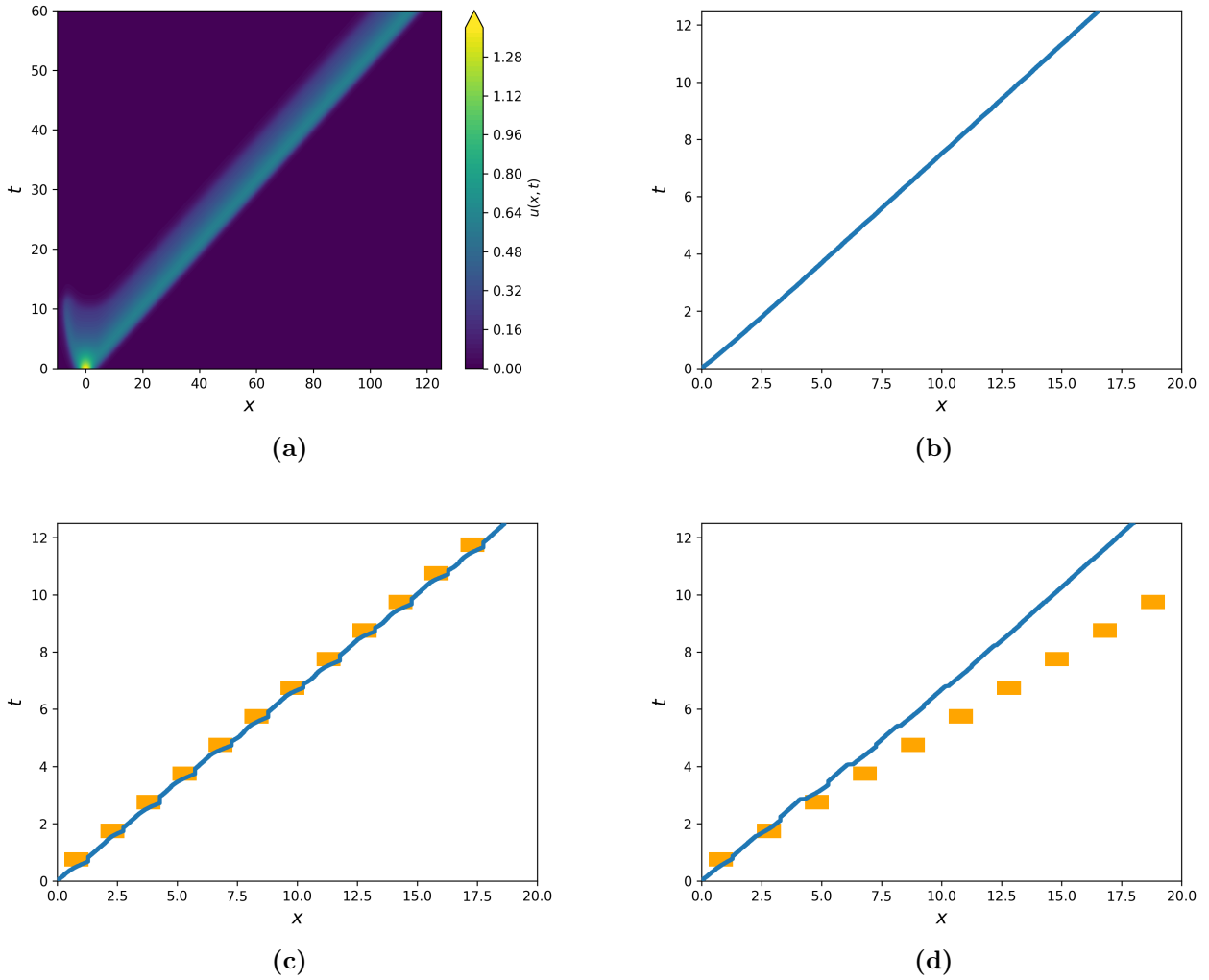


Figure 7.2: Moving pulse and entrainment: **(a,b)** Contour and leading edge plot of a moving pulse at its natural wave speed. Parameters were $\beta = 5, \tau_d = 20, \tau_q = 30$. Moreover, $c = 1, \Delta_x = 1, T_{\text{on}} = T_{\text{off}} = 0.5$. **(c)** Successful wave entrainment. Parameters were $\Delta_c = 0.5, \varepsilon = 0.2$ and otherwise the same as (a,b). **(d)** Failed wave entrainment. Parameters were $\Delta_c = 1, \varepsilon = 0.1$ and otherwise the same as (a,b).

entrainment with an apparently moving stimulus.

However, there are also some clear differences as a result of the behavior due to the next-generation model. With the parameters $\Delta_c = 1.1$ and $\varepsilon = 5$ an interesting phenomena can be observed, see *Figure 7.3(c)-(d)*. (For the other parameters we refer to the caption.) The firing rate r exhibits synchronizing behavior, which is reflected by the oscillations in the contour plot. The apparently moving stimulus is unable to entrain the pulse, since the pulse departs from the stimulus and returns to its natural wave speed. But notice that oscillations farther from the leading edge, that is, more to the left, are at some point unable to follow the leading oscillations themselves. This results in a break within the pulse, yielding two separate pulses. The trailing pulse eventually propagates at the natural wave speed c . Of course, this behavior is not captured in plot (d) which only shows the leading edge.

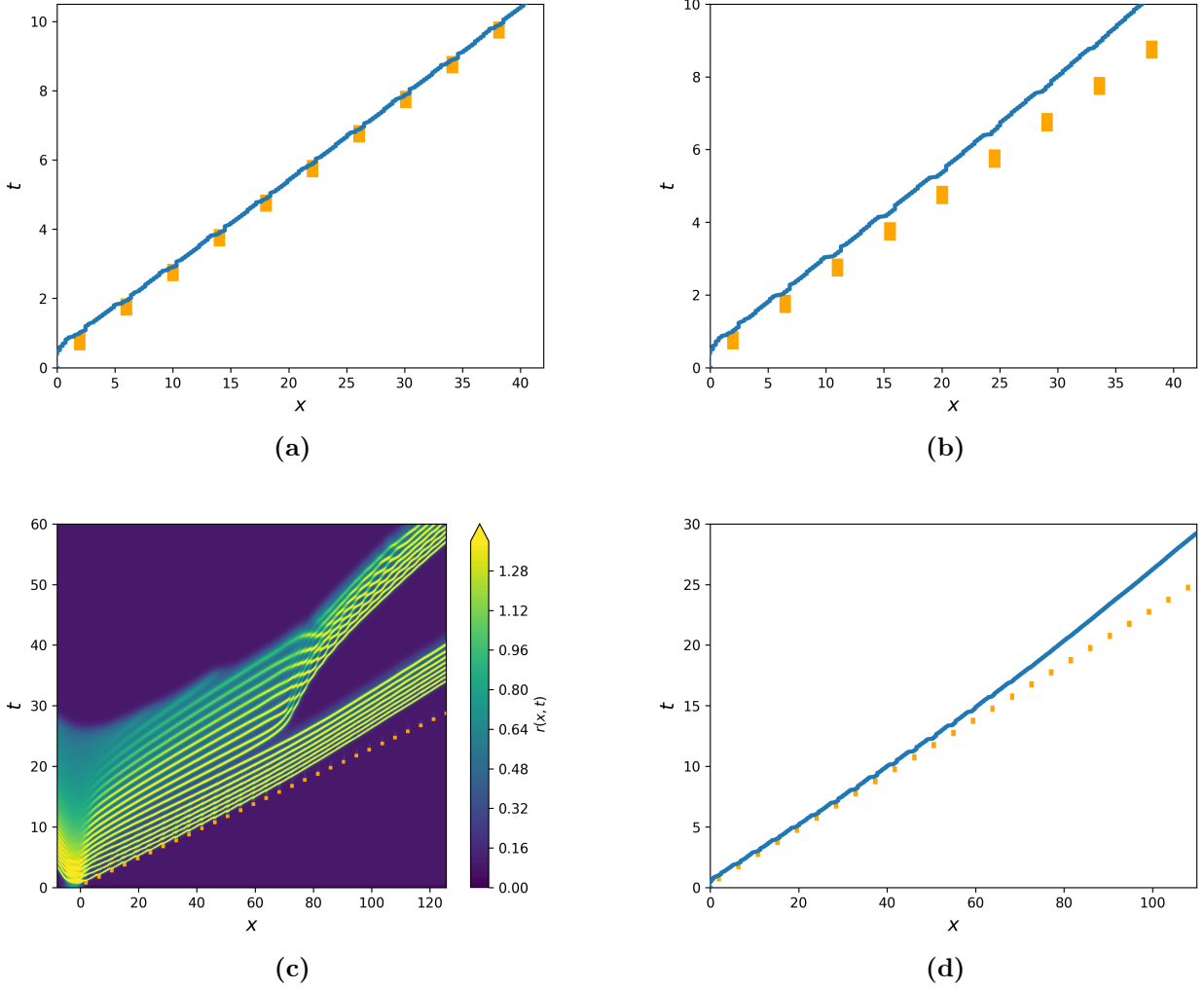


Figure 7.3: Parameters in (a) were $\Delta = 1$, $J = 20$, $\bar{\eta} = -5$, $\alpha = 0.9$, $\beta = 0.2$, $q_0 = 2$, $\tau_d = 20$, and $\tau_q = 30$. Moreover, the stimulus has parameters $c = 3.32$, $\Delta_c = 0.7$, $\Delta_x = 1$, $\varepsilon = 5$, $x^* = 0.5c + 0.8$, and $T_{\text{on}} = T_{\text{off}} = 0.5$. In (b) we set $\Delta_c = 1.1$ and $\varepsilon = 4$, and $T_{\text{on}} = 0.4$ and $T_{\text{off}} = 0.6$, leading to failed entrainment. In (c) the stimulus has parameters $\Delta_c = 1.1$, $\varepsilon = 5$, and otherwise the same as (a).

Chapter 8

Discussion and Conclusion

In this thesis, we have looked at mathematical models of the brain, with a particular focus on bumps representing persistent neural activity in working memory (WM). Our investigation began with the foundational neural field model by Wilson and Cowan [19]. This phenomenological population model has been widely successful and is well described in literature. It was expanded to include the dynamics of short-term synaptic plasticity (STSP), to obtain a model describing neural activity including changes in synaptic strength in an activity-dependent manner.

Later, we pursued the next-generation neural models. The two main reasons for doing so was the missing considerations of intrinsic noise and correlations in the Wilson and Cowan-type model, and to move away from derivations from heuristic arguments. To that end, we turned our attention to the firing-rate equations (FREs) outlined in [36] by Montbrió *et al.*. The firing rate function, known experimentally to be a sigmoid-shaped function, no longer needs to be provided. In the FREs the firing rate becomes an emergent property. These equations moreover exhibit the synchronizing and desynchronizing behavior of neurons. Furthermore, they are derived directly from a network of quadratic integrate-and-fire (QIF) neurons. This system is hence more biophysically grounded. A rather intriguing consequence is that it allows for a dual perspective. Namely, a macroscopic (FREs) state of a neuronal population and corresponding microscopic (QIF) view where individual neuron spikes can be observed.

As a next step, we reconciled the previous two by admitting STSP effects to the FREs. This endeavor is, to the best of our knowledge, not described in existent literature. The results showed how the formation of bumps is altered upon active STSP. Namely, bumps are more pronounced (wider and higher) when facilitation is dominant, while bumps are less pronounced (narrower and lower) in case of dominant depression. With regards to stability, we concluded that the phase diagram was preserved in some aspects with the including of STSP, featuring regions of low-activity stable nodes, generally high-activity stable foci, and bistability. However, the size of these regions altered when changing synaptic rate parameters. For instance, the stable focus region expanded considerably, transforming what was originally a stable node region. This mostly occurred when looking at non-zero equal rate parameters, and otherwise was more pronounced

when depression was dominant. This highlights the interplay effect of depression and facilitation. Notice that in a stable focus region the spectrum of the associated Jacobian matrix contains a complex conjugate pair. However, the complex part was generally small, resulting in only very weak spiraling in these expanded regions.

We also looked at the effects of synaptic depression in being able to achieve the high activity state in bistability regions. Sufficient depression makes this high state unattainable compared to no synaptic effects. In a particular instance pertaining this, we considered an intermittently turned on uniform stimulus in bistability circumstances. The results were that synaptic depression can ultimately cause the system to be locked in the low activity state, even if the first few trials were successful, if the system cannot restore properly between pulses.

We further explored the capabilities of the next-generation model with STSP through an application to wave entrainment. The model successfully captured both successful and failed entrainment of a traveling pulse by an apparently moving stimulus. As an important result, we highlight that the oscillatory behavior of the FREs (synchronizing and desynchronizing of neurons) can lead to internal (partial) entrainment failure under particular parameter regimes. The waves produced by the FREs themselves as part of the traveling pulse can thereby split apart, where the trailing part returns to its natural speed, while the front might still be entrained to an apparently moving object.

Now, we make some remarks on the work, general limitations, and outlooks for further resource. Notice that in this work we have not considered introducing noise in the system, and also drift instabilities could be looked into for further developments. Also, from a broader perspective, it should be acknowledged that the cortex, which has been our focus in the models, is obviously not representing the entire brain. Many vital functions depend on functions of and interactions with other structures like the thalamus or cerebellum. Secondly, notice that the work presented here highlights the effects due to STSP in the framework of WM. But, one could consider to incorporate yet other dynamics, making them more specific to other brain features. Furthermore, note that the FREs are derived from the QIF model, which is itself an abstraction of microscopic dynamics. Future research could potentially derive macroscopic descriptions from more complex and even more biophysically detailed models.

Appendix A

Simulation Code

General Code

The following serves as a basis for many simulations. It encompasses the circulant matrix, the dynamical system (in `rhs` function), and the its numerical solution using `solve_ivp` from the SciPy library. For particular simulation instances, we can change the parameters, or the definitions of `wFun` and `rhs` for instance.

```
import numpy as np
import matplotlib.pyplot as plt
from scipy.integrate import solve_ivp

# Functions and parameters
def wFun(x):
    return (1 - np.abs(x)) * np.exp(-np.abs(x))

def fFun(u):
    return 1 / (1 + np.exp(-mu * (u - theta)))

mu = 50
theta = 0.2

# Spatial grid and circulant matrix
n = 2*10
L = 10 * np.pi
h = 2 * L / n
x = -L + np.arange(n) * h

M = np.zeros((n, n))
y = wFun(x) * h
iShift = np.arange(-n // 2, n // 2)
for i in range(n):
    M[i, :] = np.roll(y, int(iShift[i]))

# Right-hand side of the dynamical system
def rhs(t, u):
    u = -u + np.dot(M, fFun(u))
```

```

    return u

# Solving
u0 = 0.01 * np.random.randn(n)
tspan = (0, 100)
sol = solve_ivp(rhs, tspan, u0, method='RK45',
               t_eval=np.linspace(tspan[0], tspan[1], 1000))

# Plotting (contour plot)
fig = plt.figure()
X, T = np.meshgrid(x, t)
plt.contourf(X, T, U, cmap='viridis')
plt.xlabel('$x$')
plt.ylabel('$t$')
plt.colorbar(label='u(x,t)')
plt.show()

```

For the next-generation neural model, the `rhs` looks like:

```

def rhs(t, x):
    r, v, d, q = x

    dr = Delta/np.pi + 2*r*v
    dv = v**2 + etabar - np.pi**2 * r**2 + J*d*(1+q)*r
    dd = 1 - d - alpha*d*(1+q)*r
    dq = -q + beta*(q0-q)*r

    return (dr, dv, dd, dq)

```

Stability Analysis

The code below was used to assess the stability of the next-generation neural model with synaptic plasticity.

```

# Functions
def rhs(x):
    r, v, d, q = x

    dr = Delta/np.pi + 2*r*v
    dv = v**2 + etabar - np.pi**2 * r**2 + J*d*(1+q)*r
    dd = 1 - d - alpha*d*(1+q)*r
    dq = -q + beta*(q0-q)*r

    return (dr, dv, dd, dq)

def Jac(x):
    r, v, d, q = x

```

```

J11 = 2*v
J12 = 2*r
J13 = 0
J14 = 0

J21 = -2*(np.pi**2)*r + J*d*(1+q)
J22 = 2*v
J23 = J*r*(1+q)
J24 = J*d*r

J31 = -alpha*d*(1+q)
J32 = 0
J33 = -1 - alpha*r*(1+q)
J34 = -alpha*d*r

J41 = beta*(q0-q)
J42 = 0
J43 = 0
J44 = -1 - beta*r

return np.block([[J11, J12, J13, J14],
                  [J21, J22, J23, J24],
                  [J31, J32, J33, J34],
                  [J41, J42, J43, J44]])

def Jac_fsolve(x, J, etabar, Delta, alpha, beta)
# Omitted (same as above, but fsolve needs the parameters)

# Parameters and initial conditions
sqrt_JDelta_vals = np.linspace(0, 22, 100)
etabarDelta_vals = np.linspace(-12, 0, 100)
alpha_vals = [0, 0.01, 0.05, 0.1]
beta_vals = [0, 0.01, 0.05, 0.1]

stability_types = np.zeros((len(sqrt_JDelta_vals),
                             len(etabarDelta_vals)))
# 0: unstable
# 1: stable node
# 2: stable focus
# 3: bistable

def find_steady_states(J, etabar, Delta, alpha, beta):
    initial_conditions = [
        (2 * np.sqrt(Delta), 1e-4),
        (1e-4, -3 * np.sqrt(Delta)),
    ]

```

```

steady_states = []
for r_init, v_init in initial_conditions:
    d_init = 1
    q_init = 0
    steady_state, info, ier, mesg = fsolve(rhs,
        x0=(r_init, v_init, d_init, q_init),
        args=(J, etabar, Delta, alpha, beta),
        fprime=Jac_fsolve, maxfev=1000, full_output=True)
    if ier == 1:
        new_state = True
        for existing_state in steady_states:
            if np.allclose(steady_state, existing_state, atol=1e-5):
                new_state = False
                break
        if new_state:
            steady_states.append(steady_state)

return steady_states

all_stability_types = np.zeros((n_alpha, n_beta,
    len(sqrt_JDelta_vals), len(etabarDelta_vals)))

for a_idx, alpha in enumerate(alpha_vals):
    for b_idx, beta in enumerate(beta_vals):
        stability_types = np.zeros((len(sqrt_JDelta_vals),
            len(etabarDelta_vals)))

        for i, sqrt_JDelta in enumerate(sqrt_JDelta_vals):
            J = sqrt_JDelta * np.sqrt(Delta)

            for j, etabarDelta in enumerate(etabarDelta_vals):
                etabar = etabarDelta * Delta

                # Find all steady states
                steady_states = find_steady_states(J, etabar,
                    Delta, alpha, beta)

                if len(steady_states) > 0:
                    all_stable = True
                    for state in steady_states:
                        Jacobian = Jac(state)
                        eigvals = np.linalg.eigvals(Jacobian)
                        if np.any(np.real(eigvals) > 0):
                            all_stable = False
                            break

                    if all_stable and len(steady_states) == 2:

```

```

        stability_types[i,j] = 3 # bistable

    else:
        # Check if at least one steady state is stable
        stable_found = False
        for state in steady_states:
            Jacobian = Jac(state)
            eigvals = np.linalg.eigvals(Jacobian)
            if np.all(np.real(eigvals) < 0):
                stable_found = True
                if np.any(np.abs(np.imag(eigvals))
                           > 1e-10):
                    # stable focus
                    stability_types[i,j] = 2
                else:
                    # stable node
                    stability_types[i,j] = 1

        if not stable_found:
            stability_types[i,j] = 0
    else:
        print("Warning")
        stability_types[i,j] = 0

all_stability_types[a_idx, b_idx] = stability_types

```

Spiking Neuron Model

In the simulation, we are modeling N all-to-all coupled QIF neurons modeled. We follow the thought of [48]. Recall that according to the QIF model:

$$\dot{V}_j = V_j^2 + I_j \quad \text{if } V_j \geq V_p, \text{ then } V_j \leftarrow V_r,$$

Whenever V_j reaches V_p (peak voltage), a spike is emitted by neuron j after which $V_j = V_r$ (reset voltage). The voltage is held at V_r for a time period $\tau_{\text{ref}} = 2/V_p$ (refractory period). The N neurons are spatially arranged in $\Omega = (-L/2, L/2]$ evenly, i.e.

$$\{x_i = iL/n - L/2 \mid i = 1, \dots, N\}.$$

Moreover, the input current is

$$I_j = \eta_j + Js(t) + I(t).$$

Numerically, the model is integrated using the Euler method:

$$V_j^{n+1} = V_j^n + (V_j(t)^2 + I_j(t))h.$$

The mean synaptic activation $s_i(t)$ is the most challenging to incorporate numerically:

$$s_i(t) = \frac{1}{N} \sum_{j=1}^N w(x_i, x_j) \sum_{k|t_j^k < t} \int_{-\infty}^t dt' \alpha_\tau(t - t') \delta(t - t_j^k),$$

where t_j^k is the time of the k th spike of neuron j . Doing multiplications directly is computationally very expensive. However, we can use the DFT (discrete Fourier transform).

Note that the function $\alpha_\tau(t')$ describes the synaptic current response to a single presynaptic spike with τ being the timescale. We have:

$$\alpha_\tau(t) = \frac{1}{\tau} [\mathcal{H}(\tau - t)], \quad H \text{ Heaviside.}$$

This means a spike from neuron j at time t_j^k contributes $1/\tau$ to $s(t)$ for the mentioned time:

$$s_i(t) = \frac{1}{N\tau} \{\text{\#spikes that occurred in the sliding time window}\} w(x_i, \cdot)$$

The sliding time window is of size δt .

On the next page is the pseudo code for the simulation of the spiking neuron network, see *Algorithm 2*. Using this, we can elaborate some more.

The synaptic activation $s_i(t)$ is weighted by a kernel $w(x_j, x_k)$, which purely depends on the distance between neuron x_j and x_k (w shows translational symmetry). This is exploited in order to circumvent long computational times. The modified kernel `wFun(delx)` (see *Algorithm 2*) is an array (or, it can still be seen as a function with the indices, being quantized distances, as input) containing the synaptic weights for increasing distances. Note that `delx` is the vector (only) containing the distances. As a toy example, assume that $N = 8$, then:

- `delx[0 : N//2 + 1] = [0, h, 2h, 3h, 4h]`
- `delx[N//2 + 1 :] = [-3h, -2h, -h]`

$\implies \text{delx} = [0, h, 2h, 3h, 4h, -3h, -2h, -h]$. Then, `wdelx = wFun(delx)` where `wFun(x)` is the synaptic kernel function. Finally, we take $\mathcal{F}(\text{wdelx})$ which is then used in the convolution allowing us to find $s_i(t)$.

For $s_i(t)$, notice we had said it includes the term

$$\{\text{\#spikes that occurred in the sliding time window}\}$$

where the sliding window is of size $\delta t = 2 \times 10^{-2}$. Notice that the step size is $\Delta t = 10^{-4}$, and hence we define `window_steps =  $\delta t / \Delta t = 200$` . So, a window is fully filled after 200 simulation steps. Since we only need the spiked neurons *in* the window, we save the spiked neurons as follows:

$$\text{spike_history[:, buffer]} = \text{spiked} \quad \text{buffer} = \text{buffer} + 1 \mod \tau_{\text{steps}}$$

meaning `sum spike_history` always returns the spiked neurons within the sliding window.

Algorithm 2 Python simulation for spiking neuron network

```

1: Input:
2:    $N = 10^4$ : #neurons
3:    $L = 20$ : length of grid
4:    $V_p = 100 = -V_r$ : spike threshold and reset value
5:    $J = 15\sqrt{2}$ ;  $\Delta = 2$ ;  $\bar{\eta} = -10$ : synaptic coupling strength; width; mean external drive
6:    $\tau = 10^{-3}$ : synaptic time constant
7:    $T = 20$ : total simulation time
8:    $dt = 10^{-4}$ ;  $\tau_{\text{steps}} = 10$ : time step for Euler integration;  $\tau/dt$ 
9:
10:   $h = 2L/n$ 
11:   $x = -L + \text{np.arange}(N + 1) * h$ 
12:   $x = x[:-1]$  ▷ remove last point (periodic)
13:
14:   $\eta = \bar{\eta} + \Delta \tan[\pi/2(2j - N - 1)/(N + 1)]$ 
15:   $\text{np.random.shuffle}(\eta)$ 
16:   $v = \text{np.ones}(N) * (-0.1)$ 
17:
18: Init: ▷ ensuring periodicity; setup for convolution
19:   $\text{indices} = [0, 1, \dots, N - 1]$ 
20:   $\text{delx}[0 : N/2 + 1] = \text{indices}[0 : N/2 + 1] \cdot h$ 
21:   $\text{delx}[N/2 + 1 :] = (\text{indices}[N/2 + 1 :] - N) \cdot h$ 
22:   $\text{wdelx} = \text{wFun}(\text{delx})$  ▷ where wFun is synaptic kernel
23:   $\text{ft} = \mathcal{F}(\text{wdelx}).\text{real}$ 
24:
25: for  $i$  in range( $T/dt$ ) do
26:    $t = i \cdot dt$ 
27:    $\text{buffer} = i \bmod \tau_{\text{steps}}$ 
28:
29:    $\text{sum\_recent} = \sum \text{spike\_history}$ 
30:   if any recent spikes then
31:      $\text{fft\_sum\_recent} = \mathcal{F}(\text{sum\_recent})$ 
32:      $\text{convolution} = \mathcal{F}^{-1}(\text{fft\_sum\_recent} * \text{ft}).\text{real}$ 
33:   else
34:      $\text{convolution} = 0$ 
35:    $s_i(t) = \frac{1}{N\tau} \text{convolution}$ 
36:
37:    $\text{active} = \{\text{neuron indices that are active, i.e. where } t \geq t_{\text{refractory\_end}}\}$ 
38:    $\text{rhs}[\text{active}] = v[\text{active}]^2 + \eta[\text{active}] + Js_i[\text{active}] + I_{\text{external}}[\text{active}]$ 
39:    $v = v + \text{rhs} \cdot dt$ 
40:
41:    $\text{spiked} = \{\text{indices of neurons that are past the threshold}\}$ 
42:   for  $i$  in  $\text{spiked}$  do
43:     Register spike index and time for plotting
44:    $v[\text{spiked}] = V_r$ 
45:    $t_{\text{refractory\_end}}[\text{spiked}] = \text{spike\_time} + 2/V_p$ 
46:
47:    $\text{spike\_history}[:, \text{buffer}] = \text{spiked}$ 

```

We now provide the full code:

```

import numpy as np
import matplotlib.pyplot as plt
from numpy.fft import fft , ifft

length = 20
n = 10**5
L = length/2
h = 2*L/n

T = 20
dt = 1e-4
V_p = 100
V_r = -100
tau_ref = 2/V_p

eta_bar = -10
Delta = 2
J_val = 15*np.sqrt(2)
alpha = 0.05
beta = 0.01
q0 = 2

tau_synaptic = 1e-3
del_t=2e-2
window_steps = 200
num_to_plot = 1000

# Spatial grid
x_coords = -L + np.arange(n+1) * h
x_coords = x_coords[0:-1]
N_neurons = len(x_coords)

j_indices = np.arange(1, N_neurons + 1)
eta = eta_bar + Delta * np.tan((np.pi/2)
    * (2*j_indices - N_neurons - 1) / (N_neurons + 1))
np.random.shuffle(eta)
v = np.ones(N_neurons) * (-0.1)

def wFun(delx):
    # alternative is: 0.5 * np.exp(-np.abs(delx))
    return np.exp(-np.abs(delx)) - (1/4)*np.exp(-np.abs(delx)/2)

k_indices = np.arange(N_neurons)
delx_vals = np.zeros_like(k_indices, dtype=float)
delx_vals[0 : N_neurons//2 + 1] = k_indices[0 : N_neurons//2 + 1] * h
delx_vals[N_neurons//2 + 1 :] =

```

```

(k_indices[N_neurons//2 + 1 :] - N_neurons) * h
wdelx = wFun(delx_vals)
ft = fft(wdelx).real

spike_hist = np.zeros((N_neurons, window_steps), dtype=float)
output_act = np.zeros(N_neurons)
t_refractory_end = np.zeros(N_neurons)

# External current function
def IextFunc(t, n, x):
    I_vals = np.zeros(n)
    if 0 <= t <= 5:
        filt = (x >= -2.5) & (x <= 2.5)
        I_vals[filt] = 5
    return I_vals

spike_times_list = []
spike_neuron_indices = []
d_neurons = np.ones(N_neurons)
q_neurons = np.zeros(N_neurons)
n_steps = int(T / dt)
sim_start_time = timer.time()

for step in range(n_steps):
    t_current = step * dt
    window_index = step % window_steps

    sum_recent = np.sum(spike_hist, axis=1)
    output_act = sum_recent

    if np.any(output_act):
        fft_output_act = fft(output_act)
        conv = fft_output_act * ft
        conv2 = ifft(conv).real
    else:
        conv2 = np.zeros(N_neurons)

    s_iu = (1/(N_neurons*tau_synaptic))*conv2

    dvdt = np.zeros(N_neurons)
    dddt = np.zeros(N_neurons)
    dqdt = np.zeros(N_neurons)

    non_refractory_filt = (t_current >= t_refractory_end)
    active_indices = np.where(non_refractory_filt)[0]

    if len(active_indices) > 0:

```

```

I_ext = IextFunc(t_current, N_neurons, x_coords)

v_active = v[active_indices]
eta_active = eta[active_indices]

fullJ = J_val * s_iu[active_indices]
I_ext_active = I_ext[active_indices]

dvdt[active_indices] = eta_active + v_active**2
+ fullJ + I_ext_active
dddt[active_indices] = (1 - d_neurons[active_indices]
- alpha * d_neurons[active_indices]
* (1 + q_neurons[active_indices]) * s_iu[active_indices]) / tau_d
dqdt[active_indices] = (-q_neurons[active_indices]
+ beta * (q0 - q_neurons[active_indices])
* s_iu[active_indices]) / tau_q

v += dt * dvdt

# Handle spikes
spiked_filt = (v >= V_p) # neurons that crossed threshold
spiked_indices = np.where(spiked_filt)[0]
spiked_num = len(spiked_indices)

spiked = np.zeros(N_neurons, dtype=float)

if spiked_num > 0:
    spiketime = t_current + dt
    for idx in spiked_indices:
        spike_times_list.append(spiketime)
        spike_neuron_indices.append(idx)
        spiked[idx] = 1

    v[spiked_indices] = V_r
    t_refractory_end[spiked_indices] = spiketime + 2/V_p

spike_hist[:, window_index] = spiked

s_i_for_dvdt = np.zeros(N_neurons)
s_i_for_dvdt[active_indices] = fullJ

# Plotting
sample_indices = np.sort(np.random.choice(N_neurons,
size=300, replace=False))
filt = np.isin(spike_neuron_indices, sample_indices)
t_filtered = spike_times_list[filt]
x_filtered = x_coords[spike_neuron_indices[filt]]

```

```
fig = plt.figure(figsize=(12, 6))
if t_filtered:
    plt.scatter(x_filtered, t_filtered, s=0.5, marker='.',
                color='black', alpha=0.5)
plt.xlabel('$x$', fontsize=14)
plt.ylabel('$t$', fontsize=14)
plt.ylim(0, T)
plt.xlim(-L, L)
plt.show()
```

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